WHERE ESTIMATORS ARE EXPOSED: STATE-INDEXED INFLUENCE SECOND MOMENTS

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Standard errors average influence contributions over observations, hiding where a fixed estimator is most exposed. This paper studies state-indexed conditional second moments of influence contributions across calendar time, support, and geography. Relative to a reference moment, ridge-whitened operators, economic probes, and bounded stress transforms summarize local amplification for high-dimensional paths and surfaces. Conditional covariance is a special case obtained by state-centering. Applications to local projections, treatment effects, and event studies reveal pronounced variation across states and estimand directions. The diagnostic complements, but does not replace, HAC, cluster-robust, or other sampling-covariance calculations.

KEYWORDS: state-indexed influence second moments, influence-function second moments, operator-valued kernels, high-dimensional inference, local projections, event studies, treatment effects.

1. INTRODUCTION

Empirical economics increasingly estimates objects that are not single coefficients. Local projections produce impulse-response paths and surfaces over horizons, outcomes, and shocks. Event studies estimate dynamic treatment-effect profiles across cohorts, event times, and locations. Program-evaluation studies report global and subgroup effects for policy-relevant populations. In each case, the estimand is a structured vector, path, or surface, and the observations that shape it need not contribute uniformly across time, geography, cohort, or support.

Standard errors and confidence bands necessarily compress those contributions into an aggregate covariance object. That compression is appropriate for inference, but it hides where a fixed estimator is most exposed. A heteroskedasticity- and autocorrelation-consistent (HAC) standard error, a cluster-robust variance matrix, a bootstrap band, or a simultaneous confidence band tells the researcher how uncertainty should be reported under a chosen sampling-covariance procedure. It usually does not tell the researcher which dates, support regions, cohorts, locations, or directions of the estimand generate the squared influence being aggregated. A local-projection surface can have stable point estimates even when its response-score outer products are concentrated in a few historical episodes. An event-study estimator can have county-clustered standard errors even when its observation-level score moments are concentrated in particular region-year cells. A treatment-effect estimator can target one vector of global and subgroup ATEs even when different support regions load on different coordinates of that vector. These patterns concern the influence geometry of a chosen estimator, not alternative state-specific causal effects.

The practical value of exposing this geometry is not that every exposed state can be repaired. Some can; many cannot. Its value is that it changes post-estimation practice from undirected robustness checking to targeted empirical auditing. Without such a diagnostic, a researcher who worries about fragility may try a long list of generic checks: alternative windows, alternative controls, alternative clusters, alternative bandwidths, alternative transformations, or alternative subgroups. Some of these checks may be useful, but they are not guided by where the estimator is actually exposed. A state-indexed influence diagnostic tells the researcher where to look first, which components of a high-dimensional estimand carry the exposure, and what type of

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robustness exercise is relevant. In all cases, the diagnostic complements HAC, cluster-robust, bootstrap, and simultaneous-band procedures by revealing what those covariance calculations aggregate away.

The practical implications depend on the state variable. When the state indexes support in a cross section, high amplification may point to remediable design problems: weak overlap, small local effective sample size, large weights, or a mismatch between treated and comparison observations. The researcher may respond by changing the target population, trimming support, collecting more comparison data, modifying weights, or reporting estimand-specific fragility. In a treatment-effect application, for example, high amplification in a low-overlap region is not merely a descriptive feature of the sample. It can indicate that a global or subgroup ATE estimate is relying heavily on observations in a part of the support where the empirical comparison is thin. The appropriate response may be to inspect overlap diagnostics, report local effective sample size, restrict the estimand to common support, or make clear that some subgroup coordinates are more fragile than others.

When the state indexes calendar time in a historical macroeconomic application, the implication is different. A high-amplification month is not a request to sample more low-amplification months. In a monetary-policy local projection, the historical sample path is largely fixed; the researcher cannot generate more financial crises, more tightening cycles, or more tranquil months in a way that repairs the design. A high-amplification month is instead a warning that a fixed response surface is empirically exposed to a particular historical episode. The relevant actions are to inspect the episode, decompose the exposure across shocks, residuals, outcomes, and horizons, compare plausible alternative specifications, and report whether the main conclusions are robust to those targeted checks. Thus the diagnostic is useful even when the design cannot be repaired: it tells the researcher what part of the state space is doing the work.

This paper studies the conditional second moment of an estimator's influence contribution. If ψ_i is the influence contribution and S_i is an economically meaningful state, the basic object is

$$\mathbb{E}[\psi_i \psi_i' \mid S_i = s].$$

It measures total squared-influence intensity in state s. For iid data, averaging this object over the state distribution recovers the one-observation influence second moment and therefore the covariance component entering the usual asymptotic variance. The conditional second moment is thus a pre-aggregation object: it shows what is concealed when influence contributions are averaged before being reported as a variance matrix, a standard error, or a confidence band.

Conditional covariance is an important special case, not the general target. The conditional second moment combines within-state dispersion with the outer product of the conditional mean of the influence contribution. Removing that mean isolates within-state covariance, but it also removes systematic state-specific loading. Which object is appropriate depends on the question. Conditional centering asks how variable influence contributions are after holding state composition fixed. The uncentered second moment asks where total squared influence is located. The empirical applications below ask the latter question and therefore smooth raw or globally recentered score outer products.

This distinction also clarifies the relation to conventional inference. In an iid setting, the state-mass-weighted average of the conditional second moment has a direct sampling-covariance interpretation. Under serial or within-cluster dependence, however, valid sampling covariance also depends on cross-period or within-cluster score products. The time-series and panel applications in this paper deliberately study diagonal observation-level score second-moment fields. They are not state-dependent HAC or cluster-robust covariance estimators. HAC, cluster-robust, bootstrap, and other conventional covariance calculations remain the objects used for tests, confidence intervals, and simultaneous bands (Abadie et al., 2023, MacKinnon, Nielsen and Webb, 2023). The diagnostic sits beside those tools. It helps explain where the estimator is exposed, while conventional covariance estimators determine how uncertainty should be reported.

The diagnostic separates two questions that are often conflated. Influence intensity asks whether a state's normalized second moment is high relative to a reference moment. Influence contribution additionally accounts for how much population or sample mass the state receives. A rare state can exhibit large amplification while contributing little to an aggregate variance, whereas a common state with moderate amplification can contribute substantially. The maps developed here display intensity; state-mass weighting is required before interpreting them as contributions to an aggregate variance budget. This distinction matters for practice because high-amplification states should be audited, but they should not automatically be described as dominant contributors to the final variance unless their state mass is also accounted for.

The method takes the estimator, estimand, and state variable as given. The state may be calendar time, geography, cohort, event time, exposure intensity, a propensity-score region, a prognostic-score region, or a region-year cell. A reference second-moment operator summarizes the chosen state distribution. Ridge whitening compares each state moment with that reference while retaining all directions in the selected representation. Economic probes then summarize amplification for specified contrasts or blocks, and bounded stress transforms report how much probe weight lies in strongly amplified directions. Spectral decompositions are used only after estimation to display the response-surface patterns responsible for the estimated geometry.

This approach differs from classical observation-level influence and leverage diagnostics. Deletion and perturbation diagnostics ask which individual observations can move an estimate. The object here averages influence outer products within substantively defined states and asks where squared influence is concentrated and which directions of a fixed estimand carry it. The state is therefore part of the substantive design. Smoothing over calendar time, propensity support, or neighboring region-year cells has an economic interpretation; smoothing over an arbitrary row number does not.

The framework is especially useful for high-dimensional estimands. In local projections, recent work emphasizes joint inference, efficiency, cross-horizon restrictions, and system-based or Bayesian estimation of response surfaces (Jordà, 2005, Inoue, Jordà and Kuersteiner, 2026, Huber, Matthes and Pfarrhofer, 2026). The present diagnostic is complementary: it takes the estimated response surface as fixed and identifies dates and response-surface directions with unusually large diagonal score second moments. A high-amplification date does not by itself imply a different causal effect at that date, nor does it invalidate HAC inference. It tells the researcher where the fixed estimator is exposed and motivates a targeted audit of shocks, residuals, horizons, outcomes, lag structure, sample windows, and possible nonlinear or time-varying alternatives.

In event studies, the modern literature focuses on identification, aggregation, and inference under staggered adoption and heterogeneous treatment effects (Sun and Abraham, 2021, Callaway and Sant'Anna, 2021, Borusyak, Jaravel and Spiess, 2024, Roth, 2026). Conditional on a chosen estimator and estimand surface, this paper instead maps the region-year cells and economic blocks in which observation-level score second moments are amplified. These maps do not estimate region-year-specific treatment effects and do not replace clustered inference. They identify the cells and coordinates where the chosen event-study surface draws influence, which helps the researcher distinguish broad empirical support from reliance on a small number of substantively important cells.

In treatment-effect applications, propensity and prognostic scores index support regions for one fixed vector of global and subgroup ATE estimators; they do not define separate ATEs at each support location. This influence-based view complements the overlap and balancing-weight literature (Rosenbaum and Rubin, 1983, Li, Morgan and Zaslavsky, 2018). When amplification is high in a support region with weak overlap, large weights, or small local effective sample size, the diagnostic has direct design implications. It can motivate trimming, overlap weights, a change in the target population, additional data collection, or explicit reporting that a particular subgroup contrast is fragile. When amplification is high in a region with adequate support, the implication is different: the researcher should inspect residual behavior, local model fit, and which estimand coordinates load on that region.

The formal construction uses the asymptotic linear representation common to regular M-, Z-, GMM, smooth-extremum, semiparametric, and orthogonal estimators (Hansen, 1982, Newey and McFadden, 1994, van der Vaart, 1998, Chernozhukov et al., 2018). Outer products of influence contributions define a random operator-valued kernel, and their conditional expectations define positive operator-valued second-moment fields. The operator formulation accommodates vectors, functional parameters, and other high-dimensional estimands without requiring a population spectral cutoff. It also connects influence-function methods to operator-valued kernels and functional moment representations (Ichimura and Newey, 2015, Micchelli and Pontil, 2005, Minh, 2016, Tian, 2025).

The paper makes three contributions. First, it defines state-indexed influence second-moment geometry for a fixed estimator and separates the full conditional second moment from its within-state covariance component. Second, it develops feasible positive-semidefinite smoothers and interpretable summaries, including directional and block probes, a trace-normalized soft amplification index, and bounded stress curves. Third, it applies the framework to monetary-policy local projections, the LaLonde–Dehejia–Wahba training-data benchmark, and a hurricane event study. The applications show substantial variation across calendar time, support regions, and region-year cells, as well as across outcomes, horizons, and subgroup coordinates (Jordà, 2005, Jarociński and Karadi, 2020, LaLonde, 1986, Dehejia and Wahba, 2002, Deryugina, 2017).

The central claim is modest but useful. A standard error tells us how much uncertainty to attach to an estimate under a chosen sampling-covariance calculation. A state-indexed influence second moment tells us where the estimator's squared influence is located and which estimand directions carry it. Used together, the two tools give a more complete post-estimation workflow: valid inference from HAC, clustering, bootstrap, or simultaneous-band procedures, and a transparent exposure map showing where the empirical design is doing the work.

2. STATE-INDEXED INFLUENCE SECOND MOMENTS

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For each sample size n, let $W_{1,n}, \dots, W_{n,n}$ take values in a measurable space $(\mathcal{W}, \mathcal{A})$, and let $S_{i,n}$ take values in a researcher-specified state space $(\mathsf{S}, \mathcal{S})$. The state may encode calendar time, geography, cohort, event time, exposure, support, a graph node, or another substantively meaningful index. Let $\mathcal{H}$ be a real separable Hilbert space, $\theta_0 \in \mathcal{H}$ the target estimand, and $\widehat{\theta}_n \in \mathcal{H}$ an estimator. For $f, g \in \mathcal{H}$, write $(f \otimes g)h = f\langle g, h\rangle_{\mathcal{H}}$. Let $\mathcal{S}_1(\mathcal{H})$ and $\mathcal{S}_2(\mathcal{H})$ denote the trace-class and Hilbert-Schmidt operators, with norms $\|\cdot\|_1$ and $\|\cdot\|_{\mathrm{HS}}$; $\|\cdot\|_{\mathrm{op}}$ is the operator norm. Whenever an expression of the form $\mathbb{E}[X \mid S = s]$ is used, assume that $(\mathsf{S}, \mathcal{S})$ is standard Borel or, equivalently for the results below, assume directly that the indicated measurable conditional-expectation version exists.

ASSUMPTION 1: *There is a triangular array of strongly measurable $\mathcal{H}$-valued influence contributions $\{\psi_{i,n} : 1 \le i \le n\}$ such that*

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = n^{-1/2} \sum_{i=1}^{n} \psi_{i,n} + o_p(1) \quad \textit{in } \mathcal{H},$$

where $\mathbb{E}[\psi_{i,n}] = 0$ *and* $\sup_{i,n} \mathbb{E}\|\psi_{i,n}\|_{\mathcal{H}}^{2+\delta} < \infty$ *for some* $\delta > 0$. *For each* i, n, *there is a state-information sigma-field* $\mathcal{G}_{i,n} \subseteq \mathcal{F}$. *Any laws of large numbers or central limit theorems used below are imposed separately in the assumptions attached to the relevant result.*

This is the usual asymptotic linearity condition for regular M-, Z-, GMM, smooth-extremum, semiparametric, and orthogonal estimators. The information set $\mathcal{G}_{i,n}$ may equal $\sigma(S_{i,n})$, a temporal filtration, or a sigma-field generated by a state node and predetermined covariates.

2.1. *Random Operator-Valued Kernels*

Let $\mathcal{I}_n := \{1, \ldots, n\}$ be the finite labeled observation-index set; repeated state values are therefore permitted. An operator-valued kernel is a map $K^* : \mathcal{I}_n \times \mathcal{I}_n \to \mathcal{L}(\mathcal{H})$ satisfying

$$\sum_{r,q=1}^{N} \langle a_r, K^*(i_r, i_q) a_q \rangle_{\mathcal{H}} \geq 0 \tag{2.1}$$

for every finite collection $i_1, \ldots, i_N \in \mathcal{I}_n$ and $a_1, \ldots, a_N \in \mathcal{H}$. Following Tian (2025), a random kernel $k : \Omega \times \mathcal{I}_n \times \mathcal{I}_n \to \mathcal{L}(\mathcal{H})$ is positive definite in expectation when

$$\mathbb{E}\left[\sum_{r,q=1}^{N} \langle a_r, k(\omega, i_r, i_q) a_q \rangle_{\mathcal{H}}\right] \geq 0. \tag{2.2}$$

Write the resulting class as $rpd(\Omega, \mathcal{I}_n, \mathcal{H})$. If $k \in rpd(\Omega, \mathcal{I}_n, \mathcal{H})$, its mean is a deterministic positive operator-valued kernel; empirical averages of i.i.d. copies remain random positive kernels and converge weakly under the conditions in Tian (2025).

Write $\psi_n(i;\omega) := \psi_{i,n}(\omega)$ and define

$$k_n^{\psi}(\omega; i, j) := \psi_n(i;\omega) \otimes \psi_n(j;\omega). \tag{2.3}$$

PROPOSITION 1: *Suppose Assumption 1 holds and* $\mathbb{E}\|\psi_{i,n}\|_{\mathcal{H}}^2 < \infty$. *Then* k_n^{ψ} *is pathwise positive definite and belongs to* $rpd(\Omega, \mathcal{I}_n, \mathcal{H})$; $K_n^{\psi}(i,j) := \mathbb{E}[k_n^{\psi}(\cdot; i, j)]$ *is a deterministic positive operator-valued kernel; and, for every sub-sigma-field* $\mathcal{G} \subseteq \mathcal{F}$, $K_n^{\psi,\mathcal{G}}(i,j) := \mathbb{E}[k_n^{\psi}(\cdot; i, j) \mid \mathcal{G}]$ *is an* $\mathcal{S}_1(\mathcal{H})$*-valued conditional expectation admitting an almost-surely positive-definite version.*

PROOF: See Appendix A.1. *Q.E.D.*

The paper uses the conditional second moment of the influence contribution, $\mathbb{E}[\psi_{i,n} \otimes \psi_{i,n} \mid \mathcal{G}_{i,n}]$. This object is not generally a conditional covariance, because the conditional mean of $\psi_{i,n}$ need not be zero. To obtain the within-state covariance component, subtract that conditional mean:

$$\chi_{i,n} := \psi_{i,n} - \mathbb{E}[\psi_{i,n} \mid \mathcal{G}_{i,n}], \qquad K_{i,n}^{\psi,\mathcal{G}} := \mathbb{E}[\chi_{i,n} \otimes \chi_{i,n} \mid \mathcal{G}_{i,n}]. \tag{2.4}$$

Let $m_{i,n} := \mathbb{E}[\psi_{i,n} \mid \mathcal{G}_{i,n}]$. Since $\chi_{i,n}$ has conditional mean zero,

$$\mathbb{E}[\psi_{i,n} \otimes \psi_{i,n} \mid \mathcal{G}_{i,n}] = K_{i,n}^{\psi,\mathcal{G}} + m_{i,n} \otimes m_{i,n}, \tag{2.5}$$

and therefore $\mathbb{E}[\psi_{i,n} \otimes \psi_{i,n}] = \mathbb{E}[K_{i,n}^{\psi,\mathcal{G}}] + \mathbb{E}[m_{i,n} \otimes m_{i,n}]$.

Thus the full conditional second moment contains two pieces: the within-state covariance and the squared conditional mean. In an i.i.d. setting, averaging the full conditional second moment over states gives the usual one-observation influence covariance, because $\mathbb{E}[\psi_{i,n}] = 0$. Averaging only the centered component drops the between-state term $\mathbb{E}[m_{i,n} \otimes m_{i,n}]$, unless $m_{i,n} = 0$ almost surely.

State centering is therefore appropriate for a narrower question: how much within-state variation remains after treating differences in conditional means as fixed. The empirical applications below use the full second-moment target. If the influence contribution is already conditionally mean zero, the full and centered targets coincide.

Throughout the remaining theory, $K(s)$ denotes the positive state-indexed moment field declared by the researcher. It may be the full conditional second moment $\mathbb{E}[\psi_i \otimes \psi_i \mid S_i = s]$ or the centered conditional covariance $\mathbb{E}[\chi_i \otimes \chi_i \mid S_i = s]$. The operator results do not require conditional centering.

For the centered case, assume $\sigma(S_{i,n}) \subseteq \mathcal{G}_{i,n}$. Then $\mathbb{E}[\chi_{i,n} \mid S_{i,n}] = 0$, and, for $\mathcal{L}(S_{i,n})$-almost every s, the tower property gives

$$\mathbb{E}\Big[\mathbb{E}[\chi_{i,n} \otimes \chi_{i,n} \mid \mathcal{G}_{i,n}] \,\Big|\, S_{i,n} = s\Big] = \mathbb{E}[\chi_{i,n} \otimes \chi_{i,n} \mid S_{i,n} = s].$$

When the state is sufficient for the chosen moment field, write $K_{i,n} = K_n(S_{i,n})$; otherwise $K_n(s)$ denotes the corresponding coarser state projection.

2.2. *Full Hilbert-Space Relative Moment Geometry*

Suppress the triangular-array index in this subsection. Let $(\mathsf{S}, \mathcal{S}, \pi)$ be the target state space equipped with a probability reference measure π, let $K : \mathsf{S} \to \mathcal{S}_1^+(\mathcal{H})$ be the state moment field, and define the π-reference moment

$$C := \int_{\mathsf{S}} K(s)\, \pi(ds). \tag{2.6}$$

The measure π may be the population law of S, a design distribution, or a researcher-specified probability measure over state nodes. If $\pi \neq \mathcal{L}(S)$, then C is a reference-weighted moment and need not equal the estimator's sampling covariance; every average-one normalization below is relative to π. If $K(s)$ is the full conditional second moment, $\pi = \mathcal{L}(S)$, and observations are i.i.d., then C is the usual one-observation influence covariance. Under other reference laws, centered targets, or dependent sampling, C should be interpreted only as the declared reference moment. When $K(s)$ is defined as a conditional moment, $\pi \ll \mathcal{L}(S)$ is sufficient for it to be identified from the sampling law. If π assigns positive mass to an $\mathcal{L}(S)$-null set, the values of K on that set must instead be supplied as part of the researcher-specified reference target.

ASSUMPTION 2: *The field $s \mapsto K(s)$ is strongly measurable and Bochner integrable in $\mathcal{S}_1(\mathcal{H})$, $K(s) \succeq 0$ for π-almost every s, and* $0 < \operatorname{tr}(C) < \infty$.

LEMMA 1: *Under Assumption 2, C is positive, self-adjoint, and trace class, with* $\operatorname{tr}(C) = \int_{\mathsf{S}} \operatorname{tr}\{K(s)\}\, \pi(ds)$.

PROOF: Positivity and self-adjointness are preserved by Bochner integration, and trace is a bounded linear functional on $\mathcal{S}_1(\mathcal{H})$. *Q.E.D.*

The state geometry is represented by a Markov semigroup. This includes heat-flow smoothing on Euclidean support, graph diffusion, and one-dimensional diffusion on an ordered state.

ASSUMPTION 3: *There is a strongly continuous Markov contraction semigroup $\{P_r : r \geq 0\}$ on $L^1(\pi)$, with generator $\mathcal{L}_{\mathsf{S}}$ and transition kernels $p_r(s, du)$. It is positivity preserving, $P_r 1 = 1$, and π is invariant. Its kernel extension to $L^1(\pi; \mathcal{S}_1(\mathcal{H}))$ is also a strongly continuous contraction semigroup. For $\eta > 0$, set*

$$J_\eta F(s) := \eta(\eta I - \mathcal{L}_{\mathsf{S}})^{-1} F(s) = \eta \int_0^\infty e^{-\eta r} (P_r F)(s)\, dr, \qquad K_\eta := J_\eta K. \tag{2.7}$$

The convention $J_\infty = I$ and $K_\infty = K$ denotes the unsmoothed target.

The vector-valued continuity clause follows from the scalar assumptions: it holds first for simple tensor fields $F(s) = f(s)T$, then for finite simple fields, and finally for every Bochner-integrable field by density and contractivity. The following lemma uses the concept of relative-domination $\preceq$. The relative-domination condition says that the state moment $K(s)$ is absolutely continuous with respect to the reference moment C: directions with zero reference second moment cannot carry state-specific second moment, and directions with small reference moment can be amplified only up to the integrable envelope $M(s)$.

LEMMA 2: *Under Assumptions 2 and 3, $K_\eta(s) \in \mathcal{S}_1^+(\mathcal{H})$ almost everywhere, $\int_{\mathsf{S}} K_\eta(s)\, \pi(ds) = C$, and J_η is a contraction on $L^1(\pi; \mathcal{S}_1)$. Moreover, $\|K_\eta - K\|_{L^1(\pi; \mathcal{S}_1)} \to 0$ as $\eta \to \infty$. If $K(s) \preceq M(s)C$ for some $M \in L^1_+(\pi)$, then*

$$K_\eta(s) \preceq M_\eta(s) C, \qquad M_\eta := J_\eta M. \tag{2.8}$$

PROOF: See Appendix A.2. *Q.E.D.*

ASSUMPTION 4: *There is an $M \in L^1_+(\pi)$ such that*

$$0 \preceq K(s) \preceq M(s) C \tag{2.9}$$

for π-almost every s.

Let $\mathcal{H}_C := (\ker C)^\perp = \overline{\operatorname{ran}(C^{1/2})}$. Because $C^{-1/2}$ is generally unbounded on an infinite-dimensional $\mathcal{H}_C$, relative moment is first defined as a form.

THEOREM 1—Full Hilbert-space relative moment: *Suppose Assumptions 2-4 hold. For each $\eta \in (0, \infty]$ and almost every s, the form*

$$a_{\eta,s}(C^{1/2} g, C^{1/2} h) := \langle g, K_\eta(s) h \rangle_{\mathcal{H}} \tag{2.10}$$

extends uniquely to a bounded positive symmetric form on $\mathcal{H}_C$. Equivalently, there is a unique weakly measurable, bounded, positive self-adjoint operator $A_\eta^\infty(s) : \mathcal{H}_C \to \mathcal{H}_C$ such that

$$\langle g, K_\eta(s) h \rangle_{\mathcal{H}} = \langle C^{1/2} g, A_\eta^\infty(s) C^{1/2} h \rangle_{\mathcal{H}}, \qquad g, h \in \mathcal{H}. \tag{2.11}$$

Moreover,

$$0 \preceq A_\eta^\infty(s) \preceq M_\eta(s) I_{\mathcal{H}_C}, \qquad \|A_\eta^\infty(s)\|_{\mathrm{op}} \leq M_\eta(s), \tag{2.12}$$

and its π-average is the identity weakly:

$$\int_{\mathsf{S}} \langle x, A_\eta^\infty(s) y \rangle_{\mathcal{H}}\, \pi(ds) = \langle x, y \rangle_{\mathcal{H}}, \qquad x, y \in \mathcal{H}_C. \tag{2.13}$$

PROOF: See Appendix A.3. *Q.E.D.*

The relative moment is the state moment expressed in the units of the reference moment. The reference operator C records the average second-moment geometry of the influence contribution, while $K_\eta(s)$ records the corresponding state-s geometry after smoothing. If C were finite-dimensional and nonsingular, the comparison would simply be the whitened matrix $C^{-1/2}K_\eta(s)C^{-1/2}$. The operator $A_\eta^\infty(s)$ is the Hilbert-space version of that same comparison. It shows, after putting all estimand directions on a common reference scale, how large the local second moment is at state s. The neutral case is $A_\eta^\infty(s) = I$: state s has the same second-moment geometry as the reference. Eigenvalues or directional ratios above one indicate amplified squared influence relative to the reference; values below one indicate attenuation. The directions with the largest amplification may be mixtures of the original coordinates rather than individual coefficients, horizons, outcomes, or subgroups. This would be captured by off-diagonal terms in $A_\eta^\infty(s)$.

In infinite dimension, this comparison cannot be formed by naively inverting C. Some directions may have zero or arbitrarily small reference moment. The space $\mathcal{H}_C$ keeps only the directions supported by the reference moment, and the domination condition below guarantees that the state moment can be compared to C without creating an unbounded object. The theorem formalizes this comparison. Its average-identity conclusion says that, once measured in reference units, the relative moment averages to the neutral geometry over the reference state law π.

The operator $A_\eta^\infty(s)$ is therefore the full relative second-moment geometry at state s. It is an operator-valued comparison, not yet a scalar index. In infinite dimension it need not be compact or trace class, so $\operatorname{tr}(A_\eta^\infty(s))$ is not a canonical summary. Scalar summaries require a choice of economic direction, block, or weighting over directions. The trace-class probes below provide that choice.

COROLLARY 1—Probe principle: *Under Theorem 1, every $Q \in \mathcal{S}_1^+(\mathcal{H}_C)$ with $\operatorname{tr}(Q) = 1$ defines*

$$\tau_{\eta,Q}(s) := \operatorname{tr}\{QA_\eta^\infty(s)\}, \qquad 0 \le \tau_{\eta,Q}(s) \le M_\eta(s), \qquad \int_{\mathsf{S}} \tau_{\eta,Q}(s)\,\pi(ds) = 1. \tag{2.14}$$

For the block case below, let $\mathcal{H}_B$ be a separable Hilbert space, let $S_B \in \mathcal{L}(\mathcal{H}, \mathcal{H}_B)$ be bounded, and set $d_B := \operatorname{tr}(S_B C S_B^*)$. Important special cases are:

$$Q_g := \frac{C^{1/2}g \otimes C^{1/2}g}{\langle g, Cg\rangle}, \qquad r_\eta(s;g) := \tau_{\eta,Q_g}(s) = \frac{\langle g, K_\eta(s)g\rangle}{\langle g, Cg\rangle}, \tag{2.15}$$

$$Q_B := \frac{C^{1/2}S_B^* S_B C^{1/2}}{d_B}, \qquad \tau_{\eta,B}(s) := \tau_{\eta,Q_B}(s) = \frac{\operatorname{tr}\{S_B K_\eta(s) S_B^*\}}{\operatorname{tr}(S_B C S_B^*)}, \tag{2.16}$$

whenever the displayed denominators are positive. For $\rho > 0$, let $S_\rho := (C + \rho I)^{-1/2}C^{1/2}$, $d_\rho := \operatorname{tr}\{C(C+\rho I)^{-1}\}$, and $Q_\rho := S_\rho^2/d_\rho$. Then

$$A_{\eta,\rho}(s) := (C+\rho I)^{-1/2}K_\eta(s)(C+\rho I)^{-1/2} = S_\rho A_\eta^\infty(s) S_\rho, \tag{2.17}$$

and the soft amplification index

$$\tau_{\eta,\rho}^{\mathrm{soft}}(s) := \frac{\operatorname{tr}\{A_{\eta,\rho}(s)\}}{d_\rho} = \tau_{\eta,Q_\rho}(s) \tag{2.18}$$

is well defined and has π-average one.

PROOF: See Appendix A.4; the soft-whitening calculation is in Appendix A.6. *Q.E.D.*

The probes in Corollary 1 are best understood as economic lenses applied to the relative moment operator. The operator $A_\eta^\infty(s)$ describes how the state-s influence second moment differs from the reference moment after putting all estimand directions on a common reference scale. A probe Q chooses which directions of the estimand to look at. Because Q is positive and has trace one, it acts like a probability weighting over reference-scaled directions. The scalar $\tau_{\eta,Q}(s)$ is therefore the probe-weighted average amplification at state s: values above one mean that the probed directions have larger local squared influence than under the reference moment, while values below one mean attenuation. The rank-one probe Q_g asks this question for a single economic contrast g, such as a particular horizon, treatment-effect contrast, or linear combination of outcomes. The block probe Q_B asks the same question for a whole group of coordinates, such as an outcome block, horizon range, subgroup family, or event-time window. The soft probe Q_ρ is a default full-space summary: it averages over all reference-supported directions while smoothly downweighting directions whose reference moment is small. The ridge parameter ρ controls how aggressively those weak-reference directions are downweighted.

For $\lambda > 0$, define the moment resolvent, inverse-amplification transform, and bounded stress transform as

$$\begin{aligned} R_{\eta,\lambda}(s) &:= \{\lambda I + A_\eta^\infty(s)\}^{-1}, \qquad\qquad B_{\eta,\lambda}(s) := \lambda R_{\eta,\lambda}(s), \\ Y_{\eta,\lambda}(s) &:= I - B_{\eta,\lambda}(s) = A_\eta^\infty(s)\{\lambda I + A_\eta^\infty(s)\}^{-1}. \end{aligned} \tag{2.19}$$

PROPOSITION 2—Resolvent and Yosida diagnostics: *Under Theorem 1, $R_{\eta,\lambda}(s)$ is positive and self-adjoint with norm at most λ^{-1}, while $0 \preceq B_{\eta,\lambda}(s), Y_{\eta,\lambda}(s) \preceq I$. In addition, $-A_\eta^\infty(s)$ generates the contraction semigroup $e^{-rA_\eta^\infty(s)}$ and*

$$R_{\eta,\lambda}(s) = \int_0^\infty e^{-\lambda r} e^{-rA_\eta^\infty(s)}\, dr. \tag{2.20}$$

Consequently, for nonzero x and trace-one $Q \succeq 0$, define $q_{\eta,\lambda}(s;x) := \langle x, Y_{\eta,\lambda}(s)x\rangle / \|x\|^2$ and $q_{\eta,\lambda,Q}(s) := \mathrm{tr}\{QY_{\eta,\lambda}(s)\}$; both lie in $[0,1]$.

PROOF: See Appendix A.5. *Q.E.D.*

Theorem 1 and Proposition 2 rely heavily on relative dominance as written. However, relative domination is sufficient, not necessary. The stronger condition turns the relative moment into a bounded operator: there are no domain choices, no unbounded inverse of C, and no additional measurability conditions beyond those already imposed. The closed-form extension below is mainly useful in infinite-dimensional or refining-discretization settings where the comparison with C remains meaningful but is not uniformly bounded. This can occur when C assigns very small reference second moment to localized directions and a state moment assigns relatively larger mass to those same directions. Examples include weak-overlap or vanishing-comparison probability regions in treatment-effect or difference-in-differences designs, and response-surface applications in which score energy concentrates in narrow horizon, maturity, or event-time bands. In such cases every fixed reported contrast may still have finite variance, and $K_\eta(s)$ may remain trace class, while no finite scalar $M(s)$ satisfies $K_\eta(s) \preceq M(s)C$. The proposition therefore records how to define a possibly unbounded relative moment and still obtain bounded resolvent and stress summaries.

PROPOSITION 3—Closed-form extension: *Fix η and a state s such that $K_\eta(s) \in \mathcal{S}_1^+(\mathcal{H})$. Suppose* $\ker C \subseteq \ker K_\eta(s)$ *and the form* (2.10) *is closable on* $\operatorname{ran}(C^{1/2})$. *Its closure determines a unique, possibly unbounded, positive self-adjoint operator $\mathfrak{A}_\eta(s)$. Then $-\mathfrak{A}_\eta(s)$ generates a strongly continuous contraction semigroup, and both $\lambda(\lambda I + \mathfrak{A}_\eta(s))^{-1}$ and $\mathfrak{A}_\eta(s)(\lambda I + \mathfrak{A}_\eta(s))^{-1}$ are positive contractions. Under Assumption 4, $\mathfrak{A}_\eta(s) = A_\eta^\infty(s)$ and is bounded.*

PROOF: See Appendix A.7. *Q.E.D.*

The price of this generality is that the relative moment may be unbounded. Linear summaries such as $\operatorname{tr}\{Q\mathfrak{A}_\eta(s)\}$ are then not automatically available for every trace-class probe Q. The bounded resolvent and stress transforms remain well defined, but using Proposition 3 over a collection of states requires an additional measurability condition. Specifically, impose strong resolvent measurability: for every $\lambda > 0$ and $x \in \mathcal{H}_C$, the map $s \mapsto (\lambda I + \mathfrak{A}_\eta(s))^{-1}x$ is strongly measurable on the relevant state set. Under this condition, $s \mapsto q_{\eta,\lambda,Q}(s)$ is measurable for every trace-class probe Q. For this reason, the we work under relative domination whenever possible. It gives a bounded, weakly measurable relative moment with the average-identity normalization, and it makes the probe formulas in Corollary 1 immediate.

2.3. *Bounded Stress Diagnostics*

The linear probe $\tau_{\eta,Q}(s)$ reports the average relative moment under an economically specified weighting Q. A complementary question is whether a nonnegligible part of that weighting is exposed to severe amplification. This question becomes important when the estimand is functional or its discretization is refined. The largest generalized eigenvalue can then be unstable or infinite, and $A_\eta^\infty(s)$ need not have a trace, even though every fixed estimand contrast has finite variance.[1]

To cover both the bounded and closed-form cases, write $\mathcal{A}_\eta(s)$ for $A_\eta^\infty(s)$ under Theorem 1 and for $\mathfrak{A}_\eta(s)$ under Proposition 3. Let $E_{\eta,s}$ denote its spectral resolution. Every positive trace-class probe Q with $\operatorname{tr}(Q) = 1$ induces the probability measure

$$\nu_{\eta,s,Q}(D) := \operatorname{tr}\{QE_{\eta,s}(D)\}, \qquad D \subseteq [0,\infty) \text{ Borel.} \tag{2.21}$$

The associated stress curve is

$$\begin{aligned} q_{\eta,\lambda,Q}(s) &:= \operatorname{tr}\left[Q\mathcal{A}_\eta(s)\{\lambda I + \mathcal{A}_\eta(s)\}^{-1}\right] \\ &= \int_{[0,\infty)} \frac{a}{\lambda + a}\, \nu_{\eta,s,Q}(da), \qquad \lambda > 0. \end{aligned} \tag{2.22}$$

The scale λ is measured in units of reference moment. A direction with amplification $a = \lambda$ contributes one half; a direction with $a \ll \lambda$ contributes little; and a direction with $a \gg \lambda$ contributes nearly one. Thus $q_{\eta,\lambda,Q}(s)$ is a smooth tail summary of relative amplification. It lies in $[0,1]$, is continuous and nonincreasing in λ, and remains well defined when $\|\mathcal{A}_\eta(s)\|_{\mathrm{op}} = \infty$. Under the neutral geometry $\mathcal{A}_\eta(s) = I$, the benchmark is $q_{\eta,\lambda,Q}(s) = (1+\lambda)^{-1}$ for every Q.

The probe identifies the economic object being stressed. The rank-one probe Q_g in (2.15) gives stress for a specified contrast g; Q_B in (2.16) gives stress for an outcome, horizon, cohort, or event-time block; and Q_ρ gives a soft reference-moment-weighted summary over

[1] Appendix B gives two concrete examples of situations where this might arise: (1) a macro local projections context and (2) the case of vanishing comparison probabilities in a treatment effect or difference-in-difference context.

all empirically supported directions. A rank-R basis may instead be formed after estimation to display which response-surface patterns account for a chosen stress measure. Such a finite-dimensional reduction is often easier to economically interpret without much cost if the kernel is truly low-rank. Appendix E gives the corresponding Galerkin approximation results.

A finite-dimensional display can be certified rather than assumed. The following bound applies to any fitted positive regularized relative moment operator, including the estimator in (2.26).

PROPOSITION 4—Ritz-certified stress approximation: *Let $\widehat{A}(s) \succeq 0$ be bounded, let P be a finite-rank orthogonal projection, and put $f_\lambda(a) := a/(\lambda + a)$. Define the compression $\widehat{A}_P(s) := P\widehat{A}(s)P\Big|_{P\mathcal{H}}$ and, for every positive trace-one probe Q,*

$$\widehat{q}_{\lambda,Q}(s) := \operatorname{tr}\{Qf_\lambda(\widehat{A}(s))\}, \qquad \widehat{q}^{P}_{\lambda,Q}(s) := \operatorname{tr}\Big\{QPf_\lambda(\widehat{A}_P(s))P\Big\}.$$

Then

$$\boxed{\left|\widehat{q}_{\lambda,Q}(s) - \widehat{q}^{P}_{\lambda,Q}(s)\right| \le \operatorname{tr}\{Q(I-P)\} + \frac{\|(I-P)\widehat{A}(s)P\|_{\mathrm{op}}}{\lambda}.} \tag{2.23}$$

PROOF: See Appendix A.10. *Q.E.D.*

The first term in (2.23) is the economic-probe mass omitted by the display space. The second is its block Ritz residual, scaled by the stress threshold. Thus a rank-R representation is adequate for a specified probe, stress scale, and collection of states when both terms are small. Explained reference-moment variation alone does not provide this certificate: a leading-eigenvector space based on C can omit low-reference-moment directions with high relative amplification. Appendix E.2 gives individual Ritz-pair, spectral-gap, and generalized-residual diagnostics.

For large sparse problems the candidate space need not come from a full eigendecomposition. After forming the moment estimators below, set $D_n := \widehat{C}_n + \rho I$. The eigenproblem for $\widehat{A}_{\eta,\rho,n}(s)$ is equivalent to the generalized Hermitian pencil

$$\widehat{K}_{\eta,n}(s)v = aD_n v.$$

FEAST and related contour-filtered subspace iterations approximate the spectral projector

$$\mathcal{P}_{\mathcal{I}} = \frac{1}{2\pi\mathrm{i}} \oint_{\Gamma} \{zD_n - \widehat{K}_{\eta,n}(s)\}^{-1} D_n \, dz$$

for a selected amplification interval $\mathcal{I}$; rational Krylov and shift-and-invert Lanczos methods provide closely related alternatives based on shifted linear solves (Polizzi, 2009, Tang and Polizzi, 2014, Ruhe, 1984). Because $f_\lambda(a) \ge c$ if and only if $a \ge c\lambda/(1-c)$, these methods can target the directions that materially contribute to a chosen stress threshold rather than those that merely explain the most reference-moment variation. Rayleigh-Ritz extraction, residual checks, and, for interval methods, eigenvalue-count diagnostics then assess whether the search space is sufficiently large. These are numerical certificates conditional on the fitted regularized operator; sampling uncertainty remains governed by the consistency theory below or by bootstrap.

2.4. *Feasibility and Consistency*

Unless a result states otherwise, retain Assumptions 2-4 and the population notation of Sections 2.2-2.3. The additional sampling assumptions below concern feasibility and stochastic rates.

Reintroduce sampling notation. Let $\xi_{i,n} = \chi_{i,n}$ for the conditional covariance target and $\xi_{i,n} = \psi_{i,n}$ for the conditional second-moment target. Let $\widehat{\xi}_{i,n}$ be its feasible estimate and write

$$Y_{i,n} := \xi_{i,n} \otimes \xi_{i,n}, \qquad \widehat{Y}_{i,n} := \widehat{\xi}_{i,n} \otimes \widehat{\xi}_{i,n}.$$

For a target state s, let $w_{\eta,n,i}(s)$ be nonnegative normalized weights approximating J_η, so that $\sum_{i=1}^n w_{\eta,n,i}(s) = 1$. Let $a_{\pi,n,i} \geq 0$, $\sum_i a_{\pi,n,i} = 1$, be reference weights that approximate the declared law π. Define

$$\widehat{K}_{\eta,n}(s) := \sum_{i=1}^n w_{\eta,n,i}(s)\widehat{Y}_{i,n}, \qquad \widehat{C}_{\pi,n} := \sum_{i=1}^n a_{\pi,n,i}\widehat{Y}_{i,n}. \tag{2.24}$$

The special case $a_{\pi,n,i} = 1/n$ corresponds to the empirical sampling law. Grid quadrature, design weights, and importance weights produce other reference laws. This distinction is necessary because the average-one normalization is relative to π, not automatically to the unweighted sample distribution. For brevity, write $\widehat{C}_n := \widehat{C}_{\pi,n}$ below.

ASSUMPTION 5—Feasible moment rates: *For every state s under consideration, there are deterministic rates $r_{K,n}(s), r_{C,n} \to 0$ such that*

$$\|\widehat{K}_{\eta,n}(s) - K_\eta(s)\|_{\mathrm{HS}} = O_p\{r_{K,n}(s)\}, \qquad \|\widehat{C}_{\pi,n} - C\|_{\mathrm{HS}} = O_p(r_{C,n}). \tag{2.25}$$

ASSUMPTION 6—Primitive moment-rate conditions: *Fix a state s, write $w_i = w_{\eta,n,i}(s)$ and $a_i = a_{\pi,n,i}$, and suppose $w_i, a_i \geq 0$, $\sum_i w_i = \sum_i a_i = 1$, $N_{\eta,n}(s) \to \infty$, and $N_{\pi,n} \to \infty$. Let $K_{i,n} := K_n(S_{i,n})$, $Y_{i,n} := \xi_{i,n} \otimes \xi_{i,n}$, and $\delta_{i,n} := \widehat{\xi}_{i,n} - \xi_{i,n}$. Assume:*

(i) For deterministic $b_{K,n}(s), b_{C,n} \to 0$,

$$\left\|\sum_i w_i K_{i,n} - K_\eta(s)\right\|_{\mathrm{HS}} = O_p\{b_{K,n}(s)\},$$
$$\left\|\sum_i a_i K_{i,n} - C\right\|_{\mathrm{HS}} = O_p(b_{C,n}).$$

(ii) The oracle moment noise satisfies

$$\frac{\mathbb{E}\left[\left\|\sum_i w_i(Y_{i,n} - K_{i,n})\right\|_{\mathrm{HS}}^2 \,\middle|\, S_{1,n}, \dots, S_{n,n}\right]}{\sum_i w_i^2} = O_p(1),$$

and

$$\frac{\mathbb{E}\left\|\sum_i a_i(Y_{i,n}-K_{i,n})\right\|_{\mathrm{HS}}^2}{\sum_i a_i^2}=O(1).$$

(iii) The weighted oracle energies obey

$$\sum_i w_i\|\xi_{i,n}\|_{\mathcal{H}}^2=O_p(1),\qquad \sum_i a_i\|\xi_{i,n}\|_{\mathcal{H}}^2=O_p(1).$$

(iv) For deterministic $d_{K,n}(s), d_{C,n}\to 0$,

$$\sum_i w_i\|\delta_{i,n}\|_{\mathcal{H}}^2=O_p\{d_{K,n}(s)^2\},\qquad \sum_i a_i\|\delta_{i,n}\|_{\mathcal{H}}^2=O_p(d_{C,n}^2).$$

PROPOSITION 5—Primitive sufficient conditions: *Under Assumption 6,*

$$\|\widehat{K}_{\eta,n}(s)-K_\eta(s)\|_{\mathrm{HS}}=O_p\big\{b_{K,n}(s)+N_{\eta,n}(s)^{-1/2}+d_{K,n}(s)\big\},$$

and

$$\|\widehat{C}_n-C\|_{\mathrm{HS}}=O_p\{b_{C,n}+N_{\pi,n}^{-1/2}+d_{C,n}\}.$$

Thus Assumption 5 holds with the displayed rates, and the same rates hold in operator norm.

PROOF: See Appendix A.11. *Q.E.D.*

The full-space estimator uses soft whitening rather than a spectral cutoff. For $\rho>0$, define

$$\widehat{A}_{\eta,\rho,n}(s):=(\widehat{C}_n+\rho I)^{-1/2}\widehat{K}_{\eta,n}(s)(\widehat{C}_n+\rho I)^{-1/2}. \tag{2.26}$$

The population counterpart $A_{\eta,\rho}(s)$ was defined in (2.17). For $\lambda>0$, define the sample transforms

$$\widehat{B}_{\eta,\lambda,\rho,n}(s):=\lambda\{\lambda I+\widehat{A}_{\eta,\rho,n}(s)\}^{-1},\qquad \widehat{Y}_{\eta,\lambda,\rho,n}(s):=I-\widehat{B}_{\eta,\lambda,\rho,n}(s).$$

When operators are compared on $\mathcal{H}$, identify $A_\eta^\infty(s)$ with its zero extension from $\mathcal{H}_C$ to $\mathcal{H}$ and define its transforms from that extension by functional calculus. Thus $B_{\eta,\lambda}(s)$ is the identity and $Y_{\eta,\lambda}(s)$ is zero on $\ker C$.

THEOREM 2—Full-space regularized consistency: *Suppose Assumption 5 holds and $\rho_n\downarrow 0$ satisfies $r_{K,n}(s)/\rho_n\to 0$ and $r_{C,n}/\rho_n^2\to 0$. Then*

$$\|\widehat{A}_{\eta,\rho_n,n}(s)-A_{\eta,\rho_n}(s)\|_{\mathrm{op}}=o_p(1), \tag{2.27}$$

while $A_{\eta,\rho}(s)\to A_\eta^\infty(s)$ strongly on $\mathcal{H}_C$ as $\rho\downarrow 0$. For every fixed $\lambda>0$ and $x\in\mathcal{H}_C$, the sample inverse-amplification and Yosida transforms converge in probability to $B_{\eta,\lambda}(s)x$ and $Y_{\eta,\lambda}(s)x$, respectively. The same holds for $\operatorname{tr}\{QY\}$ for every fixed trace-class probe $Q\succeq 0$.

PROOF: See Appendix A.12. *Q.E.D.*

REMARK: Forming $(\widehat{C}_n + \rho I)^{-1/2}$ is a dense cubic operation in the estimand dimension p. Directional and resolvent diagnostics can often be computed by solving linear systems instead, avoiding an explicit inverse square root. Let $D_n := \widehat{C}_n + \rho_n I$ and solve

$$\{\widehat{K}_{\eta,n}(s) + \lambda D_n\}\widehat{h}_n(s;g) = D_n g. \tag{2.28}$$

Set

$$\widehat{p}_{\eta,\lambda,n}(s;g) := \lambda \frac{\langle D_n g, \widehat{h}_n(s;g)\rangle}{\langle g, D_n g\rangle}, \qquad \widehat{q}_{\eta,\lambda,n}(s;g) := 1 - \widehat{p}_{\eta,\lambda,n}(s;g). \tag{2.29}$$

COROLLARY 2—Directional resolvent consistency: *Under Theorem 2, take any g satisfying $\langle g, Cg\rangle > 0$. Then*

$$\widehat{q}_{\eta,\lambda,n}(s;g) \to_p q_{\eta,\lambda}(s; C^{1/2} g).$$

PROOF: See Appendix A.13. *Q.E.D.*

The trace-normalized soft index is $\widehat{\tau}^{\mathrm{soft}}_{\eta,\rho,n}(s) := \operatorname{tr}\{\widehat{A}_{\eta,\rho,n}(s)\}/\operatorname{tr}\{\widehat{C}_n(\widehat{C}_n + \rho I)^{-1}\}$.

COROLLARY 3—Soft-index consistency: *For fixed $\rho > 0$, if*

$$\|\widehat{K}_{\eta,n}(s) - K_\eta(s)\|_1 = o_p(1), \qquad \|\widehat{C}_n - C\|_1 = o_p(1),$$

then $\widehat{\tau}^{\mathrm{soft}}_{\eta,\rho,n}(s) - \tau^{\mathrm{soft}}_{\eta,\rho}(s) \to_p 0$. If $\rho_n \downarrow 0$, the same conclusion with $\rho = \rho_n$ holds under

$$\begin{aligned} \|\widehat{K}_{\eta,n}(s) - K_\eta(s)\|_1 &= o_p(\rho_n), \\ \|\widehat{C}_n - C\|_1 &= o_p(\rho_n), \\ \|\widehat{C}_n - C\|_{\mathrm{op}} &= o_p(\rho_n^2). \end{aligned}$$

PROOF: See Appendix A.14. *Q.E.D.*

THEOREM 3—Form-based resolvent consistency: *Let a_n and a be closed, densely defined, nonnegative symmetric forms on $\mathcal{H}_C$, with associated positive self-adjoint operators A_n and A. Regard each quadratic form as an extended-real functional equal to $+\infty$ outside its domain. If a_n converges to a in the Mosco sense, then for every $\lambda > 0$ and $x \in \mathcal{H}_C$, $(\lambda I + A_n)^{-1}x \to (\lambda I + A)^{-1}x$ strongly. The associated inverse-amplification and Yosida transforms converge strongly as well.*

PROOF: See Appendix A.18. *Q.E.D.*

When relative domination fails, Theorem 3 provides the appropriate route: establish convergence of the estimated closed forms, and infer consistency of the bounded resolvent and stress transforms rather than operator-norm consistency of the unbounded relative moment itself.

2.5. *Practical Arithmetic Graph-Laplacian Smoother*

The preceding estimators already target a state-smoothed arithmetic moment field. A practical low-dimensional implementation below aggregates outer products at state nodes and applies a graph-Laplacian penalty. It uses every coordinate in the chosen finite representation of the estimand, introduces no spectral cutoff, and preserves positive semidefiniteness by construction. The only parametric component is the graph-smoothing parameter κ, or a small vector of such parameters for separable state geometries.

Fix $\rho > 0$, identify the working space with $\mathbb{R}^p$, let $\mathsf{S}_n^\circ = \{s_{1,n}, \ldots, s_{m_n,n}\}$ be the state nodes, and let $r(i)$ map observation i to a node. Write

$$n_{j,n} := \sum_{i=1}^{n} 1\{r(i) = j\}, \qquad N_n := \mathrm{diag}(n_{1,n}, \ldots, n_{m_n,n}).$$

Let $a_{jk,n} = a_{kj,n} \geq 0$ denote the graph edge weights and let L_n be the standard weighted graph Laplacian, $(L_n)_{jk} = -a_{jk,n}$ for $j \neq k$ and $(L_n)_{jj} = \sum_{k \neq j} a_{jk,n}$. Thus $L_n\mathbf{1} = 0$ and $L_n \succeq 0$. Assume that every connected component contains at least one occupied node. Empty nodes are permitted; their node averages below may be set to zero because the corresponding column of N_n is zero.

Set $\widehat{D}_{\rho,n} := \widehat{C}_n + \rho I_p$ and define the feasible ridge-whitened contributions

$$\widehat{u}_{i,n,\rho} := \widehat{D}_{\rho,n}^{-1/2}\widehat{\xi}_{i,n}.$$

For each occupied node, define the unwhitened and whitened arithmetic node averages

$$\begin{aligned} \widehat{\bar{K}}_{j,n} &:= \frac{1}{n_{j,n}} \sum_{i:r(i)=j} \widehat{\xi}_{i,n}\widehat{\xi}'_{i,n}, \\ \widehat{H}_{j,n,\rho} &:= \frac{1}{n_{j,n}} \sum_{i:r(i)=j} \widehat{u}_{i,n,\rho}\widehat{u}'_{i,n,\rho} = \widehat{D}_{\rho,n}^{-1/2}\widehat{\bar{K}}_{j,n}\widehat{D}_{\rho,n}^{-1/2}. \end{aligned} \tag{2.30}$$

For $\kappa > 0$, set

$$W_n(\kappa) := (N_n + \kappa L_n)^{-1}N_n, \qquad \widehat{W}_n := W_n(\widehat{\kappa}_n), \tag{2.31}$$

where $\widehat{\kappa}_n > 0$ may be fixed in advance or selected by cross validation, marginal likelihood, or another low-dimensional tuning rule. The positivity-preserving arithmetic estimator is

$$\begin{aligned} \widehat{A}_{j,n,\rho} &:= \sum_{k=1}^{m_n} (\widehat{W}_n)_{jk}\widehat{H}_{k,n}, \\ \widehat{K}_{j,n,\rho} &:= \widehat{D}_{\rho,n}^{1/2}\widehat{A}_{j,n,\rho}\widehat{D}_{\rho,n}^{1/2} = \sum_{k=1}^{m_n} (\widehat{W}_n)_{jk}\widehat{\bar{K}}_{k,n}. \end{aligned} \tag{2.32}$$

The final equality is exact. Thus the decoded moment field is an arithmetic average of the original outer products and is algebraically independent of ρ; the ridge affects only the relative-moment geometry used for amplification and stress diagnostics.

For every $\kappa > 0$, $N_n + \kappa L_n$ is a symmetric positive-definite Z-matrix and therefore a nonsingular M-matrix. Consequently $W_n(\kappa)$ is elementwise nonnegative. Moreover,

$$W_n(\kappa)\mathbf{1} = \mathbf{1}, \qquad \mathbf{1}'N_nW_n(\kappa) = \mathbf{1}'N_n. \tag{2.33}$$

Each fitted matrix in (2.32) is therefore a convex combination of positive-semidefinite node matrices and remains positive semidefinite. The second identity preserves the count-weighted average. Under the empirical reference $a_{\pi,n,i} = 1/n$, it gives

$$n^{-1}\sum_{j=1}^{m_n} n_{j,n}\widehat{K}_{j,n,\rho} = n^{-1}\sum_{i=1}^{n}\widehat{\xi}_{i,n}\widehat{\xi}'_{i,n} = \widehat{C}_n.$$

For weighted samples, the same statement holds after replacing counts by the corresponding node masses.

The graph smoother also has a direct Gaussian Markov Random Field (GMRF) interpretation. The matrices $\{\widehat{A}_{j,n,\rho}\}_{j=1}^{m_n}$ are the unique minimizer of

$$\sum_{j=1}^{m_n} n_{j,n}\|B_j - \widehat{H}_{j,n,\rho}\|_F^2 + \widehat{\kappa}_n \sum_{1\le j<k\le m_n} a_{jk,n}\|B_j - B_k\|_F^2 \tag{2.34}$$

over $B_1,\ldots,B_{m_n}\in\mathbb{S}^p$. Coordinate by coordinate, (2.34) is the posterior mean and mode under a Gaussian working observation model with node precision $n_{j,n}$ and an intrinsic GMRF prior with precision proportional to L_n. Gaussian correctness is not used to define the target or to establish consistency; it is only a computational and tuning interpretation of the linear arithmetic smoother.

To state the target, let $K_{i,n} := \mathbb{E}[Y_{i,n}\mid \mathcal{G}_{i,n}]$ denote the relevant conditional second moment or, after state centering, conditional covariance, or its coarser state projection, and define the arithmetic node field

$$K^{\circ}_{k,n} := \frac{1}{n_{k,n}}\sum_{i:r(i)=k} K_{i,n}$$

on occupied nodes. For a deterministic smoothing sequence $\kappa_n > 0$, set

$$\begin{aligned} K^{\mathrm{G}}_{\kappa_n,n}(s_{j,n}) &:= \sum_{k=1}^{m_n} W_n(\kappa_n)_{jk}K^{\circ}_{k,n}, \\ A^{\mathrm{G}}_{\kappa_n,\rho,n}(s_{j,n}) &:= (C+\rho I_p)^{-1/2}K^{\mathrm{G}}_{\kappa_n,n}(s_{j,n})(C+\rho I_p)^{-1/2}. \end{aligned} \tag{2.35}$$

This is the arithmetic graph-resolvent moment field. If the moment is constant within each node, then $K^{\circ}_{k,n} = K_n(s_{k,n})$. When N_n is invertible, $W_n(\kappa_n) = \{I + \kappa_n N_n^{-1}L_n\}^{-1}$, so with $\eta_n = \kappa_n^{-1}$ it is exactly the discrete resolvent associated with the Markov generator $-N_n^{-1}L_n$.

ASSUMPTION 7—Arithmetic GMRF rates: *For a deterministic sequence $\kappa_n > 0$, the following hold:*

(i) The selected smoothing matrix is stable,

$$\max_{j\le m_n}\sum_{k=1}^{m_n}|W_n(\widehat{\kappa}_n)_{jk} - W_n(\kappa_n)_{jk}| = o_p(1).$$

This condition is automatic when $\widehat{\kappa}_n = \kappa_n$.

(ii) Uniformly over occupied nodes,

$$\max_{k:n_{k,n}>0}\|\widehat{\widehat{K}}_{k,n} - K^{\circ}_{k,n}\|_{\mathrm{op}} = o_p(1), \qquad \max_{k:n_{k,n}>0}\|K^{\circ}_{k,n}\|_{\mathrm{op}} = O_p(1).$$

(iii) $\|\widehat{C}_n - C\|_{\mathrm{op}} = o_p(1)$.

The nodewise rate in Assumption 7(ii) follows, for example, by applying Assumption 6 uniformly to the within-node weights $1\{r(i)=k\}/n_{k,n}$, including the corresponding first-stage mean-square condition.

For $\lambda > 0$, define

$$\widehat{Y}_{j,\lambda,\rho} := \widehat{A}_{j,n,\rho}\{\lambda I_p + \widehat{A}_{j,n,\rho}\}^{-1},$$
$$Y^{\mathrm{G}}_{\kappa_n,j,\lambda,\rho} := A^{\mathrm{G}}_{\kappa_n,\rho,n}(s_{j,n})\{\lambda I_p + A^{\mathrm{G}}_{\kappa_n,\rho,n}(s_{j,n})\}^{-1}.$$

PROPOSITION 6—Positivity-preserving arithmetic GMRF consistency: *For every sample size and every positive smoothing parameter, $W_n(\kappa)$ satisfies* (2.33)*; hence $\widehat{A}_{j,n,\rho} \succeq 0$ and $\widehat{K}_{j,n,\rho} \succeq 0$ for every node. Under Assumption 7,*

$$\max_{j \le m_n} \|\widehat{K}_{j,n,\rho} - K^{\mathrm{G}}_{\kappa_n,n}(s_{j,n})\|_{\mathrm{op}} = o_p(1),$$

and

$$\max_{j \le m_n} \|\widehat{A}_{j,n,\rho} - A^{\mathrm{G}}_{\kappa_n,\rho,n}(s_{j,n})\|_{\mathrm{op}} = o_p(1).$$

For every fixed $\lambda > 0$,

$$\max_{j \le m_n} \|\widehat{Y}_{j,\lambda,\rho} - Y^{\mathrm{G}}_{\kappa_n,j,\lambda,\rho}\|_{\mathrm{op}} = o_p(1),$$

and therefore, for every fixed positive trace-one probe Q,

$$\max_{j \le m_n} \left|\mathrm{tr}\{Q\widehat{Y}_{j,\lambda,\rho}\} - \mathrm{tr}\{QY^{\mathrm{G}}_{\kappa_n,j,\lambda,\rho}\}\right| = o_p(1).$$

If, in addition, the ordinary graph-smoothing approximation satisfies

$$\max_{j \le m_n} \|K^{\mathrm{G}}_{\kappa_n,n}(s_{j,n}) - K_\eta(s_{j,n})\|_{\mathrm{op}} = o_p(1),$$

then $\widehat{K}_{j,n,\rho}$ is uniformly consistent for the moment path $K_\eta(s_{j,n})$, and $\widehat{A}_{j,n,\rho}$ is uniformly consistent for $A_{\eta,\rho}(s_{j,n})$.

PROOF: See Appendix A.19. *Q.E.D.*

The last clause is an ordinary smoothing-bias condition linking a discrete graph resolvent to the desired continuous or unsmoothed path. Directional, block, soft-index, and trace-class-probe stress summaries can be computed from $\widehat{K}_{j,n,\rho}$ and $\widehat{Y}_{j,\lambda,\rho}$ before any basis is selected. Ordered eigenvectors or rank-R response-surface plots may then be formed as interpretive decompositions of the fitted moment field.

2.6. *Temporal and State-Indexed Interpretations*

The random-OVK construction and relative moment factorization are index-agnostic. What changes across applications is the chosen moment target, the interpretation of the state, and the geometry used by the smoother.

Substantive state variables. In a cross section or panel, take $\mathcal{G}_i = \sigma(S_i)$ or a larger predetermined sigma-field. The full target is $K(s) = \mathbb{E}[\psi_i \otimes \psi_i \mid S_i = s]$. If the contribution is first state-centered, the target becomes $K(s) = \mathbb{E}[\chi_i \otimes \chi_i \mid S_i = s]$, the within-state covariance component. In either case, $K(s)$ and $A_\eta^\infty(s)$ are deterministic state-indexed population objects. Support scores, geography, cohorts, event-time cells, and region-year nodes fit this case; smoothing must follow the geometry of S, not the arbitrary row label.

Temporal moments and predictable covariance. In a temporal application, set $i = t$ and $\mathcal{G}_t = \mathcal{F}_{t-1}$. The state-centered contribution and its predictable covariance are

$$\chi_t := \psi_t - \mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}], \qquad K_t^{\psi,\mathcal{F}} := \mathbb{E}[\chi_t \otimes \chi_t \mid \mathcal{F}_{t-1}]. \tag{2.36}$$

If S_t is sufficient for this object, write $K_t^{\psi,\mathcal{F}} = K(S_t)$. A smoother over deterministic rescaled calendar time generally targets a local mean of the predictable covariance, not its realized path; recovery of the realized path requires filtering, stochastic-volatility, GARCH, or another latent-state restriction. The arithmetic GMRF in Section 2.5 becomes a chain-graph Laplacian smoother when L_n is temporal. With a symmetric chain Laplacian it remains a two-sided smoother; real-time recovery requires a one-sided or filtering adaptation.

The empirical time-series application instead smooths the full diagonal score outer products $\psi_t \otimes \psi_t$. Neither that field nor the predictable covariance in (2.36) replaces HAC inference. Writing $\mu_t^\psi := \mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}]$ and $\psi_t = \mu_t^\psi + \chi_t$, the long-run covariance also contains serial cross-products generated by the predictable component and its covariance with prior innovations. A state-dependent long-run covariance requires those cross-products to be included explicitly. Analogously, an observation-level panel field is not cluster robust unless within-cluster score cross-products are part of the target.

Graph and unidentified state geometry. For spatial, grouped, or network states, a graph resolvent averages moments over substantively adjacent nodes while preserving positivity; different graphs therefore define different targets. If no economically meaningful ordering, metric, grouping, or state variable exists, a varying diagonal moment field is not identified from an arbitrary observation label. The coherent objects are then the reference moment C, its spectral geometry, and unconditional directional or block summaries.

3. EMPIRICAL EXAMPLES

We illustrate the diagnostic in three contexts. The first uses standard local projections to study monetary-policy effects in the United States and then applies the same construction to a proxy-IV specification. The second uses the LaLonde (1986) and Dehejia and Wahba (2002) National Supported Work benchmark in a cross section. The third uses the hurricane event-study setting of Deryugina (2017). The time-series and panel displays below are diagonal score second-moment fields. Their uncertainty procedures assess estimation of those fields, while HAC and county-clustered covariance calculations remain separate inferential objects.

3.1. *Monetary Policy Local Projections*

Consider the standard formulation for analyzing an exogenous shock in a local projections framework. Specifically, let t index observation time, h indicate horizon, M_t indicate the exogenous shock of interest, y_t indicate a possibly vector-valued endogenous variable of interest, and x_t any additional control variables (perhaps including lags of the endogenous variables). Then the local projections model is

$$y_{t+h} - y_{t-1} = \mu_h + \sum_{p=1}^{P} \beta_{h,p}' x_{t-p} + \theta_h M_t + u_{t+h}, \qquad h \in \{0, \dots, H\}$$

The object of interest is typically θ_h and is generally estimated with OLS (a Z-estimator). If there are N endogenous variables of interest, then $\mathrm{vec}\{\theta_h : h = 0, \dots, H\} \in \mathbb{R}^{N(H+1)}$ which can be quite large for typical empirical macroeconomic studies.

Huber, Matthes and Pfarrhofer (2026) provide a Bayesian Gaussian-process approach to these coefficients and show how it can accommodate heteroskedasticity, autocorrelation, missing data, and proxy-IV identification. We instead estimate the local-projection coefficients by OLS and apply the positivity-preserving arithmetic GMRF smoother to the diagonal response-score outer products. The resulting field is a month-indexed score second moment. It is not a state-dependent HAC operator because it does not include cross-products of scores from different months. In principle, any first stage that delivers suitable influence contributions could be used.

For this example we use monthly data spanning 1991-10 through 2024-01. Our endogenous variables are industrial production, CPI, unemployment rate, yield on 2Y US Treasuries, and the BAA-10Y yield spread. For the monetary policy shock measure we use the MP_Median from Jarociński and Karadi (2020). We include 12 lags of the endogenous variables as controls. Given these specifications there are 125 coefficients of interest.

Figure 1 (top) plots the full-coordinate soft diagonal response-score second-moment index $\widehat{\mathcal{T}}_t^{\mathrm{soft}}$. The solid black line is the sample estimate, the shaded region is the 90 percent pointwise moving-block bootstrap band, and the dashed curves form the 90 percent log-simultaneous moving-block bootstrap band. The latter is calibrated jointly over the entire path on the log scale and is therefore substantially wider and asymmetric in levels. The index is normalized so that one, shown by the horizontal line, represents the reference second-moment scale. Values above one indicate that the regularized full-coordinate diagonal second moment of the monetary-policy response scores is amplified relative to this reference; values below one indicate attenuation. Because the statistic aggregates over all response-score coordinates using soft regularization, it does not by itself attribute the amplification to a particular outcome, horizon, or response-score direction.

Consider the local peak around 2008. It indicates that the estimated diagonal response-score second moment was elevated relative to the reference geometry during that period, whereas the estimate around 2002 was below the reference level. The displayed bands measure bootstrap variability obtained by resampling the serially dependent response-score sequence in blocks and recomputing the second-moment path. The moving-block bootstrap accommodates serial dependence in estimation of the displayed path, but it does not convert the target into a state-dependent HAC operator. The comparison is not a sequence of date-specific standard errors for a time-varying causal effect: the underlying LP response estimates remain fixed while the diagonal score outer-product field varies over the sample.[2]

The aggregate amplification index does not reveal how amplification is distributed across estimand directions. It is a probe average over reference eigen-directions. If $Cv_j = \mu_j v_j$ is the reference eigendecomposition, then the soft probe assigns weight $\{\mu_j/(\mu_j + \rho)\}/d_\rho$ to direction v_j. The stress curve from Proposition 2 can reveal whether the amplification is concentrated in certain other directions of the coefficient space. To separate spectral shape from aggregate scale,

[2]Because the moving-block bootstrap is applied to the estimated response-score sequence, the bands condition on the first-stage shock and LP estimates and on fixed regularization choices. They do not include uncertainty from repeating the full estimation procedure for those components.

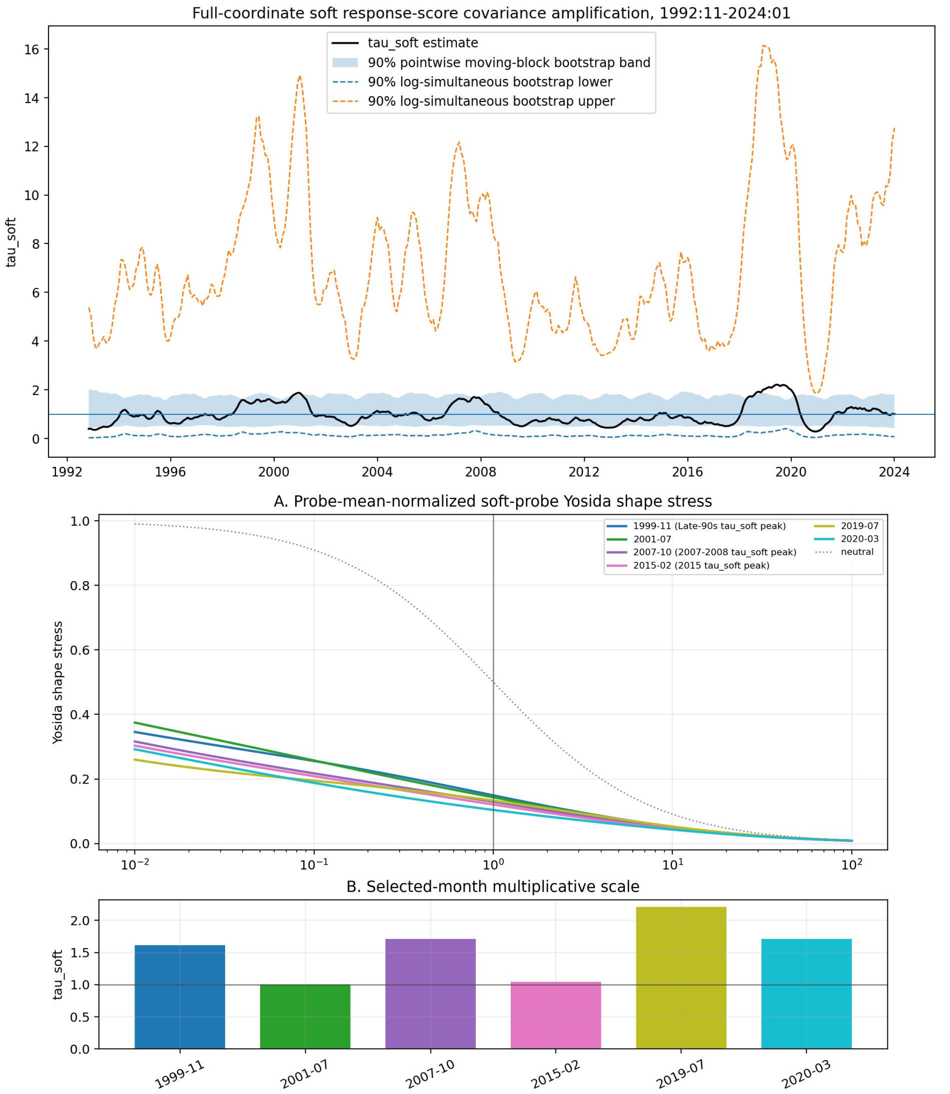


FIGURE 1.—Full-coordinate soft diagonal score second-moment amplification and selected-month scale-stress diagnostics. The top panel plots the full-coordinate soft amplification index $\widehat{\tau}^{\text{soft}}_{t,\rho} = \dfrac{\operatorname{tr}\left[(\widehat{C}_\pi + \rho I)^{-1}\widehat{K}_t\right]}{\operatorname{tr}\left[(\widehat{C}_\pi + \rho I)^{-1}\widehat{C}_\pi\right]}$, where $\widehat{K}_t$ is the estimated month-specific diagonal second moment of the monetary-policy LP response-score vector and $\widehat{C}_\pi$ is the reference second moment. The horizontal line at one marks the reference-weighted average second-moment scale. The shaded region is the 90 percent pointwise moving-block bootstrap band, and the dashed curves give the lower and upper limits of a 90 percent log-simultaneous moving-block bootstrap band. Values above one indicate diagonal score second-moment amplification relative to the reference geometry, whereas values below one indicate attenuation. The bottom panel examines selected months from the amplification path. In its upper subpanel, the colored curves plot the full-coordinate soft-probe Yosida stress for the scale-normalized month-specific diagonal score second-moment geometry over a range of multiplicative probe scales; the dotted neutral curve is the benchmark associated with a purely proportional rescaling of the reference moment. Deviations from that neutral curve therefore reflect changes in second-moment shape rather than overall scale alone. In the lower subpanel, the bars report the corresponding selected-month values of $\widehat{\tau}^{\text{soft}}_{t,\rho}$. Taken together, the two panels separate aggregate second-moment-scale amplification from differences in the shape of the relative diagonal influence geometry. Neither panel estimates a state-dependent HAC operator.

let $\tau_{t,Q} = \int a\nu_{t,Q}(da)$. Define $b = a/\tau_{t,Q}$ and $\bar{\nu}_{t,Q}$ be the induced law of b. Then $\mathbb{E}_{\bar{\nu}_{t,Q}}[b] = 1$ and the scale-normalized stress curve is

$$\bar{q}_{t,\eta,\lambda,Q} = \int \frac{b}{\lambda + b}\bar{\nu}_{t,Q}(db) \tag{3.1}$$

Figure 1 (bottom) shows the normalized stress curve and corresponding amplification index value for a selection of months. In other words, the shape of the second moment relative to the reference is shown in (A) whereas the relative scale is shown in (B). The dotted line represents the stress curve which would prevail if all of the amplification were evenly distributed among the coefficients such that all directions had the reference-level second moment. Note that the neutral profile forms a pointwise upper envelope by construction.

The following is a convenient method to interpret the stress diagram. Consider the domain point $\lambda = 1$ and any $b \geq 1$. Using the fact that $\frac{a}{1+a} \geq \frac{1}{2}$ it must be true that $\int_{[0,\infty)} \frac{a}{1+a}\nu_{\eta,s,Q}(da) \geq \frac{1}{2}\int_{[1,\infty)} \nu_{\eta,s,Q}(da)$. However, the second term is just $\Pr_Q(a \geq 1)$ for the particular trace class probe Q which induces the probability measure $\nu_{\eta,s,Q}(da)$. In other words, $\Pr_Q(a \geq 1) \leq 2q_{\eta,\lambda,Q}$. For the neutral curve this implies $\nu([1,\infty)) \leq 2 * \frac{1}{2} = 1$, which shows that this interpretation strategy is a loose upper bound.[3] The observed stress curves lie within the range 0.1 to 0.15 when crossing unity in the domain. Thus a loose upper bound is that $\Pr_Q(a \geq 1) \leq 0.3$ for each of the observed probes. In other words, the probe-weighted spectral law is highly dispersed across amplification scales, while average diagonal second-moment amplification is concentrated in a minority high-amplification tail. Furthermore, practitioners may use $-\frac{d}{d\log a}\int_{[0,\infty)} \frac{a}{\lambda+a}\nu_{\eta,s,Q}(da)$ to identify modes of the Q-weighted log spectrum. Appendix Figure 6 shows that the selected-month distributions are diffuse and that most modal points lie well below unity.

It is quite possible that we have misspecified the model. In particular, a competing model may be a time-varying parameter model for the local-projection coefficients. Under this misspecification, the time-varying diagonal score second moment could be elevated in periods of large parameter movements rather than because the score geometry of a fixed parameter is changing. Because TVP models have become quite common in empirical macroeconomics and have been shown to improve in-sample and out-of-sample fit in many contexts, we compare several candidate models.

Table I evaluates whether predictive time variation in the monetary-policy LP application is better captured by movement in the center of the response-score surface or by movement in its covariance geometry. Let $\theta_t \in \mathbb{R}^{125}$ stack the five outcomes over horizons $0, \ldots, 24$, and let $\hat{\theta}$ denote the fixed estimated LP surface. M_0 fixes the center at $\hat{\theta}$ and holds covariance constant. M_1 retains that fixed center but allows covariance to evolve through a causal arithmetic recursion applied to the full outer product of the one-step response-score innovation. M_2 holds covariance constant but permits the center to move in every response-score coordinate through a common-persistence recursive update, while M_3 allows both the full-coordinate center and covariance to evolve. The selected mean and covariance half-lives are 12 and 96 months, respectively.[4] The results show that covariance dynamics account for nearly all of the predictive improvement. M_1 strongly outperforms the fixed benchmark, whereas the $M_2 - M_0$ interval includes zero, providing no clear evidence that a moving center alone improves predictive fit. Allowing the center to move in M_3 produces no significant incremental gain over M_1, with

[3]For the neutral case, $\bar{q}_{t,\eta,\lambda} = (1+\lambda)^{-1}$ which equals $1/2$ at $\lambda = 1$.

[4]The moving-center specification is a regularized TVP alternative, because the coordinates share a common persistence parameter rather than following unrestricted coordinate-specific processes.

TABLE I

PREDICTIVE LOG-SCORE COMPARISONS FOR THE MONETARY-POLICY LP SURFACE

Model comparison	Average log-score difference per evaluation date [90% paired block-bootstrap interval]
Against fixed benchmark	
$M_1 - M_0$	4,746
	[1,101, 9,470]
$M_2 - M_0$	3.289
	[−72, 83]
$M_3 - M_0$	4,745
	[1,100, 9,471]
Nested model contrasts	
$M_3 - M_1$	−0.938
	[−2.590, 0.955]
$M_3 - M_2$	4,742
	[1,135, 9,373]
$M_2 - M_1$	−4,742
	[−9,375, −1,137]

Notes: The table reports one-step-ahead Gaussian predictive log-score comparisons for the five baseline outcomes—industrial production, CPI, unemployment, the two-year Treasury yield, and the BAA–10Y credit spread—over horizons $0, \ldots, 24$. The monthly LP response-score surface therefore has $p = 125$ coordinates. All comparisons use the complete full-coordinate LP response-score vector and a common ridge-whitened geometry. M_0 has a fixed center and constant covariance. M_1 retains the fixed center and allows covariance to evolve through a one-sided arithmetic outer-product recursion. M_2 allows a full-coordinate, common-persistence recursive moving center while holding covariance constant. M_3 allows both the moving center and the arithmetic covariance recursion. The fitted mean and covariance half-lives are 12 and 3 months, respectively. For a contrast $M_a - M_b$, the reported average log-score difference is

$$\overline{\Delta\ell}_{a,b} = \frac{1}{T_{\text{eval}}} \sum_{t \in \mathcal{T}_{\text{eval}}} \left[\log p_a(\theta_t \mid \mathcal{I}_{t-1}) - \log p_b(\theta_t \mid \mathcal{I}_{t-1})\right],$$

where $\theta_t \in \mathbb{R}^{125}$ is the complete monthly LP response-score vector and $p_m(\cdot \mid \mathcal{I}_{t-1})$ is model m's one-step-ahead predictive density, conditional on full-sample score construction. The evaluation period is 2007:11–2024:01, so $T_{\text{eval}} = 195$. Positive values favor the first model in the comparison. Bracketed values are the 5th and 95th percentiles from paired circular moving-block bootstrap draws using 12-month blocks. The same resampled evaluation dates are applied jointly to all model scores in each draw. These intervals summarize serial dependence in the fitted score-difference sequence and condition on the estimated LP response scores, reference geometry, selected persistence parameters, and other fitted tuning choices. Because this table evaluates all 125 coordinates, its numerical magnitudes are not directly comparable with those from a ten-dimensional reduced-basis specification.

the $M_3 - M_1$ bootstrap interval including zero. The results therefore indicate that, within this predictive model class, the dominant form of one-step predictable variation in the response-score process is second-moment variation rather than a hidden time-varying LP response surface. This comparison concerns one-step predictive geometry, not a state-dependent HAC operator, and should not be generalized beyond the specific TVP alternatives evaluated here.

Figure 2(a) displays the full-coordinate shape of the diagonal response-score second-moment field across the 125 outcome-horizon cells after removing the common soft scale. Each heatmap reports the log amplification of a cell relative to its reference diagonal moment and the common scale, so zero is the scale-neutral benchmark and red (blue) denotes relative over-amplification (under-amplification). July 2001 is the reference-neutral date and exhibits comparatively limited cross-cell dispersion. At the date of maximum aggregate soft scale, July 2019, amplification is concentrated at short and intermediate horizons for IP and unemployment, while several long-horizon IP and credit-spread cells are under-amplified. The maximum shape-dispersion date, March 2012, instead features broad under-amplification across the surface, particularly for IP and unemployment. In March 2020, amplification shifts sharply toward short horizons across all outcomes, whereas many longer-horizon cells remain below the common scale. These colors describe relative diagonal score second-moment amplification, not the signs or magnitudes of the underlying impulse responses and not HAC variances.

Figure 2(b) traces the corresponding block-shape probes and concentration measures over time. The top panel compares macro and financial outcomes, while the middle panel compares short-, medium-, and long-horizon blocks. A value of one indicates that a block is amplified at the common soft scale; values above or below one indicate relative over- or under-amplification. These probes are not allocation shares and need not sum to one. Financial outcomes exhibit the larger relative amplification around the late 1990s, whereas macro outcomes dominate during the 2007-08 episode and around 2020. The horizon probes show pronounced long-horizon amplification around the late 1990s and 2007-08, followed by an exceptionally large short-horizon amplification around 2020. The bottom panel reports normalized effective support across cells, outcomes, and horizons. Effective support is generally lowest at the cell level, indicating that shape variation is concentrated in particular outcome-horizon pairs even when it remains relatively diffuse after aggregation.

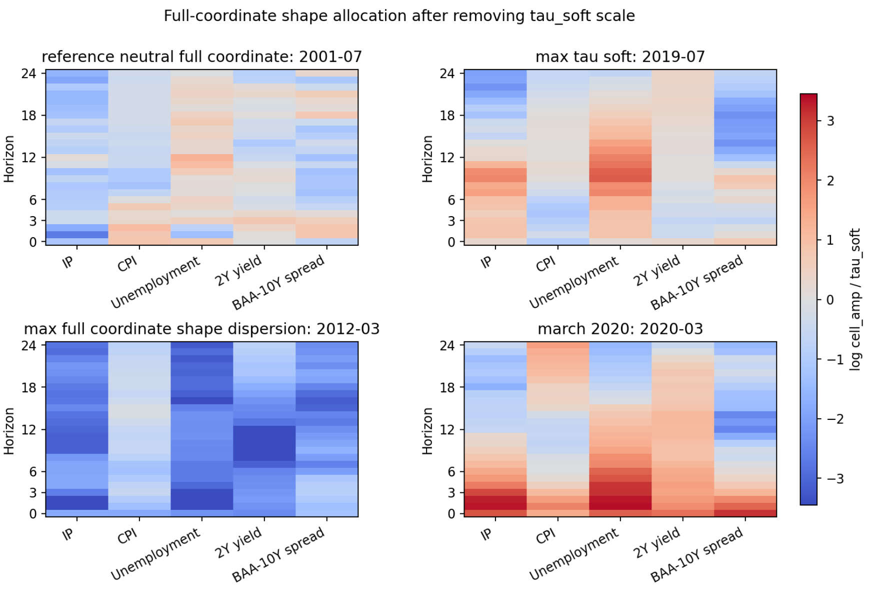
This tool is also useful for diagnosing proxy-IV estimation results. For the following example we replace the direct monetary policy shock in the local projections with one derived by proxying the yield on the US Treasury 1Y constant maturity with the monetary surprises shock of Bauer and Swanson (2023) maintained by Federal Reserve Bank of San Francisco (2025). We leverage the exact product form of the influence function in a just-identified scalar proxy context. Let M_t be the shock series used to proxy, X_t the variable being proxied for, $\kappa := \mathbb{E}[M_t X_t]$, $\hat{\psi}_t$ the estimated influence function, $D_\rho = \hat{C} + \rho I_p$, $d_\rho = \operatorname{tr}\left\{\hat{C} D_\rho^{-1}\right\}$, and u_t the second stage residuals. Then,

$$\frac{\|D_\rho^{-1/2}\hat{\psi}_t\|^2}{d_\rho} = \underbrace{\left(\frac{M_t}{\hat{\kappa}}\right)^2}_{\text{proxy/first-stage exposure}} \cdot \underbrace{\frac{\|D_\rho^{-1/2}u_t\|^2}{d_\rho}}_{\text{ridge-whitened residual energy}}$$

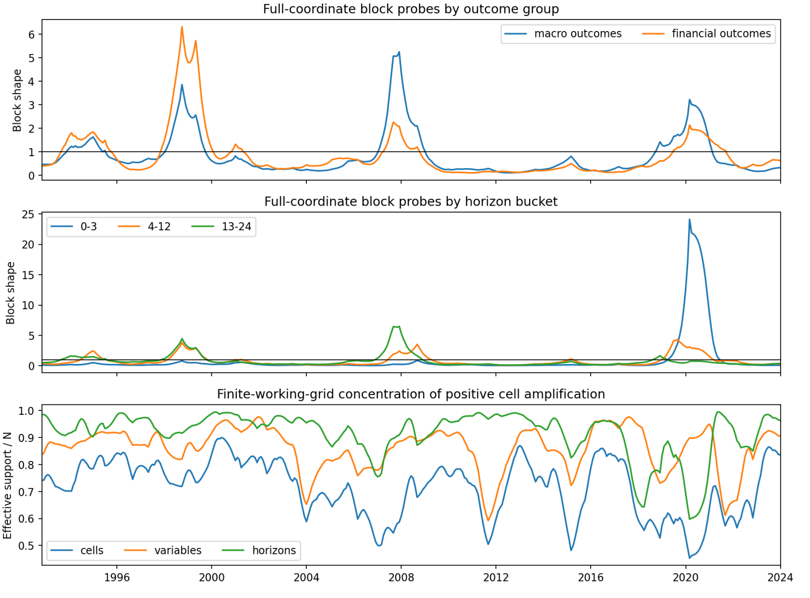
$$\hat{\tau}_t^{soft} = \sum_{r=1}^{T} \hat{w}_{tr} \frac{\|D_\rho^{-1/2}\hat{\psi}_r\|^2}{d_\rho}$$

Figure 3 characterizes months of high full-coordinate proxy-IV diagonal score second-moment amplification. The black line in the upper panel plots $\widehat{\tau}_t^{\text{soft}}$, with one denoting the reference scale. Using the smoothed weights, we also report the factors of $\widehat{\tau}_t^{\text{soft}}$ attributable to proxy exposure (blue) and residual energy (green). By construction both are strictly positive and their product equals $\widehat{\tau}_t^{\text{soft}}$. Colored markers classify high-τ months according to whether proxy/first-stage exposure exceeds its 75th percentile, ridge-whitened residual-response energy exceeds its 75th percentile, or neither condition holds. The factorization associates several recessionary peaks with unusually large proxy exposure, including episodes around the dot-com recession, the global financial crisis, and the COVID-19 recession. The factor curves are full-sample smoothed objects, whereas the markers and lower panel use month-specific factors.

The lower panel locates all months in the percentile ranks of these two candidate drivers; the solid lines mark the 75th-percentile classification thresholds and the dotted lines mark the



(a) Selected full-coordinate cell-shape maps



(b) Full-coordinate block-shape probes and effective support

FIGURE 2.—Full-coordinate diagonal score second-moment shape after removing the soft aggregate scale. Panel (a) reports the log amplification of each outcome-horizon cell relative to its reference diagonal moment and the common soft scale, using all 125 coordinates. Red (blue) cells are amplified more (less) than the common scale; colors do not indicate the sign of the impulse response. The four maps show the reference-neutral month, the dates of maximum soft scale and maximum shape dispersion, and March 2020. Panel (b) reports corresponding block probes for macro versus financial outcomes and for three horizon ranges. The horizontal benchmark of one denotes amplification equal to the common scale; these probes are not shares and need not sum to one. The bottom plot shows normalized entropy-based effective support across cells, variables, and horizons, with values near one indicating diffuse allocation and lower values indicating concentration. All series are point estimates.

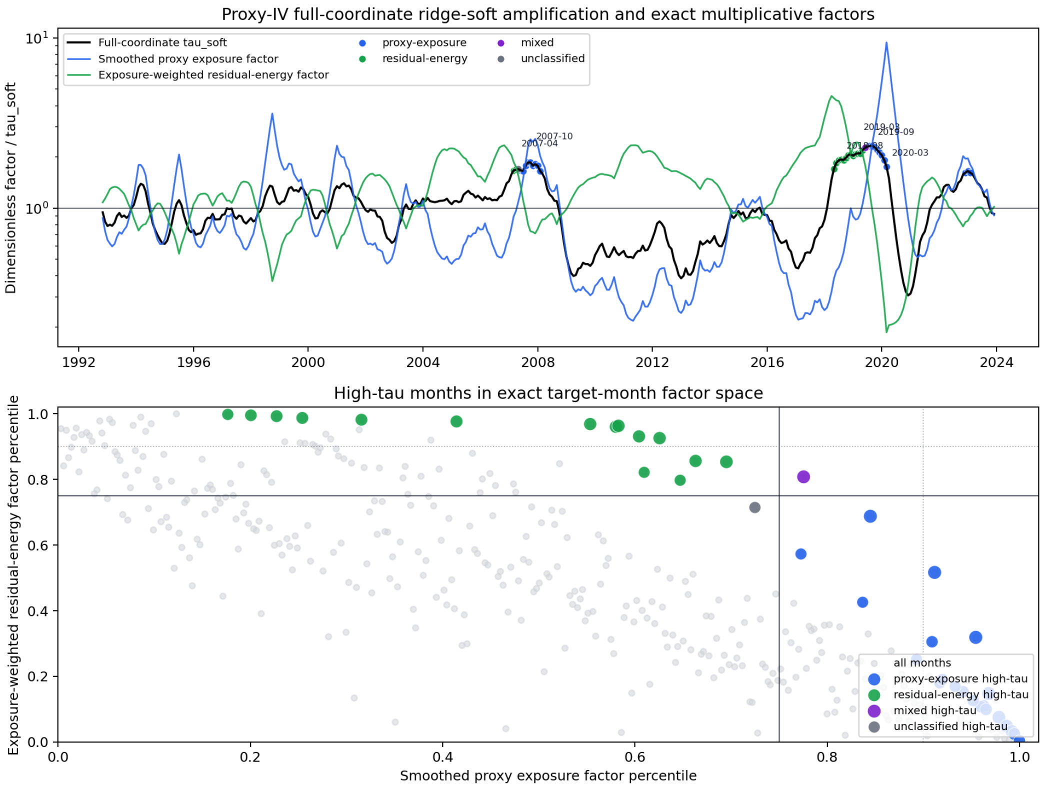


FIGURE 3.—Proxy-IV full-coordinate ridge-soft diagonal score second-moment amplification and its exact multiplicative factorization. The upper panel plots the full-coordinate soft amplification index $\widehat{\tau}_t^{\rm soft}$ in black, the smoothed proxy-exposure factor in blue, and the exposure-weighted residual-energy factor in green, on a logarithmic scale. The two factor series are dimensionless and their pointwise product equals $\widehat{\tau}_t^{\rm soft}$ at every month; the horizontal line at one denotes the reference scale. Colored markers identify high-τ months, defined as the top decile of $\widehat{\tau}_t^{\rm soft}$, according to whether the exact target-month proxy-exposure factor, residual-energy factor, both factors, or neither factor exceed their respective 75th-percentile thresholds. The lower panel plots all months by the within-sample percentile ranks of these two target-month factors. High-τ months are highlighted and sized by $\widehat{\tau}_t^{\rm soft}$. Solid lines mark the 75th-percentile classification thresholds, and dotted lines mark the 90th percentiles. Blue, green, purple, and gray markers denote proxy-exposure, residual-energy, mixed, and unclassified high-τ episodes, respectively.

90th percentiles. The classification varies across episodes. The 2007 episode contains both proxy-exposure and residual-energy months, the large and persistent 2018-19 amplification episode is predominantly associated with elevated residual-response energy, and March 2020 and the later amplification peak are associated with elevated proxy exposure. There is a single period which is mixed in 2018. Thus, high proxy-IV amplification does not require both drivers to be simultaneously extreme, and its proximate correlate differs across episodes. These labels are diagnostic associations rather than an additive or causal decomposition of $\widehat{\tau}_t^{\rm soft}$.

3.2. *Average Treatment Effects Example*

To illustrate the cross-sectional case, we turn to the studies of LaLonde (1986) and Dehejia and Wahba (2002). The original LaLonde critique showed that nonexperimental estimators can fail when treated and comparison units are not comparable, while the Dehejia-Wahba response emphasized propensity-score methods as a way to reduce the dimensionality of the comparability problem. Our application estimates one fixed vector of global and subgroup average treatment effects. It does not estimate a separate ATE at each propensity score or prognostic score. The support variables instead index the conditional second moments of the fixed estimator's influence vector, showing how observations in different support regions contribute to its global and subgroup estimand directions.

We consider two state specifications. The first uses the cross-fitted logit propensity score,

$$K_j(s) = \mathbb{E}\left[\psi_i \psi_i' \mid s = \log\left(\frac{\hat{e}(X_j)}{1-\hat{e}(X_j)}\right)\right],$$

and the second uses the joint logit-propensity and control-prognostic scores,

$$K_j(s_1, s_2) = \mathbb{E}\left[\psi_i \psi_i' \mid s_1 = \log\left(\frac{\hat{e}(X_j)}{1-\hat{e}(X_j)}\right), s_2 = \hat{m}(X_j)\right].$$

Here ψ_i denotes the influence contribution, $\hat{e}(X) = \hat{P}\{D=1 \mid X\}$, and $\hat{m}(X) = \hat{E}[Y \mid D = 0, X]$. The asymptotic benchmark uses compactly supported kernels; the empirical implementation uses Gaussian weights as a smooth practical analogue with Nadaraya-Watson smoothing. Further details are available in Appendix equations (C.35) and (C.36).

The fixed estimand vector is

$$\theta = (\theta_{\text{ATE}}, \theta_{\text{Black}}, \theta_{\text{NonBlack}}, \theta_{\text{NoDegree}}, \theta_{\text{Degree}}, \theta_{\text{RE74=0}}, \theta_{\text{RE75=0}}, \theta_{\text{Young}}, \theta_{\text{Older}}, \theta_{\text{LowEduc}}, \theta_{\text{HighEduc}})'.$$

Each coordinate is a global or subgroup ATE. Figure 4 overlays the influence-geometry diagnostics on the familiar support comparison. Panel (A) shows that treated and comparison observations occupy different regions of the empirical propensity-score distribution. Panel (B) reports the full-coordinate soft influence second-moment index when the state is the propensity score alone. The path is nonlinear and does not mechanically mirror the marginal histograms. Because Figure 4 does not display local effective sample size or maximum-weight diagnostics, this visual comparison alone does not establish that data availability is unimportant. One descriptive feature is that the largest aggregate normalized second moments occur in a region where treated observations are relatively common.

The shaded region in Panel (B) is a pointwise 5th-95th percentile full-estimation bootstrap band for

$$\widehat{\tau}^{\text{soft}}_{H,\rho}(s) = \frac{\operatorname{tr}\left[(\widehat{C}_\pi + \rho I)^{-1} \widehat{K}_H(s)\right]}{\operatorname{tr}\left[(\widehat{C}_\pi + \rho I)^{-1} \widehat{C}_\pi\right]}.$$

Each bootstrap draw resamples treated and comparison observations and then re-estimates the cross-fitted propensity and prognostic scores, generated states, treatment-effect estimator and influence contributions, local Nadaraya-Watson weights, reference-measure weights and $\widehat{C}_\pi$, and the ridge-normalized reference geometry. The band therefore propagates sampling variation through the estimated state-indexed influence second-moment path. It is not a confidence band for an ATE as a function of the propensity score.

Panel (C) augments the state by conditioning jointly on the propensity and prognostic scores. Within Panel (C)'s own reference geometry, lighter cells indicate an amplified aggregate influence second moment and darker cells indicate attenuation. The two-dimensional conditioning distinguishes support cells that the one-dimensional map necessarily averages together; in particular, the lowest-propensity and mid-range-prognostic region is amplified relative to Panel (C)'s reference. Because Panels (B) and (C) have different reference laws and different $\widehat{C}_\pi$,

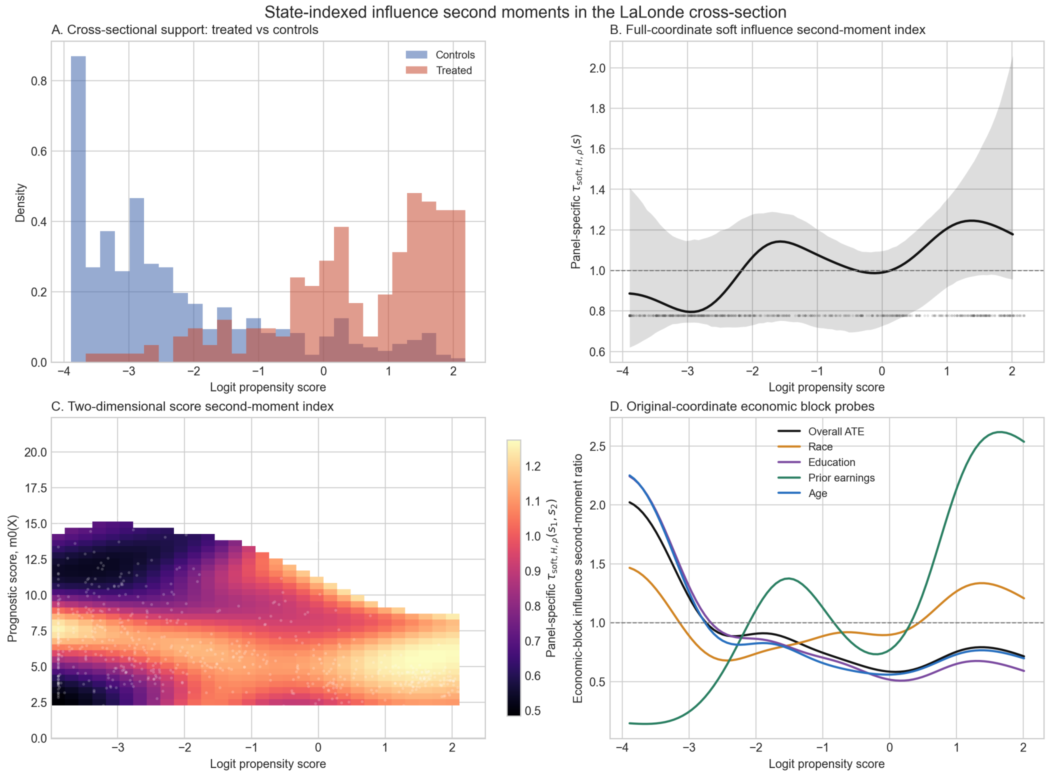


FIGURE 4.—State-indexed influence second-moment geometry in the LaLonde cross section. Panel (A) shows the familiar overlap problem: treated and comparison units occupy different regions of the propensity-score distribution. Panel (B) reports the full-coordinate soft index estimated by Nadaraya-Watson smoothing over the cross-fitted logit propensity score. The gray region is the pointwise 5th-95th percentile full-estimation bootstrap band based on 2,000 draws. The original estimate and every bootstrap draw are evaluated on the same fixed grid in raw logit-propensity-score units; within each draw, that grid is transformed using the draw-specific state scaling employed by the kernel smoother. Each draw re-estimates the generated states, influence contributions, local kernel weights, reference moment, and normalization map, conditional on the selected bandwidth and ridge parameter. Observed support points are shown in a rug plot. Panel (C) reports the corresponding two-dimensional field over the cross-fitted propensity and prognostic scores. Panels B and C use different quadrature reference laws and different $\widehat{C}_\pi$, so their numerical ranges and color scales are interpretable only within panel and cannot be compared as evidence that adding the prognostic score increases amplification. Observed support is represented by light gray points on the surface. Panel (D) reports economic block-selector probes in the original $p = 11$ estimand coordinates, as in equation (2.16). These probes are nonadditive, are not shares or a decomposition of Panel (B), and need not sum to one. None of the panels estimates a separate treatment effect at each support location.

this statement is not a comparison of their numerical ranges and does not show that adding the prognostic score increases amplification. It also does not imply that an ATE for low-propensity individuals has a large state-specific sampling variance. It means that observations near those support cells have an amplified second moment for the fixed vector of global and subgroup ATE estimators.

Panel (D) reveals a marked change in the economic directions that are locally influential across the propensity-score support. In the far-left, predominantly control region, the score second moments associated with the overall ATE and the age and education subgroup estimands are roughly twice their respective reference moments, while the prior-earnings block is strongly attenuated. Moving toward moderately low propensities, the age, education, race, and overall-

ATE ratios decline, whereas the prior-earnings block becomes locally elevated. The rotation is most pronounced in the right, predominantly treated region: the prior-earnings block, which contains the zero-1974-earnings and zero-1975-earnings subgroup estimands, rises to more than two and one-half times its reference moment, and the race block also moves above one, while the age, education, and overall-ATE directions remain attenuated.

Economically, the content of the estimator's observation-level influence therefore changes across support. Very low-propensity observations are unusually influential for age and schooling contrasts, whereas high-propensity observations are unusually influential for contrasts defined by pre-program earnings history and, to a lesser extent, race. The aggregate index in Panel (B) compresses this directional heterogeneity into a single trace ratio; Panel (D) shows that a moderate aggregate value can coexist with strong amplification in one economically specific block and attenuation in others. These curves are block-specific observation-level score second-moment ratios; not propensity-specific treatment effects or state-specific sampling variances. The selectors in equation (2.16) partition the original estimand coordinates, but because each ratio is normalized by its own block-specific reference moment, the probes are not additive, do not decompose Panel (B), and need not sum to one.

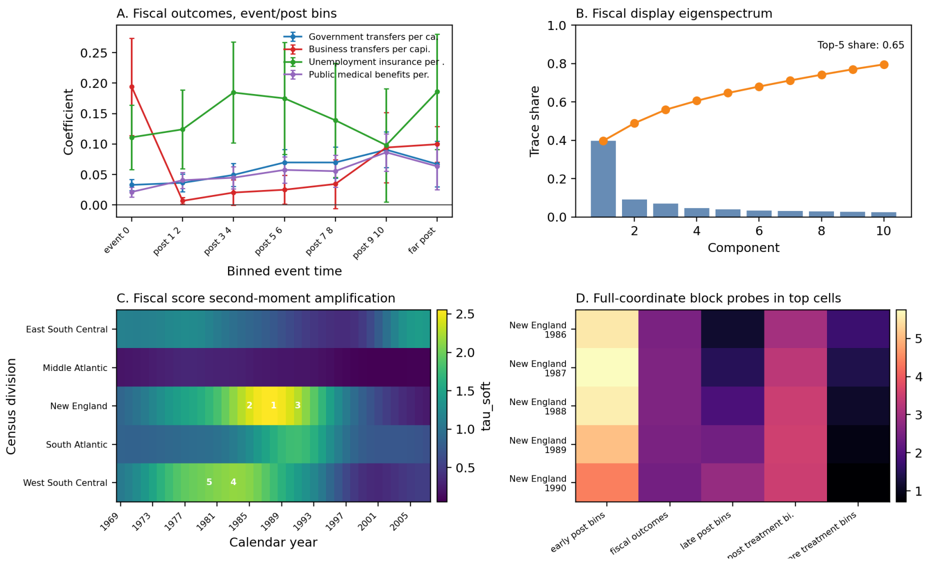
This example is not intended as a new substantive evaluation of the NSW training program. It shows that observations in different propensity-score and prognostic-score regions contribute differently to the influence second-moment geometry of the same global and subgroup ATE estimators. Any claim that a pattern is independent of local data availability should be assessed with the reported ESS and maximum-weight diagnostics.

3.3. *Panel Data Event Study*

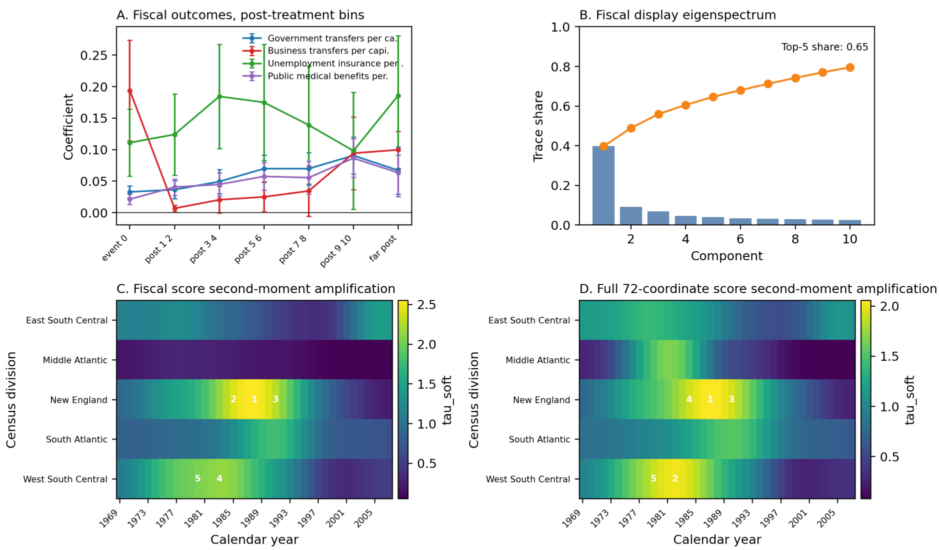
We next turn to the hurricane event-study setting of Deryugina (2017), which combines spatial variation, calendar time, and staggered treatment. The full estimand surface contains six outcomes—four fiscal and two labor-market outcomes—across 12 non-omitted event-time bins, for $p = 72$. The fiscal surface retains the four fiscal outcomes across all of the same 12 bins, including the pre-treatment bins, for $p = 48$. The state is Census division and calendar year. The displayed field smooths observation-level county-year score outer products. Because it omits within-county cross-products, it is an observation-level score second-moment field rather than a county-clustered variance operator.

Figure 5 compares state-indexed observation-level score second moments for two event-study estimand surfaces. The full surface contains six outcomes and twelve non-omitted event-time bins ($p = 72$). The fiscal surface retains the four fiscal outcomes but retains all twelve event-time bins ($p = 48$); it is therefore a fiscal-outcome surface rather than a post-treatment-only surface. Subfigure a contains the fiscal block probes, while subfigure b contains the fiscal-versus-72-coordinate comparison. Both displayed eigenspectrum panels report a top-five trace share of 0.65. The largest supported fiscal nodes occur in New England in 1988, 1985, and 1991. Because adjacent years receive heavily overlapping kernel weights, these peaks should be read as a localized episode rather than three independent findings. These values indicate local amplification of observation-level score second moments, not larger hurricane effects or larger county-clustered sampling variances. The coefficient panel displays event and post-treatment coefficients for readability; all second-moment matrices and eigenspectra use the complete 12-bin surface, including pre-treatment coordinates.

Figure 5 provides a targeted audit of the event-study design. A researcher extending this design should not read the New England peak as evidence that hurricane effects were necessarily larger in that region, nor as a replacement for county-clustered standard errors. The peak instead identifies a region-time episode in which the fixed event-study surface has unusually large



(a) Fiscal-outcome block probes



(b) Fiscal versus full-surface comparison

FIGURE 5.—Full-coordinate observation-level score second-moment diagnostics for the Deryugina hurricane event study. Region-year matrices are estimated using Gaussian Nadaraya-Watson smoothing over calendar time within Census divisions ($h = 3$ years), followed by soft normalization relative to the empirical reference moment. Subfigure (a) contains the fiscal-outcome block probes. Subfigure (b) compares the 48-coordinate fiscal-outcome surface with the 72-coordinate full surface. Both surfaces retain all 12 non-omitted event-time bins, including pre-treatment bins. Both displayed eigenspectrum panels report a top-five trace share of 0.65. Numbers mark the five largest supported cells. Values above one indicate above-reference observation-level score second moments, not larger hurricane effects or county-clustered sampling variances.

observation-level score second moments. The next step is therefore to inspect that episode. One would identify the hurricanes, counties, event-time bins, and fiscal outcomes that generate the

New England amplification; examine whether the exposure is driven by the local score mean or by within-cell dispersion; report effective support and maximum local weights; and re-estimate the main event-study surface after leaving out the high-exposure episode, the relevant division, the relevant years, or the largest contributing counties. The fiscal and full-surface comparison also suggests a useful diagnostic distinction: if the same region-year episode is amplified in both the fiscal-only and full six-outcome surfaces, the exposure reflects a broader feature of the event-study design; if it is concentrated in the fiscal surface, the audit should focus on transfer measurement, fiscal reporting, and the timing of disaster-related public payments. Similarly, because the second-moment matrices include the complete non-omitted event-time surface, including pre-treatment bins, the audit should ask whether the exposed directions load on pre-period contrasts, post-period fiscal responses, or both. If the substantive coefficients are stable after these checks, the diagnostic strengthens the study by showing that the main conclusions do not depend on the exposed New England episode. If the coefficients move materially, the appropriate conclusion is not that clustered inference was invalid, but that the fiscal-cost estimates are empirically dependent on a specific and identifiable part of the spatiotemporal design.

4. CONCLUSION

This paper develops a state-indexed second-moment diagnostic for fixed high-dimensional estimators. The object measures where total squared influence is concentrated and which estimand directions carry it before those contributions are averaged into a variance matrix, standard error, or confidence band. Conditional covariance is the state-centered special case. Under iid sampling and the population state law, averaging the full conditional second moment recovers the one-observation influence covariance; under alternative reference laws, centered targets, or dependent sampling, it remains a declared reference-moment diagnostic.

The main value of the diagnostic is practical. It converts a general concern about robustness into a targeted audit of the empirical design. A researcher does not merely learn that an estimator is "fragile" in some vague sense. The researcher learns which dates, support regions, region-year cells, outcome blocks, horizons, subgroup coordinates, or estimand directions carry amplified squared influence. That information changes what should be inspected next. It tells the researcher where to examine leverage, overlap, local effective sample size, maximum local weights, residual behavior, first-stage exposure, local score means, and sensitivity to plausible specification changes.

The appropriate response depends on what the state variable means. When the state indexes support in a cross section, high amplification can point to remediable design problems: weak overlap, small local effective sample size, large weights, or a mismatch between treated and comparison observations. In that setting, the diagnostic may motivate trimming, overlap weighting, a change in the target population, additional comparison data, or explicit reporting that particular subgroup estimands are more fragile than others. When the state indexes calendar time in a historical macroeconomic application, the implication is different. A high-amplification month is evidence that the fixed response surface is empirically exposed to a particular historical episode. The relevant response is to inspect that episode, decompose the exposure across shocks, residuals, outcomes, and horizons, compare plausible alternative specifications, and report whether the substantive conclusions survive those targeted checks. When the state indexes region-year cells in an event study, the same logic points to leave-episode, leave-region, leave-year, and leave-large-unit audits, as well as checks of whether exposure loads on pre-treatment bins, post-treatment bins, or particular outcome blocks.

This practical role is distinct from conventional inference. The local-projection and hurricane applications smooth diagonal observation-level score outer products, so the displayed fields are not state-dependent HAC or cluster-robust covariance operators. HAC, cluster-robust, bootstrap, and simultaneous-band procedures remain the relevant tools for tests, confidence intervals, and uncertainty statements about the original estimators. The bootstrap procedures used in the applications quantify uncertainty in the estimated second-moment maps rather than converting those maps into sampling-covariance estimators. The diagnostic sits beside conventional inference: it reveals what those covariance calculations aggregate away.

Large amplification should also be interpreted with care. It can arise from sparse support, leverage concentration, large residual scores, a nonzero local score mean, local nonstationarity, omitted dependence, or the interaction of these forces. A high-amplification state is not automatically a high-contribution state. Influence intensity asks whether a state's normalized second moment is large relative to a reference moment; influence contribution also depends on how much population or sample mass the state receives. Thus state mass, effective sample size, maximum local weight, and mean-versus-within decompositions are needed before translating a visible peak into a claim about variance contribution or reliability.

The applications illustrate the range of uses. In the monetary-policy local projections, the diagnostic identifies historically specific exposure in the response-score geometry and shows how that exposure rotates across horizons and outcomes. In the proxy-IV specification, the exact product form separates high-amplification periods associated with proxy or first-stage exposure from those associated with residual-response energy. In the LaLonde application, support regions do not define separate treatment effects, but they do reveal how different parts of the propensity and prognostic support load on the fixed vector of global and subgroup ATE estimators. In the hurricane event study, the region-year map identifies a New England fiscal episode as a place where the fixed event-study surface has unusually large observation-level score second moments, motivating a targeted audit rather than a claim of larger hurricane effects or larger county-clustered variances.

The limitations are correspondingly clear. The diagnostic is only as meaningful as the state variable, reference law, and smoothing geometry supplied by the researcher. A calendar-time smoother, a propensity-score smoother, and a region-year graph encode different substantive questions. If no economically meaningful state geometry exists, a varying state-indexed field is not identified from arbitrary observation labels. In addition, the current construction deliberately studies diagonal observation-level score second moments in the time-series and panel applications. Future work can extend the same relative-moment geometry to state-indexed long-run covariance operators that include serial score cross-products and to cluster-level kernels that include within-cluster cross-products. Those extensions would connect the diagnostic more directly to HAC and cluster-robust inference.

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APPENDIX A: PROOFS

A.1. *Proof of Proposition 1*

For $i \in \mathcal{I}_n$, write $\psi_n(i;\omega)$ for the corresponding influence contribution. The map $(f,g) \mapsto f \otimes g$ is continuous from $\mathcal{H} \times \mathcal{H}$ to $\mathcal{S}_1(\mathcal{H})$, so each kernel entry is strongly measurable. Since $f \otimes g$ is rank one and $\|f \otimes g\|_1 = \|f\|_{\mathcal{H}}\|g\|_{\mathcal{H}}$, Cauchy-Schwarz gives

$$\mathbb{E}\|k_n^{\psi}(\cdot;i,j)\|_1 \le \{\mathbb{E}\|\psi_n(i)\|^2\}^{1/2}\{\mathbb{E}\|\psi_n(j)\|^2\}^{1/2} < \infty.$$

Thus every kernel entry is Bochner integrable in $\mathcal{S}_1(\mathcal{H})$. Because $\mathcal{H}$ is separable, $\mathcal{K}(\mathcal{H})$ is separable and $\mathcal{S}_1(\mathcal{H}) = \mathcal{K}(\mathcal{H})^*$ is a separable dual; in particular, $\mathcal{S}_1(\mathcal{H})$ has the Radon-Nikodym property. Therefore every Bochner-integrable $\mathcal{S}_1(\mathcal{H})$-valued random variable admits a conditional expectation with respect to $\mathcal{G}$.

Fix $i_1,\dots,i_N \in \mathcal{I}_n$ and $a_1,\dots,a_N \in \mathcal{H}$. Using $(f\otimes g)h = f\langle g,h\rangle_{\mathcal{H}}$ and the fact that $\mathcal{H}$ is real,

$$\begin{aligned}\sum_{r,q=1}^{N} \langle a_r, k_n^{\psi}(\omega;i_r,i_q)a_q\rangle &= \sum_{r,q=1}^{N} \langle a_r, \psi_n(i_r;\omega)\rangle\langle a_q, \psi_n(i_q;\omega)\rangle \\ &= \left\{\sum_{r=1}^{N}\langle a_r, \psi_n(i_r;\omega)\rangle\right\}^2 \ge 0.\end{aligned}$$

This proves pathwise positive definiteness. Taking expectations proves random positive definiteness and, by linearity, positive definiteness of the deterministic mean kernel.

For the conditional statement, let

$$Q(\omega) := \sum_{r,q=1}^{N} \langle a_r, k_n^{\psi}(\omega;i_r,i_q)a_q\rangle \ge 0.$$

Then $\mathbb{E}[Q \mid \mathcal{G}] \ge 0$ almost surely. For fixed $a,b \in \mathcal{H}$, the map $L_{a,b}(T) = \langle a, Tb\rangle$ is a bounded linear functional on $\mathcal{S}_1(\mathcal{H})$ because $|L_{a,b}(T)| \le \|a\|\|b\|\|T\|_1$. Hence bounded linear functionals commute with the Banach-valued conditional expectation, and

$$\mathbb{E}[Q \mid \mathcal{G}] = \sum_{r,q=1}^{N} \left\langle a_r, \mathbb{E}[k_n^{\psi}(\cdot;i_r,i_q) \mid \mathcal{G}]a_q\right\rangle \ge 0$$

almost surely for each fixed finite collection.

Finally, $\mathcal{I}_n$ is finite and $\mathcal{H}$ is separable. Choose a countable dense subset of $\mathcal{H}$, intersect the probability-one events for all finite collections drawn from that subset, and extend the inequality to all of $\mathcal{H}$ by continuity. Modifying the finitely many conditional kernel entries on a common null set gives an almost surely positive definite version. *Q.E.D.*

A.2. *Proof of Lemma 2*

Define the resolvent transition kernel

$$q_\eta(s,du) := \eta\int_0^\infty e^{-\eta r}p_r(s,du)\,dr.$$

Because each $p_r(s,\cdot)$ is a probability measure and $\eta\int_0^\infty e^{-\eta r}dr=1$, $q_\eta(s,\cdot)$ is also a probability measure. Thus

$$K_\eta(s)=\int_{\mathsf{S}} K(u)q_\eta(s,du)$$

is a Bochner integral of positive trace-class operators and is therefore positive and trace class.

For average preservation, Fubini's theorem and invariance of π give

$$\begin{aligned}\int_{\mathsf{S}} K_\eta(s)\,\pi(ds) &= \eta\int_0^\infty e^{-\eta r}\left\{\int_{\mathsf{S}} P_r K(s)\,\pi(ds)\right\}dr\\ &= \eta\int_0^\infty e^{-\eta r}C\,dr = C.\end{aligned}$$

For the contraction property, Jensen's inequality for the Bochner norm and invariance imply

$$\begin{aligned}\|J_\eta F\|_{L^1(\pi;\mathcal{S}_1)} &\le \eta\int_0^\infty e^{-\eta r}\int_{\mathsf{S}} P_r\{\|F\|_1\}(s)\,\pi(ds)dr\\ &= \|F\|_{L^1(\pi;\mathcal{S}_1)}.\end{aligned}$$

To prove approximation of the identity, change variables $u=\eta r$:

$$\|J_\eta F-F\|_{L^1}\le\int_0^\infty e^{-u}\|P_{u/\eta}F-F\|_{L^1}\,du.$$

For each fixed u, strong continuity gives $\|P_{u/\eta}F-F\|_{L^1}\to 0$, while contractivity bounds the integrand by $2e^{-u}\|F\|_{L^1}$. Dominated convergence proves $J_\eta F\to F$ in L^1.

Finally, if $K(u)\preceq M(u)C$, then for every $g\in\mathcal{H}$,

$$\begin{aligned}\langle g,K_\eta(s)g\rangle &= J_\eta\{u\mapsto\langle g,K(u)g\rangle\}(s)\\ &\le (J_\eta M)(s)\langle g,Cg\rangle.\end{aligned}$$

The scalar inequality holds for every vector in a fixed countable dense subset of $\mathcal{H}$ on a common full-measure set. Continuity of the quadratic forms then extends it to every $g\in\mathcal{H}$, yielding $K_\eta(s)\preceq M_\eta(s)C$ almost everywhere. *Q.E.D.*

A.3. *Proof of Theorem 1*

Fix η and a state s for which $K_\eta(s)\preceq M_\eta(s)C$. First show that the form is well defined. If $C^{1/2}g=C^{1/2}g'$, then $u:=g-g'\in\ker C$, and

$$0\le\langle u,K_\eta(s)u\rangle\le M_\eta(s)\langle u,Cu\rangle=0.$$

The Cauchy-Schwarz inequality for the positive semidefinite form induced by $K_\eta(s)$ gives

$$|\langle u,K_\eta(s)h\rangle|^2\le\langle u,K_\eta(s)u\rangle\langle h,K_\eta(s)h\rangle=0.$$

Thus replacing g by g' does not change the form. The same argument applies to the second coordinate.

Positivity and symmetry follow from positivity and self-adjointness of $K_\eta(s)$. Moreover,

$$|a_{\eta,s}(C^{1/2}g, C^{1/2}h)|^2 \le \langle g, K_\eta(s)g\rangle\langle h, K_\eta(s)h\rangle$$
$$\le M_\eta(s)^2\|C^{1/2}g\|^2\|C^{1/2}h\|^2.$$

Hence $a_{\eta,s}$ is bounded on $\operatorname{ran}(C^{1/2})$, which is dense in $\mathcal{H}_C$. It extends uniquely to a bounded positive symmetric form on $\mathcal{H}_C$. By the Riesz representation theorem, there is a unique bounded operator $A_\eta^\infty(s)$ such that

$$a_{\eta,s}(x,y) = \langle x, A_\eta^\infty(s)y\rangle, \qquad x,y\in\mathcal{H}_C.$$

Symmetry and positivity of the form imply that the operator is self-adjoint and positive. The quadratic bound gives

$$0 \le \langle x, A_\eta^\infty(s)x\rangle \le M_\eta(s)\|x\|^2,$$

which proves (2.12).

For weak measurability, first take $x = C^{1/2}g$ and $y = C^{1/2}h$. Then

$$s \mapsto \langle x, A_\eta^\infty(s)y\rangle = \langle g, K_\eta(s)h\rangle$$

is measurable because K_η is strongly measurable in trace norm. For general $x,y\in\mathcal{H}_C$, choose sequences from $\operatorname{ran}(C^{1/2})$ converging to x,y. The corresponding scalar functions converge pointwise by the bound above, so the limit is measurable.

For the weak average identity, let $x = C^{1/2}g$ and $y = C^{1/2}h$. Lemma 2 gives

$$\int \langle x, A_\eta^\infty(s)y\rangle\,\pi(ds) = \int \langle g, K_\eta(s)h\rangle\,\pi(ds)$$
$$= \langle g, Ch\rangle = \langle C^{1/2}g, C^{1/2}h\rangle = \langle x,y\rangle.$$

The identity extends to all $x,y\in\mathcal{H}_C$ by density. Indeed, $|\langle x, A_\eta^\infty(s)y\rangle| \le M_\eta(s)\|x\|\|y\|$, and $\int M_\eta d\pi = \int M d\pi < \infty$ by invariance. Uniqueness follows from uniqueness in the Riesz representation theorem. *Q.E.D.*

A.4. *Proof of Corollary 1*

For $x_g = C^{1/2}g$, factorization gives

$$r_\eta(s;g) = \frac{\langle x_g, A_\eta^\infty(s)x_g\rangle}{\|x_g\|^2}.$$

The bound follows from (2.12), and the average-one identity follows from (2.13) with $x = y = x_g$.

Let $Q = \sum_{j\ge1} q_j e_j \otimes e_j$ be the spectral decomposition of the positive trace-class probe, where $q_j \ge 0$ and $\sum_j q_j = 1$. Then

$$\operatorname{tr}\{QA_\eta^\infty(s)\} = \sum_{j\ge1} q_j\langle e_j, A_\eta^\infty(s)e_j\rangle.$$

Every term is nonnegative and bounded above by $q_j M_\eta(s)$. Tonelli's theorem and (2.13) imply

$$\int \operatorname{tr}\{Q A_\eta^\infty(s)\}\,\pi(ds) = \sum_j q_j \|e_j\|^2 = 1.$$

For the block result, boundedness of S_B implies that $C^{1/2} S_B^* S_B C^{1/2}$ is trace class. Define

$$Q_B := \frac{C^{1/2} S_B^* S_B C^{1/2}}{d_B}.$$

This operator is positive trace class and

$$\operatorname{tr}(Q_B) = \frac{\operatorname{tr}(S_B C S_B^*)}{d_B} = 1.$$

Using $K_\eta = C^{1/2} A_\eta^\infty C^{1/2}$ and cyclicity of trace,

$$\operatorname{tr}\{S_B K_\eta(s) S_B^*\} = d_B \operatorname{tr}\{Q_B A_\eta^\infty(s)\}.$$

Thus $\tau_{\eta,B} = \tau_{\eta,Q_B}$, and the preceding result applies. *Q.E.D.*

A.5. *Proof of Proposition 2*

Fix s and write $A = A_\eta^\infty(s)$. The spectrum of A lies in $[0, M_\eta(s)]$. Apply the bounded spectral calculus to

$$f_\lambda(a) = \frac{1}{\lambda + a}, \qquad b_\lambda(a) = \frac{\lambda}{\lambda + a}, \qquad y_\lambda(a) = \frac{a}{\lambda + a}.$$

For $a \geq 0$, $0 < f_\lambda(a) \leq \lambda^{-1}$ and $0 \leq b_\lambda(a), y_\lambda(a) \leq 1$. This proves the positivity, self-adjointness, norm, and contraction claims.

Because A is bounded and positive, $-A$ generates the uniformly continuous contraction semigroup e^{-rA}. The scalar identity

$$(\lambda + a)^{-1} = \int_0^\infty e^{-\lambda r} e^{-ar}\,dr$$

transfers through the spectral calculus to give (2.20). The directional bounds follow from $0 \preceq Y \preceq I$. If $Q \succeq 0$ and $\operatorname{tr}(Q) = 1$, then monotonicity of trace gives

$$0 \leq \operatorname{tr}(QY) \leq \operatorname{tr}(Q) = 1.$$

Q.E.D.

A.6. *Soft-whitening part of Corollary 1*

Let $S_\rho = (C + \rho I)^{-1/2} C^{1/2}$. Since functions of C commute,

$$A_{\eta,\rho}^{\text{soft}} = S_\rho A_\eta^\infty S_\rho.$$

The operator is positive and trace class because S_ρ is Hilbert-Schmidt:

$$\|S_\rho\|_{\text{HS}}^2 = \operatorname{tr}\{C(C + \rho I)^{-1}\} = d_\rho < \infty.$$

Since $C \neq 0$, $d_\rho > 0$.

Define $Q_\rho = S_\rho^2/d_\rho$. Then Q_ρ is positive trace class with trace one, and cyclicity gives

$$\tau_{\eta,\rho}^{\mathrm{soft}}(s) = \mathrm{tr}\{Q_\rho A_\eta^\infty(s)\}.$$

Corollary 1 therefore gives the pointwise bound and average-one normalization. *Q.E.D.*

A.7. *Proof of Proposition 3*

The kernel inclusion $\ker C \subseteq \ker K_\eta(s)$ makes the form (2.10) well defined by the same representative argument used in the proof of Theorem 1. By assumption it is closable, so its closure is a densely defined, closed, nonnegative symmetric form on $\mathcal{H}_C$. The first representation theorem for closed forms gives a unique positive self-adjoint operator $\mathfrak{A}_\eta(s)$ with

$$\overline{a}_{\eta,s}(x,y) = \langle \mathfrak{A}_\eta(s)^{1/2}x, \mathfrak{A}_\eta(s)^{1/2}y\rangle.$$

A positive self-adjoint operator is maximal accretive. Equivalently, $-\mathfrak{A}_\eta(s)$ is maximal dissipative and therefore generates a strongly continuous contraction semigroup. The spectral theorem gives, for $a \geq 0$,

$$0 \leq \frac{\lambda}{\lambda + a} \leq 1, \qquad 0 \leq \frac{a}{\lambda + a} \leq 1,$$

which proves the operator inequalities. Under relative domination the form is bounded, so its representing operator is bounded. Uniqueness of the form representation then identifies it with $A_\eta^\infty(s)$. *Q.E.D.*

A.8. *Proof of Theorem 4*

For the first claim,

$$\begin{aligned}\mathbb{E}[z^{(R)}z^{(R)\prime} \mid S = s] &= \Lambda_R^{-1/2}V_R^*\mathbb{E}[\chi \otimes \chi \mid S = s]V_R\Lambda_R^{-1/2} \\ &= A_R(s).\end{aligned}$$

Integrating over s and using $\int K(s)d\pi = C$ gives

$$\int A_R(s)d\pi = \Lambda_R^{-1/2}V_R^*CV_R\Lambda_R^{-1/2} = I_R.$$

Linearity of J_η and the fact that V_R, Λ_R do not depend on s give $A_{\eta,R} = J_\eta A_R$ entry by entry.

For the compression identity, fix $1 \leq \ell, m \leq R$. Since $C^{1/2}(\lambda_\ell^{-1/2}v_\ell) = v_\ell$, factorization gives

$$\langle v_\ell, A_\eta^\infty(s)v_m\rangle = \lambda_\ell^{-1/2}\lambda_m^{-1/2}\langle v_\ell, K_\eta(s)v_m\rangle.$$

These are exactly the entries of $A_{\eta,R}(s)$, proving (E.5).

Average normalization of $\tau_{\eta,R}$ follows from $\int A_{\eta,R}d\pi = I_R$. If the retained coordinates are rotated by an orthogonal matrix O, their covariance becomes $OA_{\eta,R}O'$, whose trace is unchanged.

For reconstruction,

$$\begin{aligned}V_R\Lambda_R^{1/2}A_{\eta,R}(s)\Lambda_R^{1/2}V_R^* &= V_RV_R^*K_\eta(s)V_RV_R^* \\ &= P_RK_\eta(s)P_R.\end{aligned}$$

Let P_C denote the orthogonal projection onto $\mathcal{H}_C$. The eigenvectors associated with the positive eigenvalues of C form an orthonormal basis of $\mathcal{H}_C$, so $P_R \to P_C$ strongly on $\mathcal{H}$. By factorization,

$$K_\eta(s) = C^{1/2} A_\eta^\infty(s) C^{1/2}, \qquad K_\eta(s) = P_C K_\eta(s) P_C.$$

Therefore

$$\begin{aligned}\|K_\eta(s) - P_R K_\eta(s) P_R\|_1 &\le \|(P_C - P_R) K_\eta(s)\|_1 \\ &\quad + \|K_\eta(s)(P_C - P_R)\|_1 \to 0.\end{aligned}$$

Indeed, if T is trace class and $Q_R \to 0$ strongly with $\sup_R \|Q_R\|_{\mathrm{op}} < \infty$, then $\|Q_R T\|_1 \to 0$ and $\|T Q_R\|_1 \to 0$, by finite-rank approximation.

For the full relative moment, write $A = A_\eta^\infty(s)$. Then

$$\begin{aligned}\|P_R A P_R x - Ax\| &\le \|P_R A(P_R x - x)\| + \|(P_R - I)Ax\| \\ &\le \|A\| \|P_R x - x\| + \|(P_R - I)Ax\| \to 0.\end{aligned}$$

Finally, use $K_\eta = C^{1/2} A C^{1/2}$ and $\|A\|_{\mathrm{op}} \le M_\eta(s)$. The Schatten inequality $\|UV\|_1 \le \|U\|_{\mathrm{HS}} \|V\|_{\mathrm{HS}}$ gives

$$\begin{aligned}\|(I - P_R) K_\eta\|_1 &\le \|(I - P_R) C^{1/2}\|_{\mathrm{HS}} \|A C^{1/2}\|_{\mathrm{HS}} \\ &\le M_\eta(s) \left(\sum_{\ell > R} \lambda_\ell \right)^{1/2} \mathrm{tr}(C)^{1/2}.\end{aligned}$$

The same bound applies to $\|P_R K_\eta (I - P_R)\|_1$. Summing proves (E.6). *Q.E.D.*

A.9. *Proof of Corollary 4*

Fix s and write $A = A_\eta^\infty(s)$. Extend the finite compression to $\mathcal{H}_C$ by $\widetilde{A}_R = P_R A P_R$. Theorem 4 gives $\widetilde{A}_R x \to Ax$ for every x, and $\sup_R \|\widetilde{A}_R\| \le \|A\|$. Let $f_\lambda(a) = a/(\lambda + a)$. On the compact interval $[0, \|A\|]$, approximate f_λ uniformly by polynomials whose constant term is zero. Strong convergence of $\widetilde{A}_R$ and uniform boundedness imply strong convergence of every polynomial in $\widetilde{A}_R$, and the uniform approximation then yields

$$f_\lambda(\widetilde{A}_R)x \to f_\lambda(A)x.$$

Because $f_\lambda(0) = 0$ and the matrix of $\widetilde{A}_R$ on $P_R \mathcal{H}_C$ is $A_{\eta,R}$,

$$f_\lambda(\widetilde{A}_R) = V_R Y_{\eta,\lambda,R} V_R^*.$$

If Q is trace class, choose finite-rank Q_m with $\|Q - Q_m\|_1 \to 0$. Strong convergence gives convergence of $\mathrm{tr}\{Q_m V_R Y_{\eta,\lambda,R} V_R^*\}$ for each fixed m, and, because both Yosida transforms are contractions,

$$|\mathrm{tr}\{(Q - Q_m)(V_R Y_{\eta,\lambda,R} V_R^* - Y_{\eta,\lambda})\}| \le 2\|Q - Q_m\|_1.$$

Taking $R \to \infty$ and then $m \to \infty$ proves convergence of every fixed trace-class-probe summary. *Q.E.D.*

A.10. *Proof of Proposition 4*

Fix s, suppress it in the notation, and write $A := \widehat{A}(s)$. Define the block-diagonal operator

$$A_0 := PAP + (I-P)A(I-P).$$

Both diagonal blocks are positive, so $A_0 \succeq 0$. With $f_\lambda(a) = a/(\lambda+a) = 1-\lambda/(\lambda+a)$, the resolvent identity gives

$$f_\lambda(A) - f_\lambda(A_0) = \lambda(\lambda I + A)^{-1}(A - A_0)(\lambda I + A_0)^{-1}.$$

Since both resolvents have norm at most λ^{-1},

$$\|f_\lambda(A) - f_\lambda(A_0)\|_{\mathrm{op}} \le \frac{\|A - A_0\|_{\mathrm{op}}}{\lambda}. \tag{A.1}$$

Relative to $P\mathcal{H} \oplus (I-P)\mathcal{H}$, $A - A_0$ has off-diagonal block form

$$A - A_0 = \begin{pmatrix} 0 & PA(I-P) \\ (I-P)AP & 0 \end{pmatrix}.$$

Squaring this self-adjoint block operator shows that

$$\|A - A_0\|_{\mathrm{op}} = \|(I-P)AP\|_{\mathrm{op}}. \tag{A.2}$$

Functional calculus for the block-diagonal operator gives

$$f_\lambda(A_0) = Pf_\lambda(A_P)P + (I-P)f_\lambda(A_{P^\perp})(I-P),$$

where $A_{P^\perp} := (I-P)A(I-P)|_{(I-P)\mathcal{H}}$. Hence

$$\begin{aligned}\left|\widehat{q}_{\lambda,Q} - \widehat{q}^{P}_{\lambda,Q}\right| \le{}& |\mathrm{tr}\{Q[f_\lambda(A) - f_\lambda(A_0)]\}| \\ &+ \mathrm{tr}\{Q(I-P)f_\lambda(A_{P^\perp})(I-P)\}.\end{aligned}$$

The first term is at most $\|Q\|_1 \|f_\lambda(A) - f_\lambda(A_0)\|_{\mathrm{op}}$, and $\|Q\|_1 = \mathrm{tr}(Q) = 1$. For the second term, $0 \preceq f_\lambda(A_{P^\perp}) \preceq I$, so positivity and cyclicity of trace give

$$0 \le \mathrm{tr}\{Q(I-P)f_\lambda(A_{P^\perp})(I-P)\} \le \mathrm{tr}\{Q(I-P)\}.$$

Combining these inequalities with (A.1)-(A.2) proves (2.23). *Q.E.D.*

A.11. *Proof of Proposition 5*

Suppress n, s in the notation and write $w_i = w_{\eta,n,i}(s)$, $\delta_i = \widehat{\xi}_i - \xi_i$. Decompose

$$\begin{aligned}\widehat{K}_{\eta,n}(s) - K_\eta(s) ={}& \sum_i w_i(\widehat{Y}_i - Y_i) + \sum_i w_i\{Y_i - K(S_i)\} \\ &+ \left\{\sum_i w_i K(S_i) - K_\eta(s)\right\}.\end{aligned}$$

The last term is $O_p\{b_{K,n}(s)\}$ by Assumption 6(i). Assumption 6(ii) and conditional Markov's inequality imply

$$\left\|\sum_i w_i\{Y_i - K(S_i)\}\right\|_{\mathrm{HS}} = O_p\left\{\left(\sum_i w_i^2\right)^{1/2}\right\} = O_p(N_{\eta,n}^{-1/2}).$$

For the first-stage term,

$$\widehat{Y}_i - Y_i = \xi_i \otimes \delta_i + \delta_i \otimes \xi_i + \delta_i \otimes \delta_i.$$

Since $\|f \otimes g\|_{\mathrm{HS}} = \|f\|\|g\|$,

$$\begin{aligned}
\left\|\sum_i w_i(\widehat{Y}_i - Y_i)\right\|_{\mathrm{HS}} &\le 2\sum_i w_i\|\xi_i\|\|\delta_i\| + \sum_i w_i\|\delta_i\|^2 \\
&\le 2\left(\sum_i w_i\|\xi_i\|^2\right)^{1/2}\left(\sum_i w_i\|\delta_i\|^2\right)^{1/2} + \sum_i w_i\|\delta_i\|^2 \\
&= O_p\{d_{K,n}(s)\},
\end{aligned}$$

where the final line uses Assumption 6(iii)-(iv). Combining the terms proves the local Hilbert-Schmidt rate. Repeating the same argument with the reference weights $a_i = a_{\pi,n,i}$ gives $O_p\{b_{C,n} + N_{\pi,n}^{-1/2} + d_{C,n}\}$ for $\widehat{C}_n$. Operator-norm rates follow from $\|T\|_{\mathrm{op}} \le \|T\|_{\mathrm{HS}}$.

A simple sufficient condition for the local bound in Assumption 6(ii) is that, conditional on the state array, the $\mathcal{S}_2(\mathcal{H})$-valued variables $Y_{i,n} - K_{i,n}$ are conditionally mean zero, pairwise uncorrelated, and have uniformly bounded conditional second moments. Since $\|Y_{i,n}\|_{\mathrm{HS}} = \|\xi_{i,n}\|_{\mathcal{H}}^2$, a uniform conditional fourth-moment bound is sufficient in the conditionally independent case. Assumption 1 alone does not provide this bound when $\delta < 2$. *Q.E.D.*

A.12. *Proof of Theorem 2*

Write $K = K_\eta(s)$, $\widehat{K} = \widehat{K}_{\eta,n}(s)$, $B_\rho = (C + \rho I)^{-1/2}$, and $\widehat{B}_\rho = (\widehat{C}_n + \rho I)^{-1/2}$. Both inverse square roots have operator norm at most $\rho^{-1/2}$.

For positive operators bounded below by ρI, the integral representation

$$D^{-1/2} = \frac{1}{\pi}\int_0^\infty t^{-1/2}(D + tI)^{-1}\,dt$$

and the resolvent identity imply

$$\|D_1^{-1/2} - D_2^{-1/2}\|_{\mathrm{op}} \le c\rho^{-3/2}\|D_1 - D_2\|_{\mathrm{op}} \tag{A.3}$$

for a universal constant c. Therefore

$$\|\widehat{B}_\rho - B_\rho\|_{\mathrm{op}} \le c\rho^{-3/2}\|\widehat{C}_n - C\|_{\mathrm{op}}.$$

Now

$$\begin{aligned}
\widehat{A}_{\eta,\rho,n} - A_{\eta,\rho} = \widehat{B}_\rho(\widehat{K} - K)\widehat{B}_\rho + (\widehat{B}_\rho - B_\rho)K\widehat{B}_\rho \\
+ B_\rho K(\widehat{B}_\rho - B_\rho).
\end{aligned}$$

Consequently,

$$\|\widehat{A}_{\eta,\rho,n} - A_{\eta,\rho}\|_{\mathrm{op}} \le \rho^{-1}\|\widehat{K} - K\|_{\mathrm{op}} + c_s\rho^{-2}\|\widehat{C}_n - C\|_{\mathrm{op}}, \tag{A.4}$$

where $c_s < \infty$ depends on $\|K_\eta(s)\|_{\mathrm{op}}$. The rate conditions prove the first claim.

By factorization,

$$A_{\eta,\rho} = S_\rho A_\eta^\infty S_\rho, \qquad S_\rho = (C + \rho I)^{-1/2} C^{1/2}.$$

The spectral multiplier of S_ρ is $\sqrt{\mu/(\mu+\rho)}$. Hence $0 \preceq S_\rho \preceq I$ and $S_\rho x \to x$ for every $x \in \mathcal{H}_C$. Since A_η^∞ is bounded,

$$\|S_\rho A_\eta^\infty S_\rho x - A_\eta^\infty x\| \le \|A_\eta^\infty\|\|S_\rho x - x\| + \|(S_\rho - I)A_\eta^\infty x\| \to 0.$$

Thus $A_{\eta,\rho} \to A_\eta^\infty$ strongly on $\mathcal{H}_C$. After the zero extension, both operators vanish on $\ker C$, so the convergence is strong on $\mathcal{H}$ as well.

Let $R_n = (\lambda I + \widehat{A}_{\eta,\rho_n,n})^{-1}$, $R_{\rho_n} = (\lambda I + A_{\eta,\rho_n})^{-1}$, and $R = (\lambda I + A_\eta^\infty)^{-1}$. The resolvent identity gives

$$\|R_n - R_{\rho_n}\|_{\mathrm{op}} \le \lambda^{-2}\|\widehat{A}_{\eta,\rho_n,n} - A_{\eta,\rho_n}\|_{\mathrm{op}} = o_p(1).$$

For fixed x,

$$(R_\rho - R)x = R_\rho(A_\eta^\infty - A_{\eta,\rho})Rx,$$

which converges to zero because $A_{\eta,\rho} \to A_\eta^\infty$ strongly and $\|R_\rho\| \le \lambda^{-1}$. Thus $R_n x \to_p Rx$. Multiplication by λ gives inverse-amplification convergence, while $A(\lambda I + A)^{-1} = I - \lambda(\lambda I + A)^{-1}$ gives Yosida convergence.

For a trace-class probe Q, first suppose Q is finite rank. Strong convergence of the uniformly bounded Yosida transforms implies convergence of the finitely many matrix elements entering the trace. For general Q, choose finite-rank Q_m with $\|Q - Q_m\|_1 \to 0$. Since all Yosida transforms are contractions,

$$|\operatorname{tr}\{(Q - Q_m)(\widehat{Y} - Y)\}| \le 2\|Q - Q_m\|_1.$$

Take $n \to \infty$ and then $m \to \infty$. *Q.E.D.*

A.13. *Proof of Corollary 2*

Let $D_n = \widehat{C}_n + \rho_n I$ and $\widehat{A} = D_n^{-1/2}\widehat{K}_{\eta,n}(s)D_n^{-1/2}$. Since $\widehat{K}_{\eta,n}(s) + \lambda D_n \succeq \lambda\rho_n I$, the linear solve in (2.28) has a unique solution. Moreover,

$$\lambda I + \widehat{A} = D_n^{-1/2}\{\widehat{K}_{\eta,n}(s) + \lambda D_n\}D_n^{-1/2},$$

so

$$(\lambda I + \widehat{A})^{-1} = D_n^{1/2}\{\widehat{K}_{\eta,n}(s) + \lambda D_n\}^{-1}D_n^{1/2}.$$

With $x_n = D_n^{1/2}g$, equation (2.28) implies

$$\widehat{p}_{\eta,\lambda,n}(s;g) = \lambda\frac{\langle x_n, (\lambda I + \widehat{A})^{-1}x_n\rangle}{\|x_n\|^2}.$$

Operator-norm consistency of $\widehat{C}_n$ and $\rho_n \to 0$ imply $x_n \to_p C^{1/2}g =: x$. Let $R_n = (\lambda I + \widehat{A})^{-1}$ and $R = (\lambda I + A_\eta^\infty(s))^{-1}$. Since $\|R_n\|_{\mathrm{op}}, \|R\|_{\mathrm{op}} \le \lambda^{-1}$,

$$\begin{aligned}|\langle x_n, R_n x_n\rangle - \langle x, Rx\rangle| \le \lambda^{-1}(\|x_n\| + \|x\|)\|x_n - x\| \\ + |\langle x, (R_n - R)x\rangle| \to_p 0,\end{aligned}$$

where the last term converges by Theorem 2 on the fixed vector x. Also $\|x_n\|^2 \to_p \|x\|^2 = \langle g, Cg\rangle > 0$. Hence the displayed Rayleigh quotient converges to

$$\lambda \frac{\langle x, (\lambda I + A_\eta^\infty(s))^{-1} x\rangle}{\|x\|^2}.$$

Subtracting from one proves the claim. *Q.E.D.*

A.14. *Proof of Corollary 3*

For any $\rho > 0$, write

$$N_\rho(s) := \mathrm{tr}\{K_\eta(s)(C + \rho I)^{-1}\}, \qquad \widehat{N}_{\rho,n}(s) := \mathrm{tr}\{\widehat{K}_{\eta,n}(s)(\widehat{C}_n + \rho I)^{-1}\},$$
$$d_\rho := \mathrm{tr}\{C(C + \rho I)^{-1}\}, \qquad \widehat{d}_{\rho,n} := \mathrm{tr}\{\widehat{C}_n(\widehat{C}_n + \rho I)^{-1}\}.$$

Thus $\tau_{\eta,\rho}^{\mathrm{soft}}(s) = N_\rho(s)/d_\rho$ and $\widehat{\tau}_{\eta,\rho,n}^{\mathrm{soft}}(s) = \widehat{N}_{\rho,n}(s)/\widehat{d}_{\rho,n}$. The resolvent identity gives

$$\begin{aligned}|\widehat{N}_{\rho,n}(s) - N_\rho(s)| \le \rho^{-1}\|\widehat{K}_{\eta,n}(s) - K_\eta(s)\|_1 \\ + \|K_\eta(s)\|_1 \rho^{-2}\|\widehat{C}_n - C\|_{\mathrm{op}},\end{aligned}$$

and

$$|\widehat{d}_{\rho,n} - d_\rho| \le \rho^{-1}\|\widehat{C}_n - C\|_1 + \|C\|_1\rho^{-2}\|\widehat{C}_n - C\|_{\mathrm{op}}.$$

For fixed ρ, trace-norm consistency makes both right-hand sides $o_p(1)$, because trace-norm convergence implies operator-norm convergence. Since $C \ne 0$, $d_\rho > 0$, so $\widehat{d}_{\rho,n}$ is bounded away from zero with probability tending to one. The population index is finite, and the ratio identity below proves the fixed-ρ claim.

For $\rho = \rho_n \downarrow 0$, the three rates stated in Corollary 3 imply

$$|\widehat{N}_{\rho_n,n}(s) - N_{\rho_n}(s)| = o_p(1), \qquad |\widehat{d}_{\rho_n,n} - d_{\rho_n}| = o_p(1).$$

Because $C \ne 0$, its largest eigenvalue λ_1 is positive, and for all sufficiently large n,

$$d_{\rho_n} = \mathrm{tr}\{C(C + \rho_n I)^{-1}\} \ge \frac{\lambda_1}{\lambda_1 + \rho_n} \ge \frac{1}{2}.$$

Consequently, $\widehat{d}_{\rho_n,n} \ge 1/4$ with probability tending to one. Moreover, Corollary 1 gives

$$0 \le \frac{N_{\rho_n}(s)}{d_{\rho_n}} = \tau_{\eta,\rho_n}^{\mathrm{soft}}(s) \le M_\eta(s).$$

For either fixed or shrinking ρ, on the event that the sample denominator is positive,

$$\left|\widehat{\tau}^{\rm soft}_{\eta,\rho,n}(s) - \tau^{\rm soft}_{\eta,\rho}(s)\right| \le \frac{|\widehat{N}_{\rho,n}(s) - N_\rho(s)|}{\widehat{d}_{\rho,n}} + \tau^{\rm soft}_{\eta,\rho}(s)\frac{|\widehat{d}_{\rho,n} - d_\rho|}{\widehat{d}_{\rho,n}}.$$

The preceding bounds make the right-hand side $o_p(1)$ in both cases. *Q.E.D.*

A.15. *Eigenspace and whitening-map part of Proposition 8*

Assumption 5 and $\|T\|_{\rm op} \le \|T\|_{\rm HS}$ give

$$\|\widehat{C}_n - C\|_{\rm op} = O_p(r_{C,n}).$$

Weyl's inequality yields

$$\max_{\ell \le R+1} |\widehat{\lambda}_{\ell,n} - \lambda_\ell| = O_p(r_{C,n}).$$

Let the retained population eigenvectors be partitioned into the maximal repeated-eigenvalue clusters from Assumption 8. Each retained cluster is separated from the rest of the spectrum by a fixed positive gap. Applying Davis-Kahan separately to the corresponding sample and population cluster projectors gives an $O_p(r_{C,n})$ projector error for every retained cluster. Summing the finitely many cluster projectors gives

$$\|\widehat{P}_R - P_R\|_{\rm op} = O_p(r_{C,n}).$$

For each cluster, the orthogonal Procrustes lemma then supplies a within-cluster alignment with $O_p(r_{C,n})$ basis error. Concatenating these alignments yields a block-diagonal orthogonal matrix O_n satisfying

$$\|\widehat{V}_R O_n' - V_R\|_{\rm op} = O_p(r_{C,n}).$$

Within a repeated-eigenvalue cluster the population eigenvalue is constant, so orthogonal rotations do not alter its population inverse square root. Across distinct clusters the positive separation prevents asymptotic mixing. Since $x \mapsto x^{-1/2}$ is Lipschitz on a compact interval bounded away from zero, Weyl's bound and $\lambda_R \ge c_\lambda$ give

$$\|O_n \widehat{\Lambda}_R^{-1/2} O_n' - \Lambda_R^{-1/2}\|_{\rm op} = O_p(r_{C,n}).$$

Now add and subtract terms in $O_n \widehat{\Lambda}_R^{-1/2} \widehat{V}_R^* - \Lambda_R^{-1/2} V_R^*$ to obtain the whitening-map result. *Q.E.D.*

A.16. *Proof of Proposition 8*

Let $\widehat{W}_R^\circ = O_n \widehat{\Lambda}_R^{-1/2} \widehat{V}_R^*$ and $W_R = \Lambda_R^{-1/2} V_R^*$. Then

$$O_n \widehat{A}_{\eta,R,n}(s) O_n' = \widehat{W}_R^\circ \widehat{K}_{\eta,n}(s) \widehat{W}_R^{\circ *}, \qquad A_{\eta,R}(s) = W_R K_\eta(s) W_R^*.$$

Add and subtract $W_R\widehat{K}_{\eta,n}(s)\widehat{W}_R^{\circ *}$ and $W_R K_\eta(s)\widehat{W}_R^{\circ *}$. The eigenspace and whitening-map argument in Appendix A.15 gives $\|\widehat{W}_R^{\circ} - W_R\| = O_p(r_{C,n})$, while both whitening maps are $O_p(1)$ because λ_R is bounded away from zero. Assumption 5 gives $\|\widehat{K}_{\eta,n}(s) - K_\eta(s)\|_{\mathrm{op}} = O_p\{r_{K,n}(s)\}$ and hence $\|\widehat{K}_{\eta,n}(s)\|_{\mathrm{op}} = O_p(1)$. The displayed rate follows. Trace continuity in fixed dimension gives consistency of $\widehat{\tau}_{\eta,R,n}$. *Q.E.D.*

A.17. *Proof of Proposition 9*

Write $z_i = z_{i,n}^{(R)}$, $e_i = e_{i,n}$, $w_i = w_{\eta,n,i}(s)$, and $a_n = a_{\eta,n}(s)$. The aligned feasible estimator satisfies

$$O_n\widehat{A}_{\eta,R,n}(s)O_n' = \sum_i w_i(z_i + e_i)(z_i + e_i)'.$$

Set $\Delta_{A,n}(s) := O_n\widehat{A}_{\eta,R,n}(s)O_n' - A^{\mathrm{or}}_{\eta,R,n}(s)$. Then

$$\begin{aligned}\|\Delta_{A,n}(s)\|_F &\le 2\sum_i w_i\|z_i\|\|e_i\| + \sum_i w_i\|e_i\|^2 \\ &\le 2\left(\sum_i w_i\|z_i\|^2\right)^{1/2}\left(\sum_i w_i\|e_i\|^2\right)^{1/2} + \sum_i w_i\|e_i\|^2.\end{aligned}$$

The last two conditions in Assumption 9 imply that the first term is $o_p(a_n^{-1})$ and the quadratic term is $o_p(a_n^{-2})$. Weight normalization gives $a_n \ge 1$, so the full remainder is $o_p(a_n^{-1})$. Therefore the feasible and oracle estimators have the same first-order limit. Decompose the oracle error into its centered weighted sum and its smoothing/design bias, then apply the two convergence statements in Assumption 9 and Slutsky's theorem. *Q.E.D.*

A.18. *Proof of Theorem 3*

Fix $x \in \mathcal{H}_C$ and $\lambda > 0$. Write $a_n(v) := a_n(v, v)$ and $a(v) := a(v, v)$ on their respective form domains and set them equal to $+\infty$ outside those domains. Let

$$y_n = (\lambda I + A_n)^{-1}x, \qquad y = (\lambda I + A)^{-1}x.$$

The vector y_n is the unique minimizer of

$$F_n(v) := a_n(v) + \lambda\|v\|^2 - 2\langle x, v\rangle,$$

while y uniquely minimizes the analogous functional F built from a. Because $F_n(y_n) \le F_n(0) = 0$,

$$\lambda\|y_n\|^2 \le 2\|x\|\|y_n\|,$$

so $\{y_n\}$ is bounded. Take a weakly convergent subsequence $y_{n_k} \rightharpoonup \bar{y}$. The Mosco lower-bound condition and weak lower semicontinuity imply

$$F(\bar{y}) \le \liminf_k F_{n_k}(y_{n_k}).$$

For any v, let $v_n \to v$ be a Mosco recovery sequence. Since y_n minimizes F_n,

$$\limsup_k F_{n_k}(y_{n_k}) \le \limsup_k F_{n_k}(v_{n_k}) \le F(v).$$

Thus $F(\bar{y}) \le F(v)$ for every v, and uniqueness gives $\bar{y} = y$. Every weakly convergent subsequence has the same limit, so $y_n \rightharpoonup y$.

The preceding inequalities also imply $F_n(y_n) \to F(y)$. Combining this with the Mosco lower bound and $\langle x, y_n\rangle \to \langle x, y\rangle$ yields $\|y_n\| \to \|y\|$. Weak convergence plus convergence of norms gives $y_n \to y$ strongly. Finally,

$$A_n(\lambda I + A_n)^{-1}x = x - \lambda(\lambda I + A_n)^{-1}x,$$

so the Yosida transforms converge strongly as well. *Q.E.D.*

A.19. *Proof of Proposition 6*

Fix $\kappa > 0$ and write $M_n(\kappa) := N_n + \kappa L_n$. For every $x \in \mathbb{R}^{m_n}$,

$$x' M_n(\kappa) x = \sum_{j=1}^{m_n} n_{j,n} x_j^2 + \kappa \sum_{1 \le j < k \le m_n} a_{jk,n}(x_j - x_k)^2.$$

If this expression is zero, then x vanishes at every occupied node and is constant on each connected component. Because every connected component contains an occupied node, $x = 0$. Hence $M_n(\kappa)$ is symmetric positive definite. Its off-diagonal entries are nonpositive, so it is a nonsingular M-matrix and $M_n(\kappa)^{-1}$ is elementwise nonnegative. Therefore $W_n(\kappa) = M_n(\kappa)^{-1}N_n$ is elementwise nonnegative.

Since $L_n\mathbf{1} = 0$,

$$M_n(\kappa)\mathbf{1} = N_n\mathbf{1},$$

which gives $W_n(\kappa)\mathbf{1} = \mathbf{1}$. Symmetry of $M_n(\kappa)$ also gives $\mathbf{1}'M_n(\kappa) = \mathbf{1}'N_n$, and hence

$$\mathbf{1}'N_n M_n(\kappa)^{-1} = \mathbf{1}', \qquad \mathbf{1}'N_n W_n(\kappa) = \mathbf{1}'N_n.$$

This proves (2.33). Because every $\widehat{H}_{k,n,\rho}$ is positive semidefinite and each row of $\widehat{W}_n$ is a probability vector, $\widehat{A}_{j,n,\rho} \succeq 0$. Congruence then gives $\widehat{K}_{j,n,\rho} \succeq 0$. Moreover,

$$\widehat{D}_{\rho,n}^{1/2}\widehat{H}_{k,n,\rho}\widehat{D}_{\rho,n}^{1/2} = \widehat{\bar{K}}_{k,n},$$

which proves the exact arithmetic identity in (2.32).

For completeness, fix one matrix coordinate and stack it across nodes as $b \in \mathbb{R}^{m_n}$, with corresponding node-average vector h. The coordinate contribution to (2.34) is

$$(b-h)'N_n(b-h) + \widehat{\kappa}_n b' L_n b.$$

Its first-order condition is $(N_n + \widehat{\kappa}_n L_n)b = N_n h$, whose unique solution is $b = \widehat{W}_n h$. Applying this argument to every symmetric-matrix coordinate proves the penalized-GMRF representation.

Write $W_n := W_n(\kappa_n)$. Columns corresponding to empty nodes are zero for both $\widehat{W}_n$ and W_n. Therefore, uniformly over j,

$$\begin{aligned}
&\left\|\widehat{K}_{j,n,\rho} - K^{\mathrm{G}}_{\kappa_n,n}(s_{j,n})\right\|_{\mathrm{op}} \\
&\quad \le \sum_{k:n_{k,n}>0} (\widehat{W}_n)_{jk}\left\|\widehat{\bar{K}}_{k,n} - K^{\circ}_{k,n}\right\|_{\mathrm{op}} + \sum_{k:n_{k,n}>0}\left|(\widehat{W}_n)_{jk} - (W_n)_{jk}\right|\left\|K^{\circ}_{k,n}\right\|_{\mathrm{op}}
\end{aligned}$$

$$\leq \max_{k:n_{k,n}>0} \left\| \widehat{\widetilde{K}}_{k,n} - K^{\circ}_{k,n} \right\|_{\mathrm{op}} + \left\{ \max_{j\leq m_n} \sum_{k=1}^{m_n} \left| (\widehat{W}_n)_{jk} - (W_n)_{jk} \right| \right\} \max_{k:n_{k,n}>0} \left\| K^{\circ}_{k,n} \right\|_{\mathrm{op}}.$$

Assumption 7(i)-(ii) makes the right-hand side $o_p(1)$. Also, because W_n is row stochastic,

$$\max_{j\leq m_n} \|K^{\mathrm{G}}_{\kappa_n,n}(s_{j,n})\|_{\mathrm{op}} \leq \max_{k:n_{k,n}>0} \|K^{\circ}_{k,n}\|_{\mathrm{op}} = O_p(1).$$

Let $\widehat{R}_{\rho,n} := (\widehat{C}_n + \rho I_p)^{-1/2}$ and $R_\rho := (C + \rho I_p)^{-1/2}$. Because $\rho > 0$ is fixed, the inverse-square-root map is Lipschitz on the positive cone bounded below by ρI_p, so Assumption 7(iii) gives $\|\widehat{R}_{\rho,n} - R_\rho\|_{\mathrm{op}} = o_p(1)$. Add and subtract terms in

$$\widehat{A}_{j,n,\rho} - A^{\mathrm{G}}_{\kappa_n,\rho,n}(s_{j,n}) = \widehat{R}_{\rho,n}\widehat{K}_{j,n,\rho}\widehat{R}_{\rho,n} - R_\rho K^{\mathrm{G}}_{\kappa_n,n}(s_{j,n})R_\rho.$$

The moment-field result, the uniform $O_p(1)$ bound above, and $\|\widehat{R}_{\rho,n}\|_{\mathrm{op}}, \|R_\rho\|_{\mathrm{op}} \leq \rho^{-1/2}$ then yield the uniform relative-moment result.

For $F_\lambda(A) := A(\lambda I_p + A)^{-1} = I_p - \lambda(\lambda I_p + A)^{-1}$, the resolvent identity gives, for positive semidefinite A and B,

$$\|F_\lambda(A) - F_\lambda(B)\|_{\mathrm{op}} \leq \lambda^{-1}\|A - B\|_{\mathrm{op}}.$$

Applying this bound node by node proves uniform convergence of the stress transforms. For a fixed positive trace-one probe Q,

$$|\operatorname{tr}\{Q(F_\lambda(A) - F_\lambda(B))\}| \leq \|Q\|_1 \|F_\lambda(A) - F_\lambda(B)\|_{\mathrm{op}},$$

and $\|Q\|_1 = 1$, which proves the probe statement.

Finally, the additional graph-approximation condition and the triangle inequality give uniform consistency for $K_\eta(s_{j,n})$. Applying the same fixed-ρ congruence map gives uniform consistency for $A_{\eta,\rho}(s_{j,n})$. *Q.E.D.*

APPENDIX B: EXAMPLES

Macro example: localized amplification in a policy-response surface. Let $e_{j,k}$ be localized contrasts of a monetary-policy response surface, with larger j representing a narrower horizon or maturity interval and $k = 1, \dots, 2^j$. Suppose $Ce_{j,k} = \gamma_j e_{j,k}$ with $\gamma_j = 2^{-3j}$. An announced Treasury auction, issuance concentration, or predictable dealer balance-sheet constraint may identify a location S_t at which response-score second-moment intensity is concentrated. In the active cell, suppose

$$K(S_t) e_{j,k_j(S_t)} = 2^j \gamma_j e_{j,k_j(S_t)}.$$

If the active location ranges across cells under the reference distribution, both C and $K(s)$ remain trace class because $\sum_j 2^j \gamma_j = \sum_j 2^{-2j} < \infty$. Yet the relative amplification of the active localized contrast is $2^j \to \infty$. Its absolute second moment remains finite, but it becomes arbitrarily large relative to its reference moment. Because γ_j is small, a leading-eigenmode truncation can omit exactly these directions. Its stress contribution instead is $2^j/(\lambda + 2^j) \to 1$ and remains bounded under grid refinement.

Micro example: comparison-group erosion in difference-in-differences. Consider staggered minimum-wage adoption and a treatment-effect surface indexed by event time and baseline low-wage exposure. Let

$$\pi_t^C(x) := \Pr(G_i > t \mid X_i = x, \mathcal{F}_{t-1})$$

be the predictable probability that a jurisdiction remains available as a not-yet-treated comparison. A control component of a group-time or doubly robust DiD influence function has the schematic form

$$\chi_{i,t}^C \approx -\frac{1\{G_i > t\} \omega_t(X_i)}{\pi_t^C(X_i)} \varepsilon_{i,t},$$

so its predictable second moment contains $\omega_t(x)^2 \sigma_t^2(x)/\pi_t^C(x)$. If, in progressively narrower exposure-by-event-time cells, the remaining comparison probability is of order 2^{-j} relative to its reference level, the corresponding relative moment is of order 2^j. Every fixed cell can remain identified because $\pi_t^C(x) > 0$, while uniform overlap over the full treatment-effect surface fails. The stress transform records these directions as approaching complete stress rather than asking the researcher to estimate a diverging generalized eigenvalue. It does not restore identification when the comparison probability is exactly zero.

The two examples separate absolute from relative amplification. A state moment operator may have finite trace, and every fixed reported contrast may have a finite second moment, while increasingly localized directions exhibit unbounded amplification relative to the reference moment. The bounded stress transform remains comparable in that setting.

APPENDIX C: EMPIRICAL TARGETS AND COMPUTATIONAL DETAILS

This appendix records the estimands, influence contributions, smoothing rules, reference laws, numerical safeguards, and validation checks used in the two microeconomics applications. The purpose is to state exactly what each amplification surface estimates and to provide enough detail to reproduce the calculations. Both applications smooth complete-coordinate score second moments and normalize them with a trace-normalized soft index. They differ in their sampling units, states, kernels, reference laws, and uncertainty assessments. In the LaLonde application, an eigensystem of the reference moment is used only for a separate interpretability figure and does not enter Panels B or C or their bootstrap bands. Because the targets and reference laws differ, numerical indices are comparable only within an application and reference specification.

C.1. *A common smoothing template and the interpretation of the target*

Let $i = 1, \dots, n$ index the sampling units used by an application, let $S_i \in \mathbb{R}^d$ or $S_i \in \mathsf{S}$ be the state, let $Y_i^{\mathrm{OVK}} \in \mathbb{S}_+^q$ be the positive-semidefinite matrix contribution to be smoothed, and let $\omega_i \geq 0$ be an observation weight. For a target state s and nonnegative kernel $k_H(S_i, s)$, define

$$\Delta_H(s) := \sum_{i=1}^n \omega_i k_H(S_i, s), \qquad \ell_{i,H}(s) := \frac{\omega_i k_H(S_i, s)}{\Delta_H(s)}, \tag{C.1}$$

$$\widehat{M}_H(s) := \sum_{i=1}^n \ell_{i,H}(s) Y_i^{\mathrm{OVK}}, \qquad \mathrm{ESS}_H(s) := \left\{ \sum_{i=1}^n \ell_{i,H}(s)^2 \right\}^{-1}. \tag{C.2}$$

The normalization in (C.1) makes $\sum_i \ell_{i,H}(s) = 1$ whenever $\Delta_H(s) > 0$. Thus $\widehat{M}_H(s)$ is a Nadaraya-Watson conditional-moment estimator, not an unnormalized kernel sum.

A reference law is represented by nonnegative weights a_i satisfying $\sum_i a_i = 1$. It defines the reference moment

$$\widehat{C}_\pi := \sum_{i=1}^n a_i Y_i^{\mathrm{OVK}}. \tag{C.3}$$

When the reference is induced by a grid with normalized node weights q_g,

$$a_i = \sum_g q_g \ell_{i,H}(s_g) \quad \Longrightarrow \quad \widehat{C}_\pi = \sum_g q_g \widehat{M}_H(s_g). \tag{C.4}$$

Identity (C.4) is a useful replication check because it must hold up to floating-point error.

The target is a local second moment unless a conditional mean has explicitly been removed. In particular, if $Y_i^{\mathrm{OVK}} = u_i u_i'$, then

$$\sum_i \ell_i(s) u_i u_i' = \underbrace{\bar{u}(s)\bar{u}(s)'}_{\text{local mean component}} + \underbrace{\sum_i \ell_i(s)\{u_i - \bar{u}(s)\}\{u_i - \bar{u}(s)\}'}_{\text{local within component}}, \qquad \bar{u}(s) := \sum_i \ell_i(s) u_i. \tag{C.5}$$

Accordingly, a large amplification index can reflect a large local conditional mean of the score, a large local within-state covariance, or both. It is not by itself a standard error, a test statistic, a treatment-effect magnitude, or a measure of structural heterogeneity.

TABLE II

DESIGN CONTRAST ACROSS THE TWO MICROECONOMICS APPLICATIONS

Feature	Hurricane event study	LaLonde treatment effects
Sampling unit used by the OVK field	County-year observation-level event-study score row	Cross-sectional individual
State	Census division and calendar year	Cross-fitted logit propensity score, alone or jointly with the control prognostic score
Matrix contribution	$\psi_i\psi_i'$ in the complete coefficient space	$\widehat{E}_i\widehat{E}_i'$ in all $p=11$ estimand coordinates
Dimension	$p=48$ for the primary fiscal surface and $p=72$ for the full comparison surface	$p=11$ in Panels B-D; rank-5 PCA is display-only
Smoother	Exact-division spatial matching and Gaussian calendar-time Nadaraya-Watson weights	Gaussian Nadaraya-Watson weights with a scalar or full SPD bandwidth matrix
Default reference	Empirical observation law	Grid quadrature, with node weights proportional to the kernel denominator
Normalization	Soft whitening by $\widehat{C}_\pi+\rho I_p$ and effective dimension d_ρ	The same trace-normalized soft geometry, computed by Cholesky solves
Uncertainty assessment	Bandwidth sensitivity and leave-year-out refitting; conventional event-study inference is clustered separately by county	Multinomial unit bootstrap that repeats the full estimation procedure, with 2,000 draws

C.2. *Hurricane event-study application*

C.2.1. *Sampling unit, sample construction, and event timing*

The OVK sampling unit is an observation-level county-year score row. County clustering is used in the conventional event-study covariance calculation, but county scores are *not* summed before constructing the primary OVK field. The analysis sample contains 40,027 county-year rows from 1,051 counties, of which 317 are hurricane-treated and 734 are never treated. The event years of treated counties range from 1979 through 1996.

The event-time construction preserves the replication-design variable `sample_hurr_state` before constructing event time and does not reinterpret that variable as an event year. For treated county c, define

$$T_c := \min\{y : \texttt{hurricane}_{c,y}=1,\ 1979\le y\le 2002\}, \qquad E_{c,y} := y - T_c. \tag{C.6}$$

Never-treated counties have all event-bin indicators set to zero. The analysis keeps calendar years 1969-2012, imposes `consistent_sample`$=1$, requires latitude, longitude, the government transfer outcome, and the baseline controls, and then uses a common complete-case sample across the six headline outcomes.

The omitted category is event time -2 to -1, corresponding to `hurr_f1_2`. The remaining 12 indicators form the event-study coordinate vector. Table III reports the exact coding and treated-row counts in the analysis sample.

C.2.2. *Outcome surfaces and coefficient target*

The full event-study surface contains six standardized outcomes and 12 non-omitted event-time coefficients for each outcome, giving $p=72$. The primary fiscal surface retains four fiscal outcomes across the same 12 bins, giving $p=48$. Importantly, the primary fiscal surface includes both pre-treatment and post-treatment coordinates; it is not a post-treatment-only surface.

TABLE III

HURRICANE EVENT-TIME BINS

Code	Event times	Role	Omitted	Treated rows
far_pre	$E \leq -11$	far pre-period	no	2,121
pre_9_10	$-10 \leq E \leq -9$	pre-period	no	634
pre_7_8	$-8 \leq E \leq -7$	pre-period	no	634
pre_5_6	$-6 \leq E \leq -5$	pre-period	no	634
pre_3_4	$-4 \leq E \leq -3$	pre-period	no	634
omitted_pre_1_2	$-2 \leq E \leq -1$	baseline	yes	634
event_0	$E = 0$	event year	no	317
post_1_2	$1 \leq E \leq 2$	post-period	no	634
post_3_4	$3 \leq E \leq 4$	post-period	no	634
post_5_6	$5 \leq E \leq 6$	post-period	no	634
post_7_8	$7 \leq E \leq 8$	post-period	no	634
post_9_10	$9 \leq E \leq 10$	post-period	no	634
far_post	$E \geq 11$	far post-period	no	3,493

TABLE IV

OUTCOME COORDINATES IN THE HURRICANE SURFACES

Outcome	Prepared variable	Surface role
Government transfers per capita	log_curr_trans_ind_gov_pc	Fiscal
Business transfers per capita	log_curr_trans_ind_bus_pc	Fiscal
Average wage/salary disbursements	log_avg_wage_sal_disb	Labor/income
Employment rate	emp_rate_tot_adult	Labor/income
Unemployment insurance per capita	log_unemp_pc	Fiscal
Public medical benefits per capita	log_public_med_benefits_pc	Fiscal

Let $D^{\mathrm{raw}} \in \mathbb{R}^{n \times 12}$ contain the non-omitted event-bin indicators and let $Y^{\mathrm{raw}} \in \mathbb{R}^{n \times 6}$ contain the six outcomes. Each outcome is standardized on the final common sample:

$$Y_{im}^{\mathrm{mod}} := \frac{Y_{im}^{\mathrm{raw}} - \bar{Y}_m}{\widehat{\sigma}_m}. \tag{C.7}$$

The absorber matrix H contains county fixed effects, calendar-year fixed effects, coastal-by-year interactions, and baseline-control-by-year interactions. The event indicators and all outcomes are residualized jointly by the ridge-stabilized least-squares problem

$$\widehat{B}_H := \arg\min_B \left\| [D^{\mathrm{raw}}, Y^{\mathrm{mod}}] - HB \right\|_F^2 + \kappa_H \|B\|_F^2, \qquad [\widetilde{D}, \widetilde{Y}] := [D^{\mathrm{raw}}, Y^{\mathrm{mod}}] - H\widehat{B}_H, \tag{C.8}$$

where κ_H is a small numerical stabilizer. The coefficient matrix and residuals are

$$\widehat{Q} := n^{-1}\widetilde{D}'\widetilde{D}, \qquad \widehat{\beta} := \widehat{Q}^{+}\left(n^{-1}\widetilde{D}'\widetilde{Y}\right), \tag{C.9}$$

$$\widehat{e}_i := \widetilde{Y}_i - \widetilde{D}_i\widehat{\beta}, \qquad \widehat{d}_i := \widehat{Q}^{+}\widetilde{D}_i', \tag{C.10}$$

where $(\cdot)^{+}$ denotes the Moore-Penrose inverse, $\widetilde{D}_i$ is row i, and $\widehat{e}_i \in \mathbb{R}^6$. The observation-level coefficient influence contribution is

$$\widehat{\psi}_{i,(m,k)} := \widehat{d}_{ik}\widehat{e}_{im}, \qquad \widehat{\psi}_i = \mathrm{vec}(\widehat{e}_i\widehat{d}_i') \in \mathbb{R}^{72}. \tag{C.11}$$

The primary $p = 48$ vector is obtained by retaining the four fiscal outcome blocks. The conventional cluster-robust covariance instead aggregates these score contributions within `county_fips`; that separate calculation is not the target of the OVK heat map.

The design-matrix diagnostics are $\mathrm{cond}(\widehat{Q}) = 37.1204$, a minimum eigenvalue of 0.0011, and a maximum absolute normal-equation residual of 8.13×10^{-17}. These quantities are useful checks that an independent reconstruction is using the same sample, absorber, and coefficient ordering.

C.2.3. *Region-time Nadaraya-Watson field*

For county-year observation i, let r_i be its Census division and let t_i be calendar year. The target nodes are $s = (r, t)$. The default region kernel uses exact division matching,

$$K_R(r, r_i) := \mathbf{1}\{r = r_i\}. \tag{C.12}$$

An optional graph-resolvent kernel, $K_R^{\mathrm{graph}} = \eta_{\mathrm{space}}(\eta_{\mathrm{space}} I + L_{\mathrm{region}})^{-1}$, is available as an alternative but is not used in the reported baseline. The temporal kernel is Gaussian,

$$K_T(t, t_i; h) := \exp\left[-\frac{1}{2}\left(\frac{t_i - t}{h}\right)^2\right], \qquad h = 3.0 \text{ years in the baseline.} \tag{C.13}$$

With the analysis weights ω_i, define

$$b_i(r, t) := \omega_i K_R(r, r_i) K_T(t, t_i; h), \qquad w_i(r, t) := \frac{b_i(r, t)}{\sum_{\ell=1}^{n} b_\ell(r, t)}. \tag{C.14}$$

No cross-division borrowing occurs in the baseline.

The complete-space local score moment and its exact mean/within decomposition are

$$\widehat{K}_{\mathrm{total}}(r, t) := \sum_i w_i(r, t)\widehat{\psi}_i\widehat{\psi}_i', \qquad \widehat{\mu}(r, t) := \sum_i w_i(r, t)\widehat{\psi}_i, \tag{C.15}$$

$$\widehat{K}_{\mathrm{mean}}(r, t) := \widehat{\mu}(r, t)\widehat{\mu}(r, t)', \qquad \widehat{K}_{\mathrm{within}}(r, t) := \widehat{K}_{\mathrm{total}}(r, t) - \widehat{K}_{\mathrm{mean}}(r, t). \tag{C.16}$$

Thus the primary field is the local second moment of observation-level event-study scores. It is not a county-clustered estimator variance. All three matrices are symmetrized. $\widehat{K}_{\mathrm{total}}$ is positive semidefinite by construction. A small numerical projection of $\widehat{K}_{\mathrm{within}}$ onto the PSD cone is applied only if roundoff creates negative eigenvalues, and the correction norm is reported.

A node is retained only if

$$\sum_i b_i(r, t) > 0, \qquad \mathrm{ESS}(r, t) := \left\{\sum_i w_i(r, t)^2\right\}^{-1} \geq 30, \qquad \sum_i \mathbf{1}\{b_i(r, t) > 0\} \geq 5. \tag{C.17}$$

Support diagnostics also include the nearest positive-weight observation-year distance. Unsupported nodes are left missing rather than replaced by the identity matrix.

C.2.4. *Reference law, soft whitening, and reported indices*

The baseline uses the empirical-observation reference,

$$a_i := \frac{\omega_i}{\sum_\ell \omega_\ell}, \qquad \widehat{C}_\pi := \sum_i a_i \widehat{\psi}_i \widehat{\psi}_i'. \tag{C.18}$$

Alternative reference laws include normalized user-supplied weights and a uniform-supported-grid law. In the latter mode, normalized node weights q_j induce $a_i = \sum_j q_j w_i(s_j)$ and therefore satisfy the exact identity $\widehat{C}_\pi = \sum_j q_j \widehat{K}_{\mathrm{total}}(s_j)$.

The absolute ridge is either supplied directly or constructed from the relative ridge by

$$\rho := \rho_{\mathrm{rel}} \frac{\mathrm{tr}(\widehat{C}_\pi)}{p}, \qquad \widehat{D}_\rho := \widehat{C}_\pi + \rho I_p, \qquad \widehat{d}_\rho := \mathrm{tr}\{\widehat{C}_\pi(\widehat{C}_\pi + \rho I_p)^{-1}\}. \tag{C.19}$$

For $q \in \{\mathrm{total}, \mathrm{mean}, \mathrm{within}\}$, let

$$\widehat{A}_q(r,t) := \widehat{D}_\rho^{-1/2} \widehat{K}_q(r,t) \widehat{D}_\rho^{-1/2}. \tag{C.20}$$

The reported indices are

$$\widehat{\tau}_{\mathrm{soft}}(r,t) := \frac{\mathrm{tr}\,\widehat{A}_{\mathrm{total}}(r,t)}{\widehat{d}_\rho}, \quad \widehat{\tau}_{\mathrm{mean}}(r,t) := \frac{\mathrm{tr}\,\widehat{A}_{\mathrm{mean}}(r,t)}{\widehat{d}_\rho}, \quad \widehat{\tau}_{\mathrm{within}}(r,t) := \frac{\mathrm{tr}\,\widehat{A}_{\mathrm{within}}(r,t)}{\widehat{d}_\rho}. \tag{C.21}$$

Apart from a recorded numerical PSD cleanup, the decomposition implies $\widehat{\tau}_{\mathrm{soft}} = \widehat{\tau}_{\mathrm{mean}} + \widehat{\tau}_{\mathrm{within}}$. No post-hoc normalization of the displayed stack is applied.

Subfigure a uses economic selectors in the original fiscal estimand coordinates. For a named fiscal block $B \subseteq \{1, \ldots, 48\}$, let S_B select those outcome-event-time coordinates. The reported probe is

$$\widehat{\tau}_B(r,t) := \frac{\mathrm{tr}\{S_B \widehat{K}_{\mathrm{total}}(r,t) S_B'\}}{\mathrm{tr}\{S_B \widehat{C}_\pi S_B'\}}, \tag{C.22}$$

whenever the denominator is positive. This is the finite-dimensional block selector in equation (2.16). The fiscal probes are nonadditive diagnostics rather than shares of $\widehat{\tau}_{\mathrm{soft}}(r,t)$, and they need not sum to one.

TABLE V

HURRICANE SOFT-WHITENING CONFIGURATION

Surface	p	ρ_{rel}	ρ	$\widehat{d}_\rho$	Reference
Fiscal, all event bins	48	0.0800	1.1893	31.1158	Empirical observations
Full six-outcome surface	72	0.0800	1.1256	47.9433	Empirical observations

The display-only eigenspectrum summaries for the fiscal and full surfaces both have a top-five trace share of 0.65. They therefore do not support a claim that restricting the outcome set raises the top-five share.

Under a grid-induced reference, the q_j-weighted average of $\widehat{\tau}_{\mathrm{soft}}(s_j)$ equals one algebraically. Under the empirical-observation reference there is no primitive grid law q_j, so the reported reference-state-mass average is a diagnostic rather than a forced renormalization. In the reported fiscal specification it is close to one for all three bandwidths in Table VI.

TABLE VI

HURRICANE BANDWIDTH SENSITIVITY FOR THE $p = 48$ FISCAL SURFACE

h (years)	Supported nodes	Maximum $\widehat{\tau}_{\mathrm{soft}}$	Grid mean	Reference-mass mean
1.5	195	2.7405	0.9785	1.0038
3.0	195	2.5506	0.9803	1.0049
5.0	195	2.2619	0.9847	1.0079

Because adjacent calendar years receive highly overlapping Gaussian weights, raw rankings do not represent independent discoveries. We therefore report both the raw node ranking and episode summaries that impose a within-division separation of at least $\lceil h \rceil$ years. Under the baseline bandwidth, the three largest raw fiscal nodes are New England in 1988 (2.5506), New England in 1985 (2.4118), and New England in 1991 (2.2796). These adjacent peaks should be summarized as one episode rather than reported as three unrelated events.

C.2.5. *Replication algorithm and validation*

A direct reconstruction of the reported hurricane field proceeds as follows.

1. Apply the sample restrictions, construct T_c and $E_{c,y}$ as in (C.6), and build the 12 non-omitted event-bin indicators.
2. Form the common complete-case outcome sample, standardize the six outcomes, build the absorber matrix, and compute $(\widetilde{D}, \widetilde{Y})$ using (C.8).
3. Estimate (C.9), retain the residuals, and construct the $p = 72$ influence vectors in (C.11). Select the four fiscal blocks for the primary $p = 48$ analysis.
4. For each division-year node and bandwidth, compute (C.14), the support diagnostics in (C.17), and the local moments in (C.15)-(C.16).
5. Construct the chosen reference law, $\widehat{C}_\pi$, the ridge, $\widehat{D}_\rho^{-1/2}$, and $\widehat{d}_\rho$. Evaluate (C.21) only on supported nodes.
6. Report raw rankings, bandwidth-separated episode summaries, division maxima, whitening diagnostics, and the fiscal-versus-full-surface comparison.

The leave-year-out validation must refit every training-dependent object. For each held-out year t, remove all observations with calendar year t; recompute the training reference weights, $\widehat{C}_\pi$, ρ, $\widehat{D}_\rho^{-1/2}$, and local weights for target year t; and then score the held-out observations with their fitted division-year matrices. The procedure must not reuse a full-sample eigensystem, reference second moment, whitening map, or kernel weights. The leave-year-out exercise scores all 40,027 rows and reports an average local-versus-fixed-reference score difference of 46.7996.

The reported estimator smooths unwhitened observation-level outer products before relative normalization, uses no ESS-dependent identity shrinkage, leaves unsupported nodes missing, and applies no post-estimation rescaling of the reported index. These properties are useful checks for independent reconstruction.

C.3. *LaLonde treatment-effect application*

C.3.1. *Observed data, nuisance targets, and generated states*

For individual i, the prepared observation is $O_i = (Y_i, D_i, X_i)$, where $D_i \in \{0,1\}$ is `treat`, $Y_i = \texttt{re78}_i/1000$, and

$$\begin{aligned} X_i = (&\texttt{age}, \texttt{educ}, \texttt{black}, \texttt{hispanic}, \texttt{married}, \texttt{nodegree}, \\ &\texttt{re74\_k}, \texttt{re75\_k}, \texttt{age2}, \texttt{educ2}, \texttt{re74\_zero}, \texttt{re75\_zero}, \\ &\texttt{black\_nodegree}, \texttt{hispanic\_nodegree}, \texttt{re74\_zero\_re75\_zero}). \end{aligned} \tag{C.23}$$

The analysis uses the public MatchIt LaLonde data. Race indicators are standardized, earnings are scaled to thousands of dollars, and the polynomial, zero-earnings, and interaction variables in (C.23) are constructed before observations with missing required fields are removed.

The nuisance functions are

$$e_0(x) := \Pr(D = 1 \mid X = x), \qquad m_d(x) := \mathbb{E}[Y \mid D = d, X = x], \quad d \in \{0,1\}. \tag{C.24}$$

The population state is

$$S_i^0 := \begin{pmatrix} \operatorname{logit}\{e_0(X_i)\} \\ m_0(X_i) \end{pmatrix}. \tag{C.25}$$

The one-dimensional path in Figure 4 uses only the first coordinate; the two-dimensional surface uses both the logit propensity and the control prognostic score.

The feasible state is generated by five-fold stratified cross-fitting with seed 20260614. The same fold assignment is used for the propensity model, both prognostic models, the generated state, and the AIPW influence contribution. For held-out fold I_k, the propensity model is fit on its complement, the control prognostic model is fit only on controls in that training fold, and the treated prognostic model is fit only on treated observations in that training fold. Every weighted training fold is required to retain positive support in both treatment arms. The reported specification fixes the learner classes, preprocessing maps, solver settings, regularization parameters, and other hyperparameters rather than relying on package defaults.

Out-of-fold propensities are clipped only for numerical stability,

$$\begin{aligned} &\widehat{e}_i^{\text{clip}} := \min\{0.98, \max(0.02, \widehat{e}_i)\}, \\ &\widehat{S}_i := \begin{pmatrix} \operatorname{logit}(\widehat{e}_i^{\text{clip}}) \\ \widehat{m}_{0i} \end{pmatrix}, \qquad \widehat{S}_i^{\text{std}} := \frac{\widehat{S}_i - \widehat{\mu}_S}{\widehat{\sigma}_S}, \end{aligned} \tag{C.26}$$

where the last division is coordinatewise. Clipping does not discard observations or redefine the target population. Clipping incidence, fold diagnostics, raw and standardized states, arm-specific nuisance predictions, the learner specification, and the fitted scaler are reported as replication diagnostics. Each bootstrap draw fits its own nuisance models and state scaler; raw coordinates are retained for figure axes and cross-draw alignment.

C.3.2. *Distinct weight systems*

The analysis distinguishes among resampling, training, empirical-moment, local-kernel, quadrature-node, and induced reference weights. This distinction is essential for reproducing the bootstrap and the panel-specific reference laws.

TABLE VII
WEIGHT SYSTEMS IN THE LALONDE ANALYSIS

Name	Symbol	Definition and use
Bootstrap weight	c_i	Raw multinomial unit count in a bootstrap draw; $c_i = 1$ in the original sample.
Training weight	t_i	Weight passed to nuisance-model fitting; the bootstrap specification sets $t_i = c_i$.
Observation weight	ω_i	Normalized empirical measure for AIPW and influence second moments, $\omega_i = c_i / \sum_j c_j$.
Local kernel weight	$\ell_{ji}(s)$	Panel-specific row-normalized Nadaraya-Watson weight at target state s.
Quadrature node weight	q_{jg}	Normalized weight on supported grid node g for panel $j \in \{\mathrm{B}, \mathrm{C}\}$.
Reference weight	a_{ji}	Induced observation-level measure defining panel j's reference second moment $\widehat{C}_{\pi,j}$.

C.3.3. *Global and subgroup ATE targets and AIPW influence contributions*

The application estimates one fixed vector of $p = 11$ global and subgroup ATEs. It does not estimate a separate ATE at each propensity or prognostic score. In the canonical estimand order,

$$\begin{aligned}\mathcal{B} = \{&\text{overall ATE, Black, non-Black, no degree, degree, zero 1974 earnings, zero 1975 earnings,}\\ &\text{age} \leq 25,\ \text{age} > 25,\ \text{education} \leq 10,\ \text{education} > 10\}.\end{aligned}$$

Let b_{iB} be the indicator for target population B. For the overall ATE, $b_{iB} = 1$ for every observation. Using the out-of-fold nuisance predictions, define the AIPW pseudo-outcome

$$\widehat{\phi}_i := \widehat{m}_{1i} - \widehat{m}_{0i} + \frac{D_i}{\widehat{e}_i^{\text{clip}}}(Y_i - \widehat{m}_{1i}) - \frac{1 - D_i}{1 - \widehat{e}_i^{\text{clip}}}(Y_i - \widehat{m}_{0i}). \tag{C.27}$$

For each target population,

$$\widehat{p}_B := \sum_i \omega_i b_{iB}, \qquad \widehat{\theta}_B := \frac{\sum_i \omega_i b_{iB} \widehat{\phi}_i}{\widehat{p}_B}, \tag{C.28}$$

$$\widehat{\psi}_{iB} := \frac{b_{iB}}{\widehat{p}_B}(\widehat{\phi}_i - \widehat{\theta}_B), \qquad \widehat{\psi}_i := (\widehat{\psi}_{iB} : B \in \mathcal{B})' \in \mathbb{R}^{11}. \tag{C.29}$$

The weighted mean of each influence coordinate is zero by construction, up to floating-point error. A final numerical recentering is nevertheless applied,

$$\widehat{\beta} := \sum_i \omega_i \widehat{\psi}_i, \qquad \widehat{E}_i := \widehat{\psi}_i - \widehat{\beta}. \tag{C.30}$$

The target estimates, raw influence vectors, recentered vectors, and influence summaries are retained separately for validation.

C.3.4. *Full-coordinate influence second-moment field*

Panels B and C are computed directly in the complete $p = 11$ estimand coordinates. Define the observation-level outer product and its empirical average by

$$\widehat{\Gamma}_i := \widehat{E}_i \widehat{E}_i' \in \mathbb{S}_+^{11}, \qquad \widehat{K}_{\mathrm{bar}} := \sum_i \omega_i \widehat{\Gamma}_i. \tag{C.31}$$

The notation $\widehat{\Gamma}_i$ is used to avoid confusing this matrix with the observed earnings outcome Y_i. No eigendecomposition of the empirical reference moment, display-rank choice, retained-basis projection, or observation-level SPD lift enters Panels B-D. In particular, the numerical constant 0.03 is the soft-whitening ridge ρ introduced below; it is not an $\alpha I + (1-\alpha)ZZ'$ ridge added to each observation.

At a state s in panel $j \in \{\mathrm{B}, \mathrm{C}\}$, the local target remains an influence second moment rather than a conditional covariance unless the local influence mean vanishes. With the panel-specific local weights defined below, write $\widehat{\mu}_{E,j}(s) := \sum_i \ell_{ji}(s)\widehat{E}_i$. Then

$$\widehat{K}_{H_j}(s) = \widehat{\mu}_{E,j}(s)\widehat{\mu}_{E,j}(s)' + \sum_i \ell_{ji}(s)\{\widehat{E}_i - \widehat{\mu}_{E,j}(s)\}\{\widehat{E}_i - \widehat{\mu}_{E,j}(s)\}'. \tag{C.32}$$

Thus attenuation or amplification can reflect the local mean term, the within-state covariance term, or both. It is not an estimate of a larger or smaller treatment effect at state s.

C.3.5. *Gaussian Nadaraya-Watson smoothing and evaluation grids*

Let x_i and y_i denote the first and second coordinates of $\widehat{S}_i^{\mathrm{std}}$, and define the panel-specific standardized states

$$\widehat{S}_{i,\mathrm{B}}^{\mathrm{std}} := x_i \in \mathbb{R}, \qquad \widehat{S}_{i,\mathrm{C}}^{\mathrm{std}} := (x_i, y_i)' = \widehat{S}_i^{\mathrm{std}} \in \mathbb{R}^2. \tag{C.33}$$

For panel $j \in \{\mathrm{B}, \mathrm{C}\}$ and its symmetric positive-definite bandwidth matrix H_j, the Gaussian kernel is

$$k_{H_j}(\widehat{S}_{i,j}^{\mathrm{std}}, s) := \exp\left[-\frac{1}{2}\left\| H_j^{-1/2}(\widehat{S}_{i,j}^{\mathrm{std}} - s)\right\|_2^2\right]. \tag{C.34}$$

Using observation weights ω_i, the panel-specific denominator, local weights, full-coordinate second-moment field, and support diagnostics are

$$\Delta_{H_j}(s) := \sum_i \omega_i k_{H_j}(\widehat{S}_{i,j}^{\mathrm{std}}, s), \qquad \ell_{ji}(s) := \frac{\omega_i k_{H_j}(\widehat{S}_{i,j}^{\mathrm{std}}, s)}{\Delta_{H_j}(s)}, \tag{C.35}$$

$$\widehat{K}_{H_j}(s) := \sum_i \ell_{ji}(s)\widehat{\Gamma}_i, \qquad \mathrm{ESS}_j(s) := \left\{\sum_i \ell_{ji}(s)^2\right\}^{-1}, \qquad w_{\max,j}(s) := \max_i \ell_{ji}(s). \tag{C.36}$$

The smoother permits a full SPD H_j. Reported diagnostics include H_j, $H_j^{-1/2}$, kernel denominators, ESS, maximum local weight, and distance summaries. The one- and two-dimensional results retain these diagnostics and the supported-node indicator at every grid location. This is a Euclidean Nadaraya-Watson conditional-moment estimator, not a heat-kernel, semigroup, or resolvent smoother.

Let $\widehat{Q}_\alpha(z)$ denote the empirical α-quantile of a scalar sample z. For the one-dimensional figure path,

$$\begin{aligned} G_{\mathrm{B}} &= 90, & \mathcal{G}_{\mathrm{B}} &= \operatorname{linspace}\{\widehat{Q}_{0.02}(x), \widehat{Q}_{0.98}(x), G_{\mathrm{B}}\}, \\ h_x &= \max\{0.15, 1.06\widehat{\sigma}_x n^{-1/5}\}, & H_{\mathrm{B}} &= [h_x^2]. \end{aligned} \tag{C.37}$$

The standardized baseline nodes are also mapped back to raw logit-propensity units and used as the fixed horizontal-axis grid for the bootstrap.

For the two-dimensional propensity-prognostic surface,

$$\begin{aligned} G_{\mathrm{C}} &= 30 \times 30, & \mathcal{G}_{\mathrm{C},x} &= \operatorname{linspace}\{\widehat{Q}_{0.02}(x), \widehat{Q}_{0.98}(x), 30\}, \\ \mathcal{G}_{\mathrm{C},y} &= \operatorname{linspace}\{\widehat{Q}_{0.02}(y), \widehat{Q}_{0.98}(y), 30\}, & h_x &= \max\{0.35, 1.25\widehat{\sigma}_x n^{-1/6}\}, \\ h_y &= \max\{0.35, 1.25\widehat{\sigma}_y n^{-1/6}\}, & H_{\mathrm{C}} &= \operatorname{diag}(h_x^2, h_y^2), \\ \mathcal{G}_{\mathrm{C}} &= \mathcal{G}_{\mathrm{C},x} \times \mathcal{G}_{\mathrm{C},y}. \end{aligned} \tag{C.38}$$

A Panel (C) node with $\mathrm{ESS}_{\mathrm{C}}(s) < 20$ is invalid for the reported surface and is left missing rather than interpolated, extrapolated, or imputed. If the initial bandwidth does not meet the predeclared minimum valid-node criterion, both bandwidths are multiplied by 1.2, for at most five attempts. The minimum valid-node count, number of bandwidth expansions, final bandwidth, supported-node mask, and final valid-node count are fixed or recorded without being chosen after examining the surface.

C.3.6. *Reference measure and the soft index*

Four reference modes are available:

$$\begin{aligned} \text{empirical:} \quad & a_{ji} \propto \omega_i; \\ \text{user supplied:} \quad & a_{ji} \propto \widetilde{a}_i, \quad \widetilde{a}_i \geq 0; \\ \text{importance:} \quad & a_{ji} \propto \omega_i r_{\pi,j}(\widehat{S}_{i,j}^{\mathrm{std}}); \\ \text{quadrature:} \quad & a_{ji} = \sum_g q_{jg}\ell_{ji}(s_{jg}), \quad j \in \{\mathrm{B}, \mathrm{C}\}. \end{aligned} \tag{C.39}$$

All observation-level reference weights are finite, nonnegative, and normalized to one. The reported figures use the quadrature mode. For panel $j \in \{\mathrm{B}, \mathrm{C}\}$, let s_{jg} be grid node g, let ℓ_{ji} and Δ_{H_j} denote the local weights and denominator formed with that panel's state and bandwidth, and let $\mathcal{G}_j^+$ denote its supported nodes. Panel (C)'s supported set excludes every node with $\mathrm{ESS}_{\mathrm{C}}(s_{jg}) < 20$. The normalized node weights are

$$q_{jg} := \frac{\mathbf{1}\{s_{jg} \in \mathcal{G}_j^+\}\Delta_{H_j}(s_{jg})}{\sum_h \mathbf{1}\{s_{jh} \in \mathcal{G}_j^+\}\Delta_{H_j}(s_{jh})}, \qquad q_{jg} \geq 0, \qquad \sum_g q_{jg} = 1. \tag{C.40}$$

The induced observation weights and reference moment are

$$a_{ji} := \sum_g q_{jg}\ell_{ji}(s_{jg}), \qquad \sum_i a_{ji} = 1,$$

$$\widehat{C}_{\pi,j} := \sum_i a_{ji}\widehat{\Gamma}_i = \sum_g q_{jg}\widehat{K}_{H_j}(s_{jg}), \qquad \mathrm{ESS}_{\pi,j} := \left(\sum_i a_{ji}^2\right)^{-1}. \tag{C.41}$$

The one- and two-dimensional grids induce separate quadrature laws, separate reference weights, and generally different $\widehat{C}_{\pi,j}$ matrices and effective dimensions.

For ridge $\rho > 0$ and panel j, form

$$\widehat{D}_{\rho,j} := \widehat{C}_{\pi,j} + \rho I_p = \widehat{L}_{\rho,j}\widehat{L}'_{\rho,j}, \qquad \widehat{d}_{\rho,j} := \mathrm{tr}(\widehat{D}_{\rho,j}^{-1}\widehat{C}_{\pi,j}). \tag{C.42}$$

The aggregate index displayed in Panels B and C is the full-coordinate, trace-normalized soft influence second-moment index

$$\begin{aligned}\widehat{\tau}_{\mathrm{soft},j}(s) &:= \frac{\mathrm{tr}\{\widehat{D}_{\rho,j}^{-1}\widehat{K}_{H_j}(s)\}}{\widehat{d}_{\rho,j}} \\ &= \frac{\mathrm{tr}\{\widehat{L}_{\rho,j}^{-\top}\widehat{L}_{\rho,j}^{-1}\widehat{K}_{H_j}(s)\}}{\widehat{d}_{\rho,j}}.\end{aligned} \tag{C.43}$$

Both traces are computed by Cholesky solves without explicitly forming $\widehat{D}_{\rho,j}^{-1}$ or $\widehat{D}_{\rho,j}^{-1/2}$. Under panel j's quadrature law,

$$\sum_g q_{jg}\widehat{\tau}_{\mathrm{soft},j}(s_{jg}) = \frac{\mathrm{tr}\left\{\widehat{D}_{\rho,j}^{-1}\sum_g q_{jg}\widehat{K}_{H_j}(s_{jg})\right\}}{\widehat{d}_{\rho,j}} = 1 \tag{C.44}$$

up to floating-point error. A value above one denotes an amplified soft-whitened influence second moment relative to that panel's own quadrature reference; a value below one denotes attenuation relative to that reference. Neither statement concerns the magnitude of a treatment effect or a state-specific standard error at that support location. Because Panels B and C use different quadrature laws and different $\widehat{C}_{\pi,j}$, their numerical ranges and color scales are not directly comparable.

C.3.7. *Original-coordinate directional and economic block ratios*

Panel (D) is evaluated on Panel (B)'s one-dimensional grid and uses exactly Panel B's local matrices, quadrature node weights, induced observation weights, and reference moment $\widehat{C}_{\pi,\mathrm{B}}$. It uses the economic selectors in equation (2.16), not columns of a Cholesky factor, eigenvectors, PCA scores, or any other transformed basis. For a coordinate subset $J \subseteq \{1,\ldots,p\}$, let $S_J \in \mathbb{R}^{|J|\times p}$ select the corresponding coordinates from the original estimand vector. Provided the denominator is positive, define

$$\widehat{\tau}_J(s) := \frac{\mathrm{tr}\{S_J\widehat{K}_{H_\mathrm{B}}(s)S'_J\}}{\mathrm{tr}\{S_J\widehat{C}_{\pi,\mathrm{B}}S'_J\}}. \tag{C.45}$$

For a singleton $J = \{j\}$, this reduces to the original-coordinate directional probe

$$\widehat{\tau}_j(s) := \frac{e'_j\widehat{K}_{H_\mathrm{B}}(s)e_j}{e'_j\widehat{C}_{\pi,\mathrm{B}}e_j} = \frac{(\widehat{K}_{H_\mathrm{B}}(s))_{jj}}{(\widehat{C}_{\pi,\mathrm{B}})_{jj}}. \tag{C.46}$$

Selectors are defined from the canonical estimand names rather than from hard-coded numerical positions. The five reported economic coordinate sets are

$$\begin{aligned}
J_{\text{overall}} &= \{\text{overall ATE}\},\\
J_{\text{race}} &= \{\text{Black}, \text{non-Black}\},\\
J_{\text{education}} &= \{\text{no degree}, \text{degree}, \text{education} \le 10, \text{education} > 10\},\\
J_{\text{earnings}} &= \{\text{zero 1974 earnings}, \text{zero 1975 earnings}\},\\
J_{\text{age}} &= \{\text{age} \le 25, \text{age} > 25\}.
\end{aligned}$$

These five coordinate sets form a disjoint partition of the original $p = 11$ estimand coordinates, although the underlying target populations represented by those coordinates can overlap at the observation level. The ratios in (C.45) are separately normalized probes, not additive shares. They do not decompose the soft index in Panel (B) and need not sum to one. They use neither the ridge ρ nor a whitening map. They are invariant to an internal permutation of matrix coordinates when the estimand-name map and selectors are permuted consistently.

The five economic block curves shown in Panel (D) are computed directly from (C.45). The 11 singleton-coordinate ratios are retained as supplementary diagnostics. No Cholesky-column, eigenvector, or other transformed probe enters the figure.

C.3.8. *Display-Only Reference-Moment Eigenbasis*

An eigensystem of the empirical reference moment is computed only for interpretability tables and a separate basis-loading figure. With $\widehat{K}_{\text{bar}}\widehat{v}_r = \widehat{\lambda}_r\widehat{v}_r$, define

$$R_{\text{disp}} := \min\{5, \#\{r : \widehat{\lambda}_r > 0\}, p\}, \qquad \widehat{V}_{\text{disp}} := [\widehat{v}_1, \ldots, \widehat{v}_{R_{\text{disp}}}], \tag{C.47}$$

and the display-only whitened scores

$$\widehat{Z}_i^{\text{disp}} := \widehat{\Lambda}_{\text{disp}}^{-1/2}\widehat{V}'_{\text{disp}}\widehat{E}_i. \tag{C.48}$$

This calculation occurs only after the full-coordinate Panel (B)-(D) objects have been formed. It cannot alter their paths, surfaces, block probes, or bootstrap bands and is not part of the estimand. It is used only to report the eigenspectrum, display rank, basis loadings, and a whitening check.

C.3.9. *Bootstrap of the Full Estimation Procedure*

The bootstrap uses $B = 2{,}000$ multinomial unit-weight draws with seed 20261614. For draw b,

$$(c_1^{(b)}, \ldots, c_n^{(b)}) \sim \text{Multinomial}(n; 1/n, \ldots, 1/n), \qquad \omega_i^{(b)} = c_i^{(b)} / \sum_j c_j^{(b)}. \tag{C.49}$$

Each draw used for the Panel (B) band repeats the complete estimator:

1. Refit the out-of-fold propensity, control prognostic, and treated prognostic models using draw-specific training weights $c_i^{(b)}$.
2. Rebuild the generated state, the draw-specific state scaler, and the observation weights.
3. Recompute $\widehat{\phi}_i^{(b)}$, all 11 influence coordinates, $\widehat{\beta}^{(b)}$, $\widehat{E}_i^{(b)}$, and $\widehat{\Gamma}_i^{(b)} = \widehat{E}_i^{(b)}\widehat{E}_i^{(b)\prime}$.
4. Rebuild the bandwidth, local kernels, local second-moment field, supported-node mask, quadrature node weights $q_{\text{B},g}^{(b)}$, induced reference weights $a_{\text{B},i}^{(b)}$, and $\widehat{C}_{\pi,\text{B}}^{(b)}$.

5. Rebuild $\widehat{D}^{(b)}_{\rho,\mathrm{B}}$, $\widehat{d}^{(b)}_{\rho,\mathrm{B}}$, and $\widehat{\tau}^{(b)}_{\mathrm{soft},\mathrm{B}}$.

No generated state, state scaler, bandwidth, kernel matrix, supported mask, quadrature law, $\widehat{C}_{\pi,\mathrm{B}}$, or soft-whitening solve from the original sample is reused. Display-only PCA is not required to form the band.

Panel (B) uses the fixed raw logit-propensity grid from the original sample. If r_g is the baseline raw-grid value at identifier g, draw b evaluates the smoother at the draw-specific standardized coordinate

$$x^{(b)}_g := \frac{r_g - \widehat{\mu}^{(b)}_x}{\widehat{\sigma}^{(b)}_x}. \tag{C.50}$$

Define the draw's raw-grid-indexed value by

$$\widetilde{\tau}^{(b)}_g := \widehat{\tau}^{(b)}_{\mathrm{soft},\mathrm{B}}(x^{(b)}_g). \tag{C.51}$$

Each draw is evaluated at the same raw coordinate r_g, after applying the draw-specific standardization, without interpolation from a draw-specific raw grid. At grid identifier g, the reported 90 percent pointwise percentile band is

$$\left[\widehat{Q}_{0.05}\{\widetilde{\tau}^{(b)}_g\}^B_{b=1}, \widehat{Q}_{0.95}\{\widetilde{\tau}^{(b)}_g\}^B_{b=1}\right]. \tag{C.52}$$

C.3.10. *Reported Specification*

TABLE VIII

LALONDE ESTIMATION SETTINGS

Quantity	Setting
Sample size; treated; controls	614; 185; 429
Estimand dimension	$p = 11$
Cross-fitting folds	5
Cross-fitting seed	20260614
Propensity clipping interval	$[0.02, 0.98]$
Soft ridge	$\rho = 0.03$
Panel (B) grid size	90
Panel (C) grid size	30×30
Panel (C) support threshold	$\mathrm{ESS}_\mathrm{C}(s) \geq 20$
Bootstrap draws	2,000
Bootstrap seed	20261614

Panels B and C use separate quadrature laws and separate $\widehat{C}_{\pi,j}$ matrices. Their minima, medians, maxima, effective dimensions, and color ranges are meaningful only relative to their own reference laws and are not compared as evidence that adding the prognostic score increases amplification. Figure text describes attenuated or amplified influence second moments, not state-specific treatment-effect standard errors. ESS, maximum local weight, and kernel-denominator diagnostics are reported alongside the panel estimates, so the index alone is not used to claim that a pattern is independent of data availability.

C.3.11. *Replication Checks*

A reconstruction should satisfy the following numerical invariants:

1. the resampling, training, empirical-moment, local-kernel, quadrature-node, and induced reference weights are normalized according to their distinct definitions;
2. $\sum_i \omega_i \widehat{\psi}_i \approx 0$, $\widehat{\Gamma}_i = \widehat{E}_i \widehat{E}_i'$, and $\widehat{K}_{\rm bar} = \sum_i \omega_i \widehat{\Gamma}_i$;
3. each $\widehat{\Gamma}_i$ is symmetric positive semidefinite, every supported kernel row sums to one, and every $\widehat{K}_{H_j}(s)$ is symmetric positive semidefinite up to numerical tolerance;
4. for each panel in quadrature mode, $\sum_g q_{jg} = 1$, $\widehat{C}_{\pi,j} \approx \sum_g q_{jg} \widehat{K}_{H_j}(s_{jg})$, and $\sum_g q_{jg} \widehat{\tau}_{{\rm soft},j}(s_{jg}) \approx 1$ on the supported quadrature set;
5. Panels B and C construct separate q_{jg}, a_{ji}, and $\widehat{C}_{\pi,j}$ objects, and their numerical scales are not treated as directly comparable;
6. the aggregate soft index is computed by Cholesky solves involving $\widehat{C}_{\pi,j} + \rho I_p$;
7. each singleton Panel (D) ratio equals $(\widehat{K}_{H_{\rm B}}(s))_{jj}/(\widehat{C}_{\pi,{\rm B}})_{jj}$, and each economic block ratio equals the sum of selected local diagonal entries divided by the corresponding reference sum;
8. Panel (D) is invariant to an internal coordinate permutation when the estimand-name map is permuted consistently and depends only on original estimand coordinates;
9. omitting the display-only reference-moment eigensystem leaves Panels B-D numerically unchanged;
10. invalid two-dimensional nodes remain missing, while ESS, maximum-weight, and denominator diagnostics are retained; and
11. every bootstrap draw refits the nuisance functions, generated state, state scaler, bandwidth, supported set, quadrature law, reference moment, and soft normalization before evaluation on the common raw Panel (B) grid.

C.4. *Interpretive limits common to both applications*

The applications locate states in which the influence process of a chosen estimator has a large normalized local second moment. They do not replace the sampling covariance used for formal inference. In the hurricane application, county-clustered event-study standard errors remain the relevant conventional inferential object; the OVK field is observation-level and state indexed. In the LaLonde application, the full-estimation bootstrap quantifies uncertainty in the estimated amplification surface, but $\widehat{\tau}(s)$ is still an influence second-moment diagnostic rather than a confidence interval for a treatment effect.

The state also has a different substantive meaning in each application. A hurricane node is a division-year location in a smoothed spatiotemporal score field. A LaLonde node is a support location generated from estimated nuisance functions. High values can therefore arise from different mechanisms, including leverage concentration, sparse local support, large residual scores, poor overlap, or a nonzero local score mean. The mean/within decomposition in the hurricane output, the ESS and maximum-weight diagnostics in both applications, and the full-coordinate directional or block summaries in the LaLonde output should be examined before attaching an economic interpretation to any peak.

APPENDIX D: EXTRA FIGURES AND TABLES

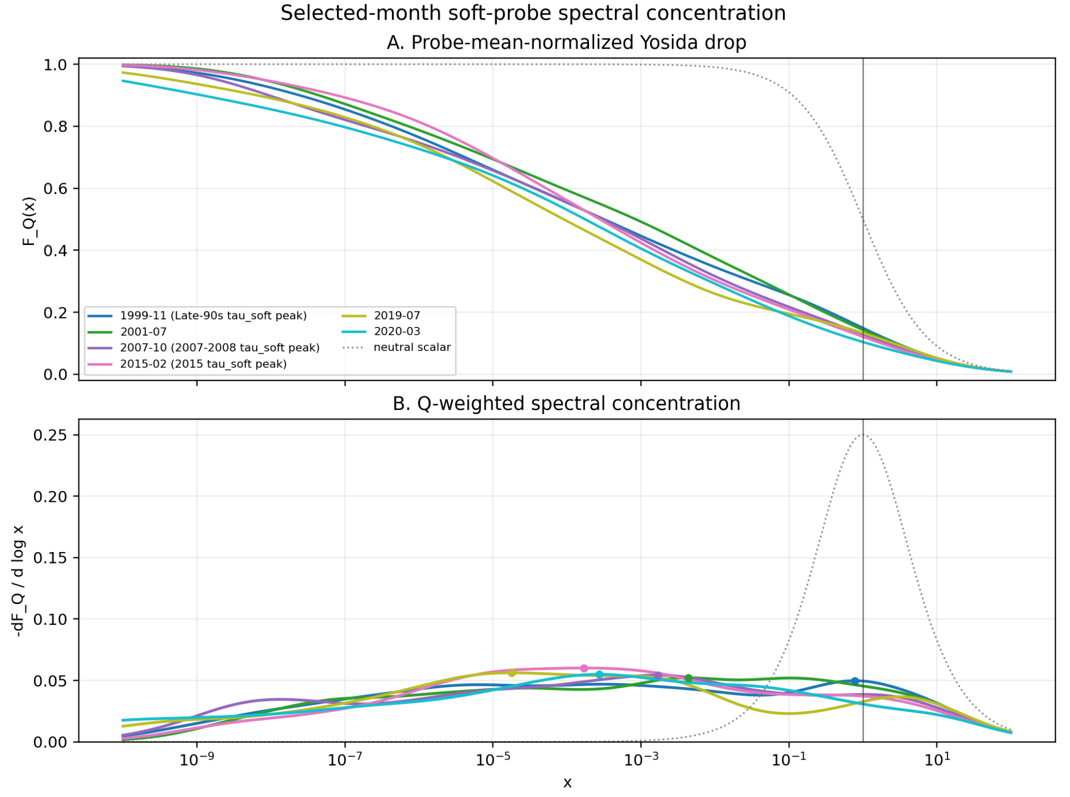
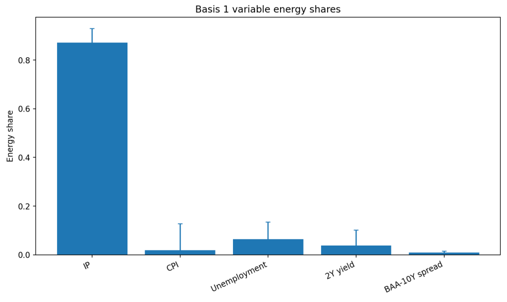
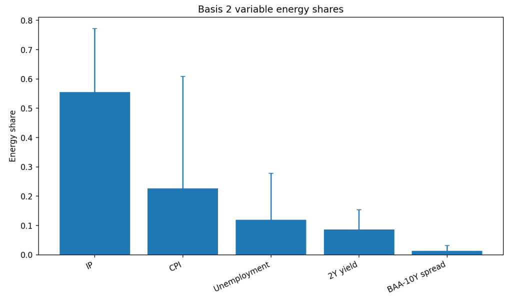


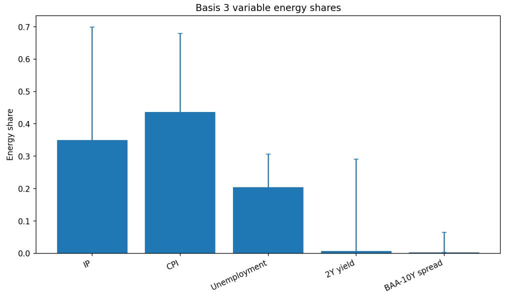
FIGURE 6.—**Selected-month soft-probe spectral concentration after removing aggregate diagonal second-moment scale.** For month t, let $A_t \succeq 0$ denote the estimated relative diagonal score second-moment operator, let $Q \succeq 0$ be the trace-one soft probe, and let $\nu_{t,Q}(D) = \operatorname{tr}\{QE_t(D)\}$ be the corresponding Q-weighted spectral probability measure, where E_t is the spectral resolution of A_t. Define the probe mean $\tau_{t,Q} = \operatorname{tr}(QA_t) = \int_{[0,\infty)} a\,\nu_{t,Q}(da)$,and normalize spectral amplification by $b = a/\tau_{t,Q}$. If $\bar{\nu}_{t,Q}$ denotes the induced law of b, then $\int b\,\bar{\nu}_{t,Q}(db) = 1$. **Panel (A)** plots the probe-mean-normalized Yosida drop $\mathrm{F}_{t,Q}(x) = \int_{[0,\infty)} \frac{b}{x+b}\,\bar{\nu}_{t,Q}(db), \qquad x > 0$.A normalized spectral value $b = x$ contributes one half, values $b \gg x$ contribute approximately one, and values $b \ll x$ contribute approximately zero. The dotted curve is the scalar-neutral benchmark $\bar{\nu}_{t,Q} = \delta_1$, for which $F_0(x) = (1+x)^{-1}$; the vertical line at $x = 1$ marks the probe-mean spectral scale. Since $b \mapsto b/(x+b)$ is strictly concave and $E_{\bar{\nu}_{t,Q}}[b] = 1$, Jensen's inequality gives $F_{t,Q}(x) \leq (1+x)^{-1}$. Thus departures below the dotted curve measure dispersion of the normalized diagonal second–moment geometry rather than differences in its overall scale. **Panel (B)** plots the corresponding log-scale derivative -$\frac{d}{d\log x}\,F_{t,Q}(x) = \int_{[0,\infty)} \frac{xb}{(x+b)^2}\,\bar{\nu}_{t,Q}(db)$.For $b > 0$, its kernel satisfies $\frac{xb}{(x+b)^2} = \frac{1}{4\cosh^2\{(\log x - \log b)/2\}}$,so the lower panel is a logistic smoothing of the Q-weighted spectrum on the logarithmic scale. Spectral mass near b_0 contributes a bump centered near $x = b_0$ and produces the associated decline in $F_{t,Q}$ at that scale. The colored markers identify the maxima of the selected-month smoothed concentration profiles. Relative to the scalar-neutral peak of $1/4$ at $x = 1$, the broad and substantially lower colored profiles indicate that the selected-month spectra are dispersed across multiple multiplicative amplification scales rather than concentrated around a single common scale. Because the derivative is smoothed, its peaks describe modes of the logistic-smoothed Q-weighted log spectrum, not individual eigenvalues or an unsmoothed spectral density. The month-specific normalization removes $\tau_{t,Q}$, so the figure compares spectral *shape*; it does not rank the months by their absolute diagonal score second-moment amplification.



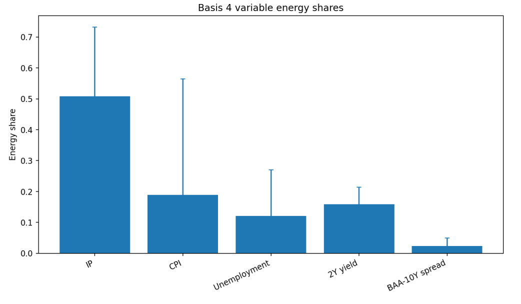
(a) Basis 1



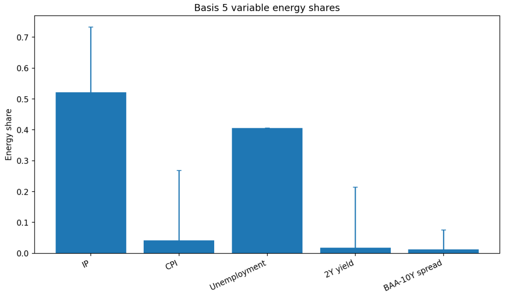
(b) Basis 2



(c) Basis 3



(d) Basis 4



(e) Basis 5

FIGURE 7.—Dominant eigenmodes of the sample-average diagonal response-score second-moment operator in the monetary policy local projections example. Each panel reports the variable-level energy decomposition for one of the first five display basis functions, with energy shares normalized within eigenmode across industrial production (IP), CPI, unemployment, the two-year yield, and the BAA-10-year credit spread. Bars give the estimated share of total modal energy attributable to each variable, and vertical whiskers denote 90% bands. The leading mode is concentrated primarily in IP, while subsequent modes assign relatively more energy to inflation, labor-market conditions, and interest-rate or credit-spread variables, indicating that the dominant display eigenmodes capture distinct macro-financial sources of score variation.

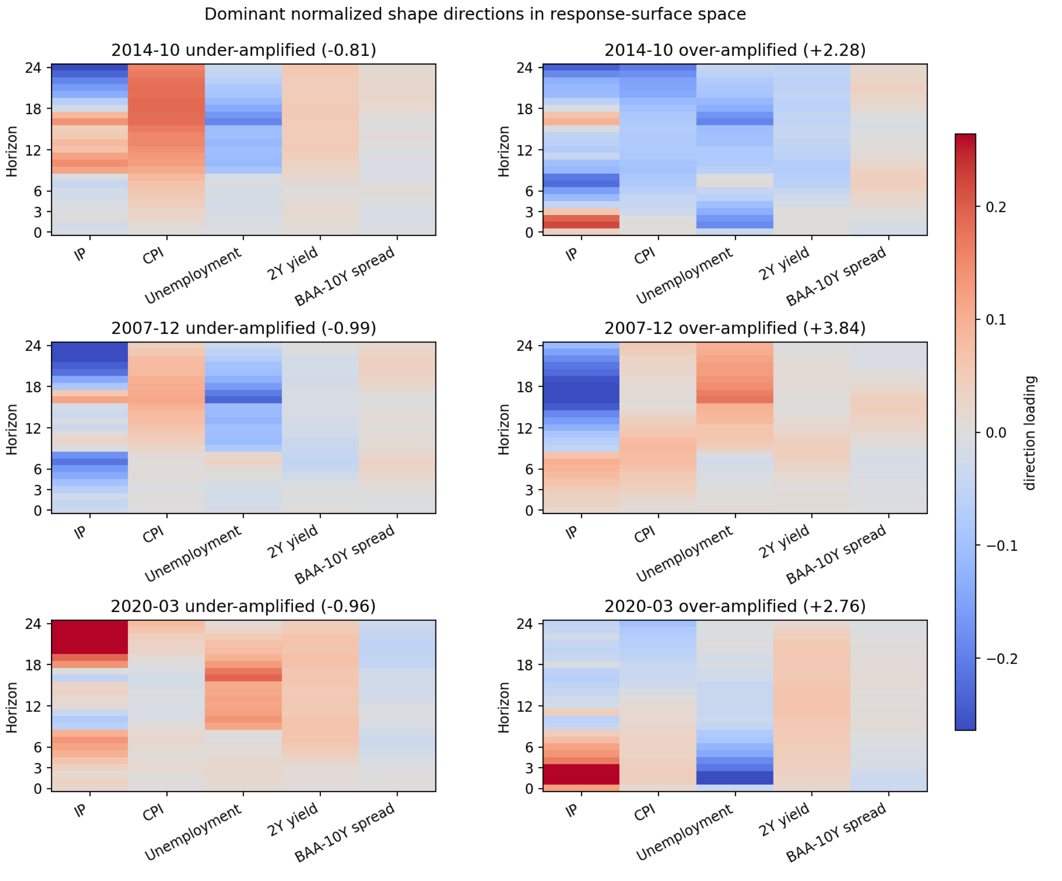


FIGURE 8.—Dominant normalized shape directions in response-surface space.

Notes: Let $V \in \mathbb{R}^{125\times 5}$ denote the rank-five display basis, with rows indexed by outcome–horizon cells, and let $\widehat{A}_{t,R} \in \mathbb{R}^{5\times 5}$ denote the estimated arithmetic diagonal score second-moment matrix at month t. Define

$$\widehat{\tau}_{t,R} = \frac{1}{5}\,\mathrm{tr}(\widehat{A}_{t,R}), \qquad \widehat{S}_{t,R} = \frac{\widehat{A}_{t,R}}{\widehat{\tau}_{t,R}}, \qquad \mathrm{tr}(\widehat{S}_{t,R}) = 5.$$

The displayed directions satisfy

$$(\widehat{S}_{t,R} - I_5)u_{tk} = \delta_{tk}u_{tk}.$$

For each selected month, the left panel shows the direction associated with the smallest eigenvalue δ_{tk}, and the right panel shows the direction associated with the largest eigenvalue. The plotted response-surface direction is Vu_{tk}, reshaped over the five outcomes and horizons $h = 0, \ldots, 24$. The labels "under-amplified" and "over-amplified" refer to the eigenvalue δ_{tk} of $\widehat{S}_{t,R} - I_5$, not to the sign of the heatmap colors. Parenthetical values report δ_{tk}; consequently, $1 + \delta_{tk}$ is the second-moment amplification along the displayed direction relative to the month-specific neutral scale. For example, $\delta_{tk} = -0.81$ corresponds to 0.19 times the neutral scale, whereas $\delta_{tk} = 3.84$ corresponds to 4.84 times the neutral scale. Heatmap colors show the signed loadings of Vu_{tk} across outcomes and horizons. Because the sign of an eigenvector is arbitrary, the colors should be interpreted only as the relative pattern of a direction within a panel; they do not represent positive or negative impulse responses. The figure therefore identifies response-surface patterns that are relatively compressed or expanded by the normalized diagonal second-moment shape, conditional on the estimated LP response scores, the chosen display basis, and the fitted arithmetic second-moment smoothing parameter.

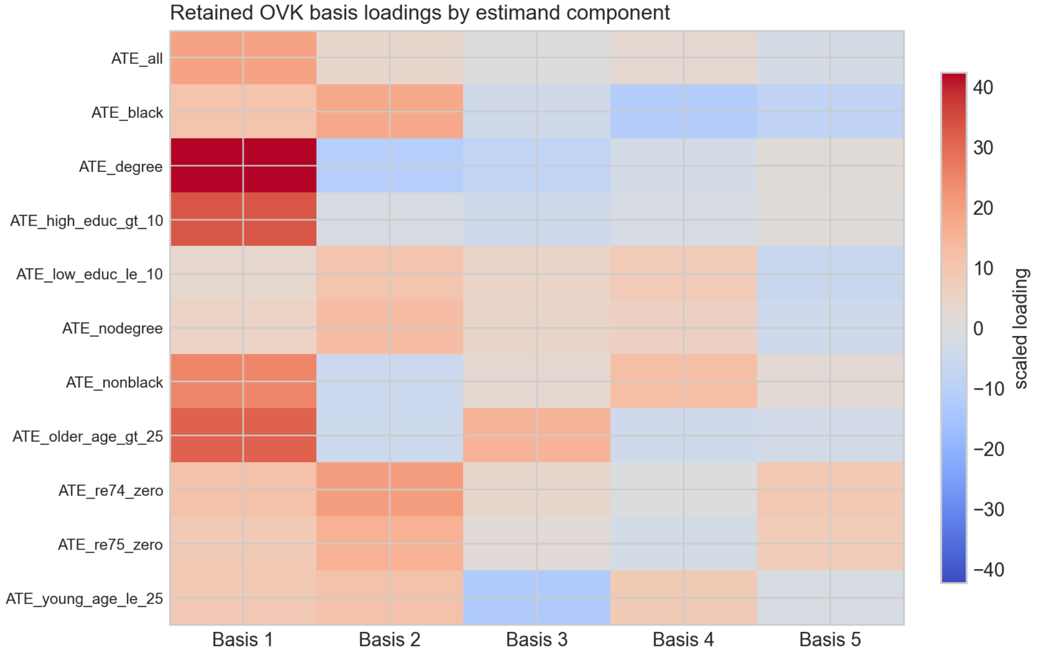


FIGURE 9.—Display-only reference-moment eigenbasis loadings for the LaLonde cross-sectional application. Rows correspond to coordinates of the fixed estimand vector, including the overall ATE and subgroup ATEs; columns correspond to the first five display eigen-directions of the average influence second-moment operator. Colors report signed scaled loadings, with larger absolute values indicating estimand coordinates that contribute more strongly to a common direction of influence variation. This does not represent treatment-effect estimates. Since eigenvector signs are arbitrary, interpretation should focus on relative magnitudes and clustering of loadings across estimand coordinates rather than the sign of any individual cell.

APPENDIX E: SPECTRAL COMPRESSION FOR INTERPRETATION

The main text estimates relative moment geometry and bounded stress without selecting a spectral rank. This appendix records the eigenbasis representation used to interpret a fitted field and to establish Galerkin convergence. It also gives fixed-rank sampling theory for the rank-R displays in Appendix Figures 7-9.

Finite-rank spectral compression supplies interpretable coordinates and shape decompositions of the full-space population object. The formulas below are written for the centered contribution χ, so that $K(s) = \mathbb{E}[\chi \otimes \chi \mid S = s]$. The same statements apply to the full conditional second-moment target after replacing χ by ψ. Fix a measurable representative of the declared conditional moment on a set of full π-measure. The absolute-continuity condition $\pi \ll \mathcal{L}(S)$ is sufficient to ensure that any two measurable versions agree π-almost everywhere. If $\pi \not\ll \mathcal{L}(S)$, values assigned on $\mathcal{L}(S)$-null sets that carry positive π-mass are part of the researcher-specified reference target, not identified conditional moments.

PROPOSITION 7: *Under Assumption 2,*

$$C = \sum_{\ell \geq 1} \lambda_\ell v_\ell \otimes v_\ell, \tag{E.1}$$

where $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$, $\sum_{\ell \geq 1} \lambda_\ell < \infty$, and the eigenvectors associated with positive eigenvalues form an orthonormal basis of $\mathcal{H}_C$.

PROOF: This is the spectral theorem for positive trace-class, hence compact, self-adjoint operators. *Q.E.D.*

For $R < \infty$ with $\lambda_R > 0$, let $V_R a := \sum_{\ell=1}^{R} a_\ell v_\ell$, $\Lambda_R := \operatorname{diag}(\lambda_1, \ldots, \lambda_R)$, and $P_R := V_R V_R^*$. For a centered influence contribution χ, define

$$z^{(R)} := \Lambda_R^{-1/2} V_R^* \chi. \tag{E.2}$$

THEOREM 4—Finite-rank spectral implementation: *Suppose Assumptions 2-4 hold and a measurable representative $K(s) = \mathbb{E}[\chi \otimes \chi \mid S = s]$ has been fixed for π-almost every s. Define*

$$A_R(s) := \Lambda_R^{-1/2} V_R^* K(s) V_R \Lambda_R^{-1/2}, \tag{E.3}$$

$$A_{\eta,R}(s) := \Lambda_R^{-1/2} V_R^* K_\eta(s) V_R \Lambda_R^{-1/2}. \tag{E.4}$$

Then $A_R(s) = \mathbb{E}[z^{(R)} z^{(R)\prime} \mid S = s]$, $\int A_R(s)\,\pi(ds) = I_R$, and $A_{\eta,R} = J_\eta A_R$ entry by entry. Moreover,

$$A_{\eta,R}(s) = V_R^* A_\eta^\infty(s) V_R, \qquad K_{\eta,R}(s) := V_R \Lambda_R^{1/2} A_{\eta,R}(s) \Lambda_R^{1/2} V_R^* = P_R K_\eta(s) P_R. \tag{E.5}$$

Thus $\tau_{\eta,R}(s) := R^{-1} \operatorname{tr}\{A_{\eta,R}(s)\}$ is rotation invariant, has π-average one, and equals the probe $\operatorname{tr}\{Q_R A_\eta^\infty(s)\}$ for $Q_R := R^{-1} P_R$. As $R \to \infty$, $\|K_{\eta,R}(s) - K_\eta(s)\|_1 \to 0$ and $P_R A_\eta^\infty(s) P_R x \to A_\eta^\infty(s) x$ for every $x \in \mathcal{H}_C$. Under (2.8)*,*

$$\|K_\eta(s) - P_R K_\eta(s) P_R\|_1 \leq 2 M_\eta(s) \left\{ \operatorname{tr}(C) \sum_{\ell > R} \lambda_\ell \right\}^{1/2}. \tag{E.6}$$

PROOF: See Appendix A.8. *Q.E.D.*

The theorem makes the relationship explicit: the eigenbasis of C provides a Galerkin compression and a particular trace-class probe of the full relative moment, not a separate population construction. There is no canonical limit of $R^{-1}P_R$ in trace norm, so $\tau_{\eta,R}$ need not converge to a "full trace divided by dimension."

COROLLARY 4—Galerkin approximation of Yosida stress: *Let*

$$Y_{\eta,\lambda,R}(s) := A_{\eta,R}(s)\{\lambda I_R + A_{\eta,R}(s)\}^{-1}$$

Then $V_R Y_{\eta,\lambda,R}(s) V_R^* x \to Y_{\eta,\lambda}(s)x$ *for every* $x \in \mathcal{H}_C$. *Hence fixed directional and trace-class-probe stress summaries converge to their full-space counterparts.*

PROOF: See Appendix A.9. *Q.E.D.*

E.1. *Post-Estimation Rank-R Displays*

After fitting the full-coordinate moment field in Section 2.5, a rank-R display may be formed without refitting the model. Let $\widehat{V}_R$ and $\widehat{\Lambda}_R$ be the leading eigenvectors and eigenvalues of $\widehat{C}_n$. Define

$$\widehat{A}_{R,j}^{\mathrm{disp}} := \widehat{\Lambda}_R^{-1/2} \widehat{V}_R^* \widehat{K}_{j,n,\rho} \widehat{V}_R \widehat{\Lambda}_R^{-1/2}. \tag{E.7}$$

Changing R changes the resolution of the display, not the fitted moment field $\widehat{K}_{j,n,\rho}$. Surface directions, scale-free allocations, and rank-R stress plots may all be constructed from $\widehat{A}_{R,j}^{\mathrm{disp}}$.

E.2. *Ritz-Residual Certification*

A candidate display space can be assessed against the fitted relative moment operator itself. Fix a state, suppress its index, and write $\widehat{A} := \widehat{A}_{\eta,\rho,n}(s)$. Let $U : \mathbb{R}^R \to \mathcal{H}$ be an isometry, set $P := UU^*$, and define the Ritz matrix and block residual

$$\Theta := U^* \widehat{A} U, \qquad \mathcal{R}_U := \widehat{A}U - U\Theta = (I-P)\widehat{A}U. \tag{E.8}$$

The equality uses the Galerkin orthogonality $U^*\mathcal{R}_U = 0$, and

$$\|\mathcal{R}_U\|_{\mathrm{op}} = \|(I-P)\widehat{A}P\|_{\mathrm{op}}.$$

Thus the second term in (2.23) is directly computable from the same residual used to assess a Rayleigh-Ritz approximation.

If $\Theta y_j = \theta_j y_j$, $\|y_j\| = 1$, and $u_j := U y_j$, then (θ_j, u_j) is a Ritz pair with residual

$$r_j := \widehat{A}u_j - \theta_j u_j = \mathcal{R}_U y_j. \tag{E.9}$$

Because $\widehat{A}$ is self-adjoint,

$$\operatorname{dist}\{\theta_j, \sigma(\widehat{A})\} \le \|r_j\|. \tag{E.10}$$

If a target spectral cluster $\Sigma_\star$ is separated from the remainder of the spectrum and $E_\star$ is its spectral projector, then, whenever

$$\delta_j := \inf_{a \in \sigma(\widehat{A}) \setminus \Sigma_\star} |a - \theta_j| > 0,$$

the spectral theorem also gives the residual-to-gap certificate

$$\|(I-E_\star)u_j\| \le \frac{\|r_j\|}{\delta_j}. \tag{E.11}$$

A small residual therefore certifies proximity to the spectrum, while a small residual together with an established gap certifies the corresponding direction. Residual size alone does not establish that every relevant eigenvalue has been found; interval eigensolvers supplement it with search-space oversampling and eigenvalue-count checks.

The certificate can be computed without explicitly forming a whitening square root. Let $D_n := \widehat{C}_n + \rho I$ and $\widehat{K} := \widehat{K}_{\eta,n}(s)$. If the columns of V are D_n-orthonormal, $V^* D_n V = I_R$, set $U := D_n^{1/2} V$. Then

$$\begin{aligned}
\Theta &= V^* \widehat{K} V, \\
\mathcal{R}_U &= D_n^{-1/2}\{\widehat{K}V - D_n V\Theta\}, \\
\|\mathcal{R}_U\|_{\mathrm{op}}^2 &= \lambda_{\max}\left[\{\widehat{K}V - D_n V\Theta\}^* D_n^{-1}\{\widehat{K}V - D_n V\Theta\}\right].
\end{aligned} \tag{E.12}$$

The last expression requires only solves with D_n. For an individual generalized Ritz pair (θ_j, v_j), define $\|r\|_{D_n^{-1}} := \langle r, D_n^{-1} r\rangle^{1/2}$; the relevant residual norm is $\|\widehat{K}v_j - \theta_j D_n v_j\|_{D_n^{-1}}$. In applications one can report the omitted probe mass and the largest residual term over the displayed states and stress scales. A small value of their sum certifies the finite-dimensional stress calculation conditional on the estimated operator; it is not a substitute for accounting for first-stage and moment-estimation uncertainty.

E.3. *Fixed-Rank Feasibility and Limit Theory*

For completeness, consider the fixed-rank reduced-coordinate estimator. Let $\widehat{z}_{i,n}^{(R)} := \widehat{\Lambda}_R^{-1/2}\widehat{V}_R^*\widehat{\chi}_{i,n}$ and $\widehat{A}_{\eta,R,n}(s) := \sum_i w_{\eta,n,i}(s)\widehat{z}_{i,n}^{(R)}\widehat{z}_{i,n}^{(R)\prime}$. The normalized Euclidean kernel special case is

$$\widehat{A}_R(s) = \frac{\displaystyle\sum_{i=1}^n \mathcal{K}\{H_n^{-1/2}(S_{i,n}-s)\}\widehat{z}_{i,n}^{(R)}\widehat{z}_{i,n}^{(R)\prime}}{\displaystyle\sum_{i=1}^n \mathcal{K}\{H_n^{-1/2}(S_{i,n}-s)\}}. \tag{E.13}$$

For the finite-rank estimator, let $\widehat{C}_n = \sum_{\ell\ge 1}\widehat{\lambda}_{\ell,n}\widehat{v}_{\ell,n}\otimes\widehat{v}_{\ell,n}$ and define $\widehat{V}_R$, $\widehat{\Lambda}_R$, and $\widehat{P}_R$ in the natural way.

ASSUMPTION 8—Fixed-rank spectral regularity: *For fixed $R<\infty$, $\lambda_R \ge c_\lambda > 0$ and $\lambda_R - \lambda_{R+1} \ge c_\delta > 0$. Partition the positive spectrum into maximal repeated-eigenvalue clusters and assume that R is a cluster boundary. Distinct retained clusters are separated by a positive gap, and alignment is performed cluster-by-cluster.*

PROPOSITION 8—Fixed-rank feasibility: *Under Assumptions 5 and 8, $\|\widehat{P}_R - P_R\|_{\mathrm{op}} = O_p(r_{C,n})$. There is an orthogonal $O_n \in \mathbb{R}^{R\times R}$ such that $\|\widehat{V}_R O_n' - V_R\|_{\mathrm{op}} = O_p(r_{C,n})$ and*

$$\left\|O_n\widehat{\Lambda}_R^{-1/2}\widehat{V}_R^* - \Lambda_R^{-1/2}V_R^*\right\|_{\mathrm{op}} = O_p(r_{C,n}).$$

Moreover,

$$\widehat{A}_{\eta,R,n}(s) = \widehat{\Lambda}_R^{-1/2}\widehat{V}_R^*\widehat{K}_{\eta,n}(s)\widehat{V}_R\widehat{\Lambda}_R^{-1/2}, \tag{E.14}$$

satisfies

$$\|O_n\widehat{A}_{\eta,R,n}(s)O_n' - A_{\eta,R}(s)\|_{\mathrm{op}} = O_p\{r_{K,n}(s) + r_{C,n}\} = o_p(1).$$

Hence $R^{-1}\operatorname{tr}\{\widehat{A}_{\eta,R,n}(s)\} \to_p \tau_{\eta,R}(s)$.

PROOF: The eigenspace and whitening-map steps are proved in Appendix A.15; the moment step is proved in Appendix A.16. *Q.E.D.*

For distributional theory, define

$$z_{i,n}^{(R)} := \Lambda_R^{-1/2}V_R^*\chi_{i,n}, \qquad A_{\eta,R,n}^{\mathrm{or}}(s) := \sum_i w_{\eta,n,i}(s)z_{i,n}^{(R)}z_{i,n}^{(R)\prime},$$

and let $a_{\eta,n}(s) := N_{\eta,n}(s)^{1/2}$.

ASSUMPTION 9—Weighted finite-dimensional CLT: *For fixed s, the weights satisfy $w_{\eta,n,i}(s) \geq 0$, $\sum_i w_{\eta,n,i}(s) = 1$, and $N_{\eta,n}(s) \to \infty$, with $a_{\eta,n}(s) = N_{\eta,n}(s)^{1/2}$. Assume*

$$a_{\eta,n}(s)\operatorname{vech}\left[A_{\eta,R,n}^{\mathrm{or}}(s) - \sum_i w_{\eta,n,i}(s)A_R(S_{i,n})\right] \Rightarrow N\{0, \Sigma_{\eta,R}(s)\},$$

while $a_{\eta,n}(s)\operatorname{vech}[\sum_i w_{\eta,n,i}(s)A_R(S_{i,n}) - A_{\eta,R}(s)] \to_p b_{\eta,R}(s)$.
With $e_{i,n} := O_n\widehat{z}_{i,n}^{(R)} - z_{i,n}^{(R)}$, *assume also*

$$a_{\eta,n}(s)^2\sum_i w_{\eta,n,i}(s)\|e_{i,n}\|^2 = o_p(1), \qquad \sum_i w_{\eta,n,i}(s)\|z_{i,n}^{(R)}\|^2 = O_p(1).$$

PROPOSITION 9—Generic state-smoother limit theory: *Under Assumption 9,*

$$a_{\eta,n}(s)\operatorname{vech}\Big[O_n\widehat{A}_{\eta,R,n}(s)O_n' - A_{\eta,R}(s)\Big] \Rightarrow N\{b_{\eta,R}(s), \Sigma_{\eta,R}(s)\}.$$

If $b_{\eta,R}(s) = 0$, the feasible estimator is asymptotically centered at the smooth state target.

PROOF: See Appendix A.17. *Q.E.D.*

APPENDIX F: VERIFYING THE PRIMITIVE MOMENT-RATE CONDITION (IV)

Assumption 6(iv) is a mean-square requirement for the estimated state-centered influence contribution. Write

$$\chi_{i,n} := \psi_{i,n} - \mathbb{E}[\psi_{i,n} \mid \mathcal{G}_{i,n}], \qquad \widehat{\chi}_{i,n} := \widehat{\psi}_{i,n} - \widehat{m}^{\psi}_{i,n}.$$

The relevant empirical norm may be global, with weights $a_{i,n} = 1/n$, or state-local, with weights $a_{i,n} = w_{\eta,n,i}(s)$.

LEMMA 3—Weighted mean-square plug-in verification: *Suppose*

$$\widehat{\chi}_{i,n} = \varphi_n(W_{i,n}; \widehat{\gamma}_n), \qquad \chi_{i,n} = \varphi_n(W_{i,n}; \gamma_{0,n}),$$

where d_n is a metric on the first-stage parameter or nuisance space. Let $a_{i,n} \geq 0$, $\sum_i a_{i,n} = 1$, and let $\mathcal{N}_n$ be a neighborhood of $\gamma_{0,n}$ such that $\mathbb{P}(\widehat{\gamma}_n \in \mathcal{N}_n) \to 1$. Define

$$L^2_{a,n} := \sup_{\gamma \in \mathcal{N}_n : \gamma \neq \gamma_{0,n}} \frac{\sum_{i=1}^n a_{i,n} \|\varphi_n(W_{i,n}; \gamma) - \varphi_n(W_{i,n}; \gamma_{0,n})\|^2_{\mathcal{H}}}{d_n(\gamma, \gamma_{0,n})^2}.$$

If

$$L_{a,n} = O_p(1), \qquad d_n(\widehat{\gamma}_n, \gamma_{0,n}) = O_p(r_n),$$

then

$$\sum_{i=1}^n a_{i,n} \|\widehat{\chi}_{i,n} - \chi_{i,n}\|^2_{\mathcal{H}} = O_p(r_n^2).$$

PROOF: On the event $\{\widehat{\gamma}_n \in \mathcal{N}_n\}$, the definition of $L_{a,n}$ gives

$$\sum_{i=1}^n a_{i,n} \|\widehat{\chi}_{i,n} - \chi_{i,n}\|^2 \leq L^2_{a,n} d_n(\widehat{\gamma}_n, \gamma_{0,n})^2 = O_p(r_n^2).$$

The complement of the event has probability tending to zero. *Q.E.D.*

Applying Lemma 3 with $a_{i,n} = 1/n$ verifies Assumption 6(iv) with the global first-stage rate $d_{C,n}$. Applying it with $a_{i,n} = w_{\eta,n,i}(s)$ verifies the state-weighted rate $d_{K,n}(s)$. The two rates need not be identical if the nuisance estimator is less accurate in some parts of the state space.

Derivative-based check

Suppose the nuisance parameter lies in a normed linear space $(\Gamma_n, \|\cdot\|_{\Gamma_n})$, let $d_n(\gamma, \gamma_{0,n}) = \|\gamma - \gamma_{0,n}\|_{\Gamma_n}$, and suppose there is a bounded linear derivative

$$\dot{\varphi}_n(W_{i,n}; \gamma_{0,n}) : \Gamma_n \to \mathcal{H}$$

such that, uniformly for $\gamma \in \mathcal{N}_n$,

$$\varphi_n(W_{i,n}; \gamma) - \varphi_n(W_{i,n}; \gamma_{0,n}) = \dot{\varphi}_n(W_{i,n}; \gamma_{0,n})[\gamma - \gamma_{0,n}] + R_{i,n}(\gamma).$$

If

$$\sum_i a_{i,n}\|\dot{\varphi}_n(W_{i,n};\gamma_{0,n})\|_{\mathrm{op}}^2 = O_p(1)$$

and

$$\sup_{\gamma\in\mathcal{N}_n:\gamma\neq\gamma_{0,n}} \frac{\sum_i a_{i,n}\|R_{i,n}(\gamma)\|_{\mathcal{H}}^2}{d_n(\gamma,\gamma_{0,n})^2} = o_p(1),$$

then $L_{a,n} = O_p(1)$. Indeed, $\|u+v\|^2 \le 2\|u\|^2 + 2\|v\|^2$ gives

$$\sum_i a_{i,n}\|\varphi_n(W_{i,n};\gamma) - \varphi_n(W_{i,n};\gamma_{0,n})\|^2$$
$$\le 2d_n(\gamma,\gamma_{0,n})^2 \sum_i a_{i,n}\|\dot{\varphi}_n(W_{i,n};\gamma_{0,n})\|_{\mathrm{op}}^2 + 2\sum_i a_{i,n}\|R_{i,n}(\gamma)\|^2.$$

Dividing by $d_n(\gamma,\gamma_{0,n})^2$ proves the claim.

Finite-dimensional parametric check

Suppose $\gamma_{0,n}\in\mathbb{R}^q$ with fixed q, $\widehat{\gamma}_n - \gamma_{0,n} = O_p(r_n)$, and let $\mathcal{N}_n$ be a convex neighborhood such that $\mathbb{P}(\widehat{\gamma}_n\in\mathcal{N}_n)\to 1$ and

$$\sup_{\gamma\in\mathcal{N}_n} \sum_i a_{i,n}\|\nabla_\gamma\varphi_n(W_{i,n};\gamma)\|_{\mathrm{op}}^2 = O_p(1).$$

On the event $\{\widehat{\gamma}_n\in\mathcal{N}_n\}$, set $\Delta_{\gamma,n} := \widehat{\gamma}_n - \gamma_{0,n}$ and $\gamma_{t,n} := \gamma_{0,n} + t\Delta_{\gamma,n}$. The integral mean-value formula and Cauchy-Schwarz give

$$\sum_i a_{i,n}\|\widehat{\chi}_{i,n} - \chi_{i,n}\|^2 \le \|\Delta_{\gamma,n}\|^2 \int_0^1 \sum_i a_{i,n}\|\nabla_\gamma\varphi_n(W_{i,n};\gamma_{t,n})\|_{\mathrm{op}}^2 dt$$
$$\le \|\Delta_{\gamma,n}\|^2 \sup_{\gamma\in\mathcal{N}_n} \sum_i a_{i,n}\|\nabla_\gamma\varphi_n(W_{i,n};\gamma)\|_{\mathrm{op}}^2 = O_p(r_n^2).$$

Thus a root-n first stage gives an $O_p(n^{-1})$ mean-square contribution for both equal and local weights whenever the weighted derivative moment is bounded.

Example: smooth Z-estimator centered influence contributions

Consider a finite-dimensional smooth Z-estimator with moment contribution $g(W_i;\theta)\in\mathbb{R}^p$ and uncentered influence contribution

$$\psi_i = -G_0^{-1}g(W_i;\theta_0), \qquad G_0 := \mathbb{E}\left[\frac{\partial g(W_i;\theta_0)}{\partial\theta'}\right].$$

The state-centered contribution is

$$\chi_i = \psi_i - m_i^\psi, \qquad m_i^\psi := \mathbb{E}[\psi_i \mid \mathcal{G}_i].$$

Let

$$\widehat{\chi}_i = \widehat{\psi}_i - \widehat{m}_i^{\psi}, \qquad \widehat{\psi}_i = -\widehat{G}^{-1} g(W_i;\widehat{\theta}), \qquad \widehat{G} = n^{-1}\sum_{i=1}^{n} \frac{\partial g(W_i;\widehat{\theta})}{\partial \theta'}.$$

For any nonnegative weights a_i summing to one,

$$\sum_i a_i \|\widehat{\chi}_i - \chi_i\|^2 \le 2\sum_i a_i \|\widehat{\psi}_i - \psi_i\|^2 + 2\sum_i a_i \|\widehat{m}_i^{\psi} - m_i^{\psi}\|^2.$$

Hence Assumption 6(iv) follows if both terms on the right have the required global or state-weighted rate.

For a deterministic sequence $r_n \to 0$, a sufficient condition for the influence plug-in term is $\widehat{\theta} - \theta_0 = O_p(r_n)$, $\|\widehat{G} - G_0\|_{\mathrm{op}} = O_p(r_n)$, $\lambda_{\min}(G_0' G_0) \ge c_G > 0$, and

$$\sum_i a_i \|g(W_i;\theta_0)\|^2 = O_p(1), \qquad \sup_{\theta \in \mathcal{N}} \sum_i a_i \left\| \frac{\partial g(W_i;\theta)}{\partial \theta'} \right\|_{\mathrm{op}}^2 = O_p(1).$$

Because $r_n \to 0$, the singular-value bound and the stated rate for $\|\widehat{G} - G_0\|_{\mathrm{op}}$ imply that $\widehat{G}$ is invertible with probability tending to one and $\|\widehat{G}^{-1} - G_0^{-1}\|_{\mathrm{op}} = O_p(r_n)$. Under these conditions,

$$\widehat{\psi}_i - \psi_i = -\widehat{G}^{-1}\{g(W_i;\widehat{\theta}) - g(W_i;\theta_0)\} - (\widehat{G}^{-1} - G_0^{-1}) g(W_i;\theta_0),$$

and the weighted empirical mean square is $O_p(r_n^2)$. If the uncentered score is already conditionally mean zero relative to $\mathcal{G}_i$, then $m_i^{\psi} = 0$ and the centering estimation condition is vacuous.

Remark on slower nuisance estimators

Neyman orthogonality can make the sample average of a score insensitive to slower nuisance estimation without automatically giving a fast observation-level mean-square rate. For a first-order Lipschitz influence map, Lemma 3 gives a weighted mean-square error of order $d_n(\widehat{\gamma}_n, \gamma_{0,n})^2$. More generally, if

$$\sum_i a_{i,n} \|\varphi_n(W_{i,n};\widehat{\gamma}_n) - \varphi_n(W_{i,n};\gamma_{0,n})\|^2 = O_p\{d_n(\widehat{\gamma}_n, \gamma_{0,n})^{2a}\}$$

for some $a \ge 1$, then a first-stage rate $d_n(\widehat{\gamma}_n, \gamma_{0,n}) = O_p(r_n^{1/a})$ is sufficient for a target mean-square rate $O_p(r_n^2)$. The required rate must be checked in the state weights actually used by the moment smoother, not only under the global empirical norm.